BT405
NANO BIOTECHNOLOGY

FINAL TERM MERGED PPTS

TOPIC : 46 TO 90

BY TABIH FATIMA
ADMIN: ALL ABOUT BIOTECH

Gene chip

Learning Objectives

What is Gene chip

Microarray

Gene chip

- There are different names for the microarrays, like DNA/RNA Chips, BioChips or GeneChips.
- The array can be defined as an ordered collection of microspots, each spot containing a single defined species of a nucleic acid.
- The microarray technique is based on hybridisation of nucleic acids.
- There exist two variants of the chips:

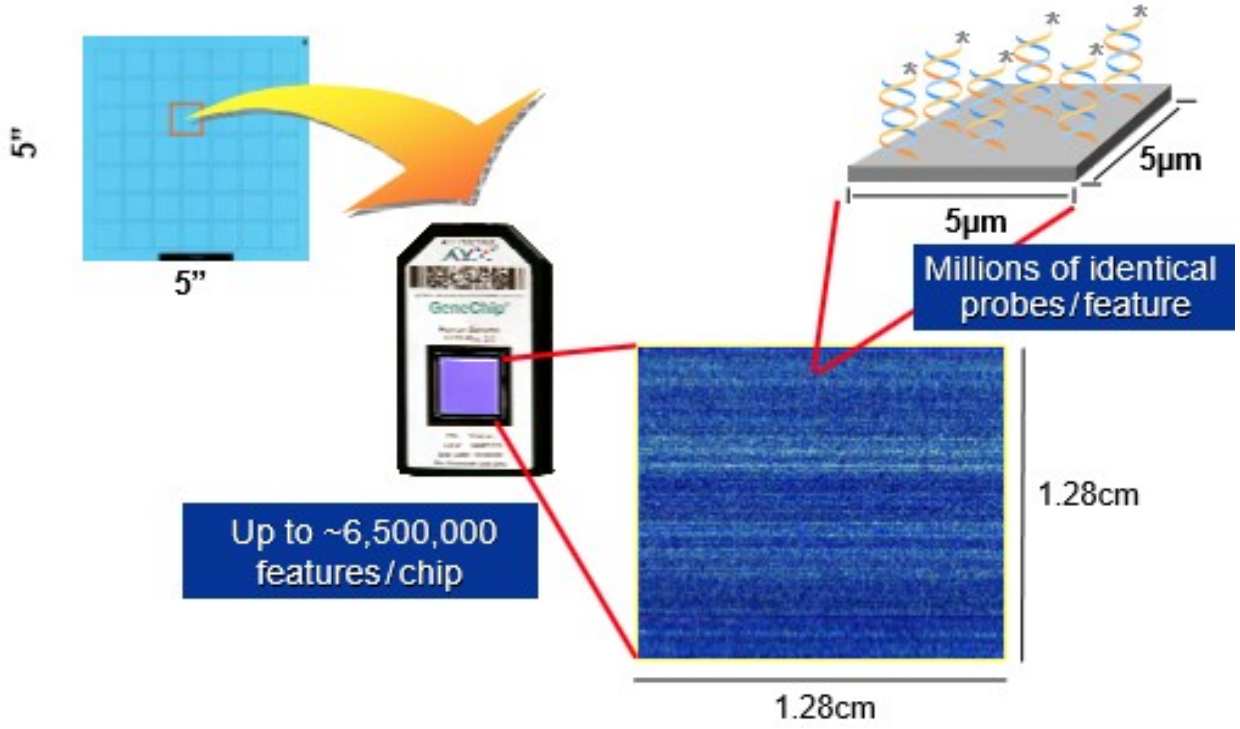
cDNA microarrays

oligonucleotide arrays

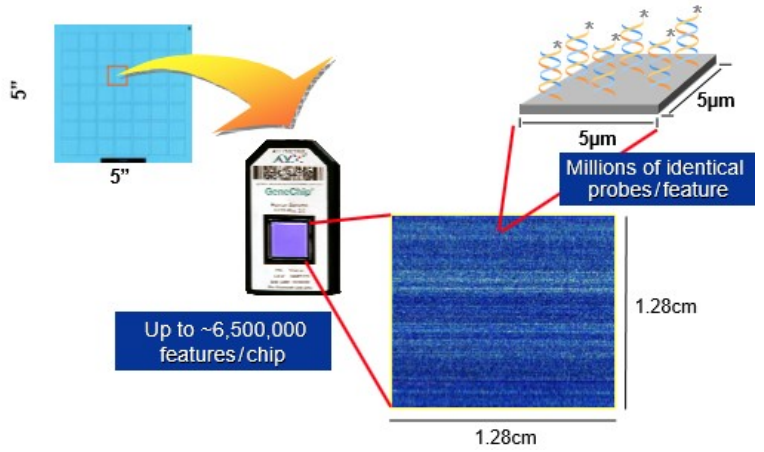
- Although both the DNA and oligonucleotide chips can be used to analyse patterns of gene expression, fundamental differences exist between these methods.
- Two commonly used types of chips differ in the size of the arrayed nucleic acids.

Gene Chip

- The Gene Chip System consists of arrays, hybridization ovens, fluidics stations, scanners, and software for data analysis.
- All of these pieces of equipment are used in a full experiment with a microarray and how the results are analyzed.



- These GeneChip microarrays are a fusion of the technology of the semiconductor industry with life sciences – this is why these arrays are called “chips”.
- The chips are manufactured on wafers. These wafers are 5-inch-by-5-inch pieces of glass.
- You can get anywhere from 49 to 900 individual chips out of each wafer, depending on the overall size of the final chip to be made
- A standard chip is 1.28 cm X 1.28 cm, the size of a thumbnail
- On each one of those chips there are now over 6.5 million sections (squares) known as features
- In each feature there are millions of identical DNA probes on them
- In the diagram above, the probes are represented by the blue spiral connected to the feature
- A probes is a 25 base pair (25-mer) length piece of DNA that is attached to the chip
- Each feature has millions of identical probes attached to them
- The probes are used to probe the sample for certain DNA or RNA segments
- The orange spirals with the stars at the end are RNA from a sample applied to the chip
- Probes are the essential part of how the GeneChip functions



Hybridization

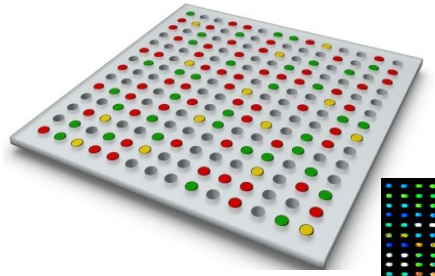
- Hybridization is the basis for the function of microarrays
- Hybridization: The process of joining two complementary strands of DNA to form a double -stranded molecule. The two strands are known as complimentary.
- This hybridization occurs because of complimentary base pairing rules (A – T, and C – G).
- The hybridization of the probes on the features to RNA in sample from a cell being studied is the key to the function of the genechip

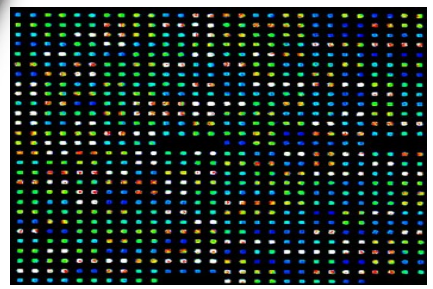
Microarrays

- A snapshot that captures the activity pattern of thousands of genes at once.
- These devices are used for diagnostic purposes such as DNA analysis (DNA microarray), protein detection (protein microarrays) as well as whole cell analysis.
- Microarrays are platforms made of hundreds of detection sites that have micron-sized dimensions and allow the specific detection of a (bio)chemical within a mixture or the simultaneous detection of many (bio)chemicals.
- The detection is related to the chemical functionality on the micron-sized spots in the array and it leads to a single chemical ‘yes/no’ reaction per spot.
- Microarrays are used as screening tools, not only for diagnostic purposes but also for screening new drugs.
- Nanotechnology can impact microarray technology by creating densely packed, smaller, nano-sized arrays (nanoarrays) that could allow faster screening of a larger number of (bio) chemicals.

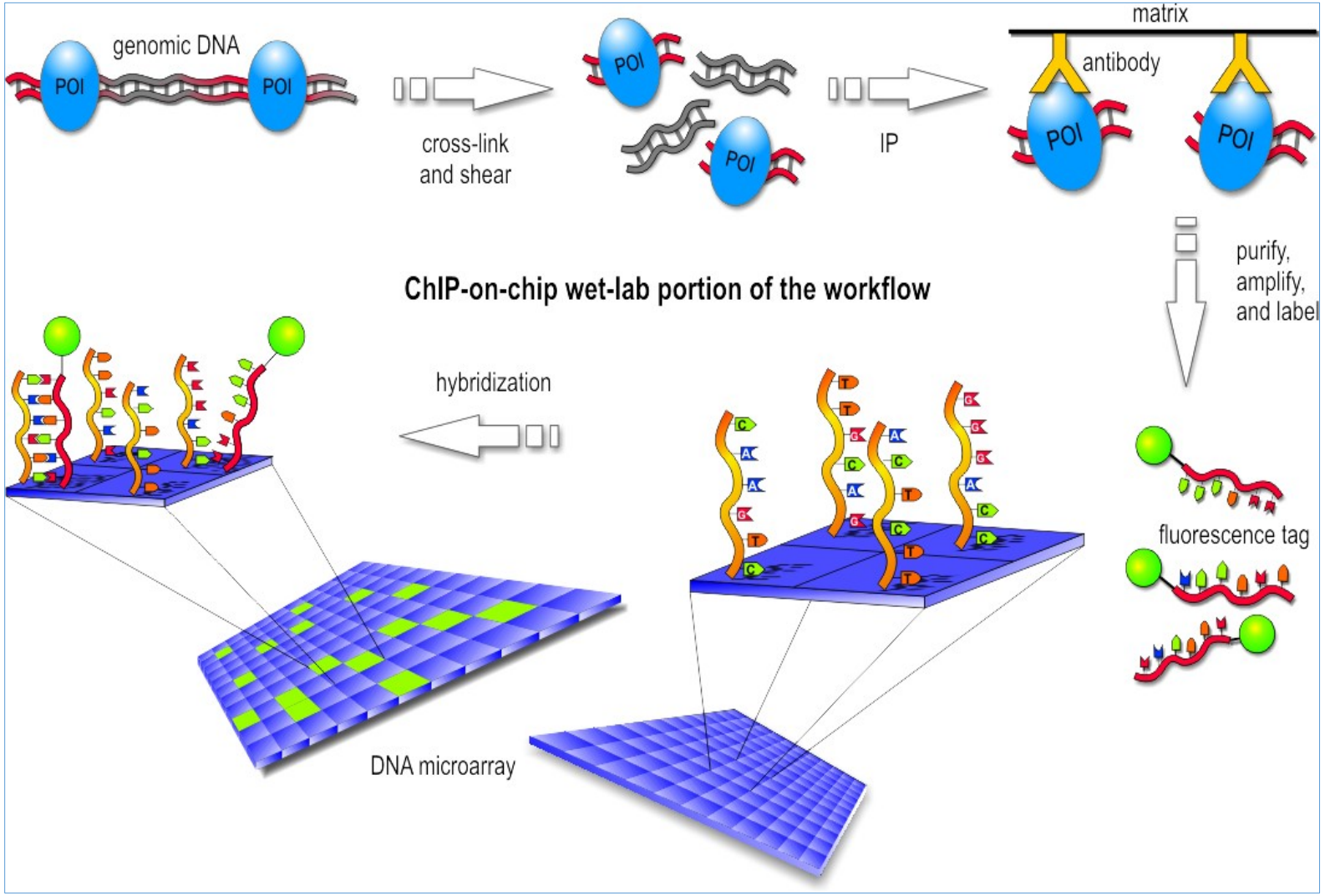
Microarrays

- This technique uses fluorescent probes made of organic molecules attached to the species to be detected (e.g. a protein or a fragment of DNA)
- When reaction occurs, this is attached to the detection spot, which becomes fluorescent in a 'colour' corresponding to the emission of the fluorescent probe.
- Nanoparticles in the form of quantum dots (QD) can be used as an alternative to conventional organic dyes, being more stable, sensitive and monochromatic.
- A substantial (tenfold) enhancement in sensitivity compared to common fluorescent markers has been accomplished through the use of gold and silver particles of uniform dimensions in the range 40 to 120 nm.
- Signal amplification is also obtained using metal nanoparticle labels, such as DNA-modified gold nanoparticles.
- These nano-sized probes have molecules attached to their surface that ensure the selectivity of the detection, while the nano-properties of the probe are responsible for enhancing the signal.






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Human Genome
U133 Plus 2.0
P/N: 520019
Lot #: SAMPLES
Exp. Date: 00/00/00
For Research Use Only


genomic DNA
POI
POI
cross-link
and shear
POI
POI
IP
matrix
antibody
POI
POI
purify,
amplify,
and label
ChIP-on-chip wet-lab portion of the workflow
hybridization
fluorescence tag
DNA microarray

Therapy

Learning Objectives

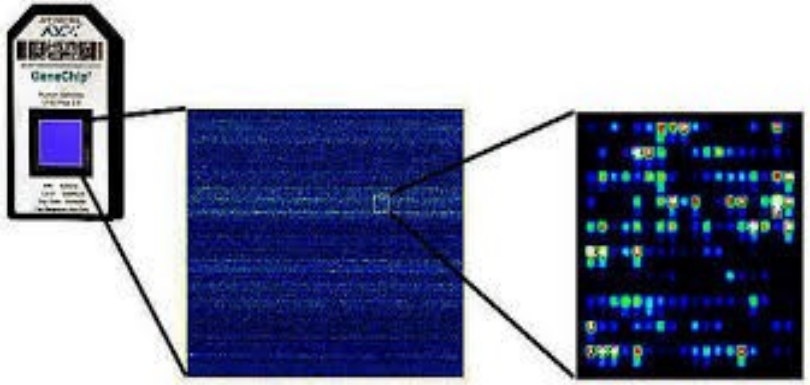
Drug development and targeted drug delivery

siRNA drug delivery

Stimuli-activated drug delivery

Gene chip

- A therapy normally involves a pharmaceutical route (drugs) to treat the disease from the inside of the body.
- Nanotechnologies are making a tremendous impact in this field, with new drugs and new types of treatments under development, some of which have already proven clinically effective and have entered the market.





Drug development and targeted drug delivery

- Advances in the field of pharmacology stem from two main concepts:
  - the development of new biologically active drugs
  - the development of new drug delivery systems able to reach the specific site of the disease.
- Drug delivery systems (DDS) are not a new concept: research in this field started in the mid 1960s and resulted in the type of drugs used today
- Drug delivery systems are in the same form as the pills that are frequently taken and that release their active component gradually in time or that dissolve based on some physiological conditions
- Drug delivery systems also exist in the form of implants, inserts or other drug-releasing systems.

Drug design and screening

- The structure of biological macromolecules defines a three-dimensional nano-environment that mediates specific functions in the cell.
- The design of new drugs requires a very detailed understanding of this nano-environment.
- Gaining insight into the structure of macromolecules on the nanoscale through electron microscopy, nuclear magnetic resonance spectroscopy (NMR) and X-ray crystallography is of fundamental importance tounderstanding biological processes and for the development of new medicines.

Targeted drug delivery

- Pharmaceutical drugs developed using conventional synthesis routes are limited by problems such as low efficacy, low solubility in water and lack of selectivity.
- Physiological barriers often prevent the drug from reaching and acting at the target site — a phenomenon called drug resistance.
- The efficacy of drugs is also dependent on the dose used, but dose-dependent side effects often limit their acceptable usage.
- Delivery systems need to be miniaturized to much smaller than the target, and specific in composition to elicit a certain response.
- With the use of nanotechnologies, targeted drugs are becoming a reality.

- The aim is to design and deliver drugs in such a way that they can recognise the bad cells at a molecular level, pene trate the cell membrane, and act inside the infected cell.
- This is often crucial for the efficacy of a drug, since most virus replication and other disease conditions take place across the cell membrane and inside the cell.
- This way, the treatment will be delivered where it is needed and will be specific, eliminating the problem of the drug killing healthy cells.
- An example of this approach is siRNA drug delivery.

siRNA drug delivery

- RNA interference is a natural, fundamental mechanism in gene regulation that occurs in both plants and animals, humans included.
- DNA in the nucleus of a cell, when genes are express, the genetic information is copied from DNA messenger RNA (mRNA), which then orchestrate the formation of proteins.
- In 1998, Andrew Fire and Craig Mello discovered that double stranded RNA (dsRNA) can interfere with and break down the mRNA for a specific gene, thus stopping the production of a specific protein.
- The gene is, therefore, 'silenced' and the production of the protein is turned off.
- RNA interference plays an important role in switching off genes during an organism's development and controlling cellular functions.

siRNA drug delivery

- Developing nanocarriers for the targeted delivery of siRNA.
- They are studying a novel chitosan-based siRNA nanoparticle delivery system for RNA interference in vitro and in vivo.
- Chitosan is a naturally occurring that has been widely used in drug delivery systems. cationic polysaccharide.
- It contains positively charged amine groups that can interact with the negatively charged backbone of siRNA and form polyplexes in the form of nanoparticles about 200 nm in size.
- The protonated amine groups allow transport across cellular membranes and subsequent endocytosis into cells.
- It has been shown that a chitosan/siRNA nanoparticle delivery system silences genes both in vivo and in vitro.
- Moreover, this delivery system has been shown to be biocompatible, non-toxic and biodegradable.

Stimuli-activated drug delivery

- Activated on reaching the target and the active component released at a controlled rate, this is called stimuli-activated drug delivery.
- In gene therapy, one of the biggest challenges is the targeted delivery of the nuclei acid load to the target (e.g. plasmid-DNA or siRNA) either to silence (RNA silencing) or to activate the expression of a protein as a way to treat a number of diseases.
- The idea is to utilize a nanocarrier that passively accumulates in the diseased tissues (e.g. tumours), followed by stimuli-induced activation at the required site.
- In the case of thermoresponsive systems, the application of heat in precise locations of the tissue can induce the deposition of the nanocarrier in the extracellular target region.

Current and future nano-drug carriers

- Nano-sized drug carriers currently under development include either materials that self-assemble, or conjugated multicomponent systems, for instance a drug linked to a protein and a polymer (called a polymer-drug conjugate.
- Numerous nanosystems are now being investigated and include micelles, nanoemulsions, nanotubes, nanofibres, liposomes, dendrimers, polymer therapeutics, nanoparticles, nanocapsules, nanospheres and hydrogels.
- Some of these nano-sized drug carriers are established in the field of drug delivery, such as liposomes.
- Many are now used for treating some forms of cancer, hepatitis, and leukaemia. An example is an anticancer drug called DOXIL.

Regenerative Medicine

Learning Objectives

Regenerative Medicine

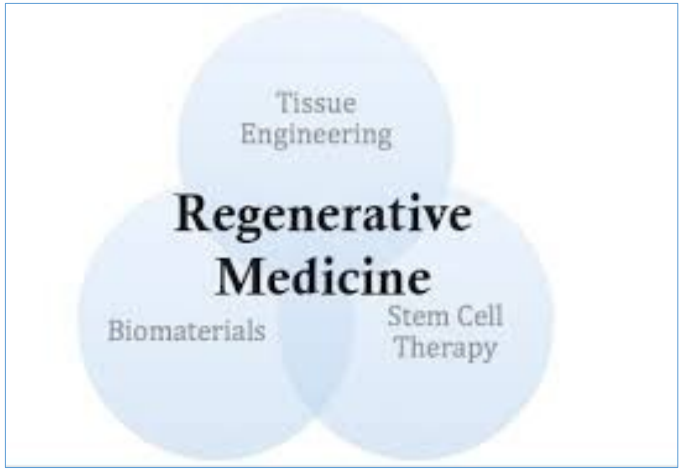
Regenerative Disease

Tissue and biomaterial engineering

Biomaterials for Tissue Engineering

Regenerative Medicine

- At times, the only way to treat a disease is the removal of the infected organ or tissue.
- Such loss can also derive from an injury or a congenital condition (e.g. vision or hearing impairment).
- To compensate for the lost or impaired body function, an artificial construct is implanted in the body.
- Depending on the type, site and extent of the loss, this construct can be in the form of an engineered tissue or an implant.



Regenerative Disease

- A degenerative disease is a type of a medical condition that causes a tissue or organ to deteriorate over time.
- There are quite a number of degenerative diseases and many of them are associated with aging, or gets worse during the aging process.
- Degenerative diseases refer to medical problems that worsen over time. These degenerative diseases may affect the central nervous system (brain and spinal cord), bones, blood vessels or heart.
- Sometimes, certain medications and therapies can treat these degenerative diseases.
- Unfortunately, some degenerative diseases have no cure.
- Degenerative diseases are classified into three main groups: cardiovascular, neoplastic, and nervous system.
- The most common cardiovascular diseases are hypertension, coronary disease, and myocardial infarction.
- Neoplastic diseases include tumors and cancer.
- Diseases that affect the nervous system include Parkinson's and Alzheimer's.

Tissue and biomaterial engineering

- Tissue engineering deals with the fabrication of artificial scaffolds to support the growth of donor cells, which differentiate and grow into a tissue that mimics the lost natural one.
- This tissue engineered construct is then implanted in the patient and, in time, resorbed by the body and fully integrated by the host tissue.
- The scaffold that supports cell growth is the core of this technique.
- In the body, cells are supported in their growth and function by a natural scaffold, called the extracellular matrix (ECM).
- This is a very complex and intricate web of nanofibres that provide the mechanical architecture for cellular growth.

Tissue and biomaterial engineering

- Moreover, the ECM is filled with small molecules (e.g. growth factors) and proteins that direct many cell processes, such as adhesion, migration, growth, differentiation, secretion and gene expression.
- The biggest challenge in regenerative medicine is the artificial replication of this perfect nano-scaffold.
- Close to the field of tissue engineering, and in many cases an integral part of it, is biomaterial engineering.
- Materials used in regenerative medicine are called biomaterials in the sense of being able to trigger and support a biological response.
- One of the distinguishing features of nanotechnologies is their ability to create new functional materials.

Biomaterials for Tissue Engineering

- Biomaterials serve as an integral component of tissue engineering.
- They are designed to provide architectural framework reminiscent of native extracellular matrix in order to encourage cell growth and eventual tissue regeneration.
- Bone and cartilage represent two distinct tissues with varying compositional and mechanical properties.
- Typically, three individual groups of biomaterials, ceramics, synthetic polymers and natural polymers, are used in the fabrication of scaffolds for tissue engineering.
- widespread use of ceramic scaffolds, such as hydroxyapatite (HA) and tri-calcium phosphate (TCP), for bone regeneration applications.
- Ceramic scaffolds are typically characterized by high mechanical stiffness (Young's modulus), very low elasticity, and a hard brittle surface.

Nanotechnologies

- Nanotechnologies are also employed in the fabrication of biomaterials that are responsive to the environment, for this reason, are called smart biomaterials.
- Finally, nano-sized sensors could be inserted inside a biomaterial (e.g. nanowire biosensors) functionalised with receptors that can monitor the presence of small organic molecules, proteins, cells (e.g. cancer cells) and viruses.
- This could be used to collect information on the implant status and activity.
- Tissue and biomaterial engineering have applications in basically all aspects of regenerative medicine (i.e. neuroprosthetics and neuron regeneration (e.g. spinal cord repair), bone restoration, hearing and vision restoration, motor restoration, etc.).

Diagnostic Imaging

Learning Objectives

Diagnostic Imaging

Nanoparticles and Diagnostic Imaging

Diagnostic Imaging

- Techniques such as X-ray, computer tomography (CT), ultrasound (US), magnetic resonance imaging (MRI) and nuclear medicine (NM) are well established imaging techniques, widely used in both medicine and biochemical research.
- Imaging techniques could only detect changes in the appearance of a tissue when the symptoms of the disease were relatively advanced.
- It is in this specific area that nanotechnologies are making their greatest contribution by developing better contrast agents for nearly all imaging techniques.
- The physiochemical characteristics of the nanoparticles (particle size, surface charge, surface coating and stability) allow the redirection and concentration of the marker at the site of interest.
- An example of nanoparticles used in research for imaging is perfluorocarbon nanoparticles employed as contrast agents for nuclear imaging, magnetic resonance imaging and ultrasound, with applications in the imaging of blood clots, angiogenesis, cancer metastases and other pathogenic changes in blood vessels.

Diagnostic Imaging

- Diagnostic imaging refers to a variety of non-invasive methods for identifying and monitoring diseases or injuries via the generation of images representing internal anatomic structures and organs of the patient's body.
- In X-ray imaging, to enhance the signal, an agent must deliver a detectable number of heavy atoms into targeted tissue without toxic effects.
- Nanoparticles of heavy metals have the highest density of surface atoms but they must be inert and stable.
- Nanoparticles of inert metals like silver and gold are too expensive and would render the technique not cost-effective.
- A solution has been proposed by General Electric in the form of nanoparticles made of heavy metal compounds encapsulated in gold shells.
- By targeting receptors unique to a certain type of cancer cell, gold nanoparticles can enhance an X-ray image of a suspected cancer tissue by many orders of magnitude.

Diagnostic Imaging

- Gold nanoshells are a promising material for the optical imaging of cancer. Optical technologies could provide high resolution, non-invasive functional imaging of tissue at competitive costs.
- Magnetic resonance imaging (MRI) uses a large magnet and radio waves to look at organs and structures inside your body.
- Health care professionals use MRI scans to diagnose a variety of conditions, from torn ligaments to tumors.
- MRIs are very useful for examining the brain and spinal cord.
- Nuclear scans use radioactive substances to see structures and functions inside your body. They use a special camera that detects radioactivity.
- X-rays are a type of radiation called electromagnetic waves. X-ray imaging creates pictures of the inside of your body.

Remediation and Mitigation

Learning Objectives

Remediation and Mitigation

Remediation using metal nanoparticles

Remediation and Mitigation

- Simply stated, remediation is the process of clean up and disinfection.
- Mitigation is structural and non-structural measures undertaken to limit the adverse impact of natural hazards, environmental degradation and technological hazards. In other words, help in reducing the cause of problem to prevent further damage.

- Soil and groundwater contamination arising from manufacturing processes are a matter of great complexity and concern.
- Affected sites include contaminated industrial sites (including lakes and rivers in their vicinity), underground storage tank leakages, landfills and abandoned mines.
- Pollutants in these areas include heavy metals (e.g. mercury, lead, cadmium) and organic compounds (e.g. benzene, chlorinated solvents, creosote).

Remediation and Mitigation

- Nanotechnology can develop techniques that will allow for more specific and cost-effective remediation tools.
- Currently, many of the methods employed to remove toxic contaminants involve laborious, time-consuming and expensive techniques.
- Nanotechnology facilitates developing technologies that can perform in situ remediation and reach inaccessible areas such as crevices and aquifers, thus eliminating the necessity for costly 'pump-andtreat' operations.
- In addition, as a result of its ability to manipulate matter at a molecular level, nanoscience can be used to develop remediation tools that are specific to a certain pollutant (e.g. metal), therefore increasing affinity and selectivity, as well as improving the sensitivity of the technique.
- Drinking water quality and its contamination from pollutants is another matter of concern.
- Mercury and arsenic are, in particular, two extremely toxic metals that pose very high health risks. Remediation methods that allow the fast, economic and effective treatment of water polluted with such contaminants is urgently needed.

Remediation using metal nanoparticles

- The use of zero-valent (Fe0) iron nanoparticles for the remediation of contaminated groundwater and soil is a good example of how environmental remediation can be improved with nanotechnology.
- When exposed to air, iron oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.
- Since iron is non-toxic and is abundant in the natural envir onment (rocks, soil, water, etc.).
- Some industries have started using an ‘iron powder’ to clean up their new industrial wastes.

Remediation using metal nanoparticles

- Nanotechnology has offered a solution to this remediation technology in the form of iron nanoparticles.
- These nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders.
- They have a larger surface area available for reacting with the organic contaminant and their small size (1–100 nm) allows them to be much more mobile, so they can be transported effectively by the flow of groundwater.
- When nano-sized iron powders are used, no toxic by-products are formed, a result of the increased reactivity and stability of the nanoparticles compared to the granular iron powder.
- Bimetallic iron nanoparticles, such as iron/palladium, have been shown to be even more active and stable than zero-valent iron nanoparticles, thus further improving this remediation technology.

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Nano-Vaccines

Learning Objectives

Vaccine

Nano based vaccines

Delivery of Nano-vaccines

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Introduction

- A vaccine is a biological preparation that improves immunity to a particular disease.

- A vaccine typically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microbe or its toxins.

- The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and "remember" it, so that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters.

Nano-Vaccine

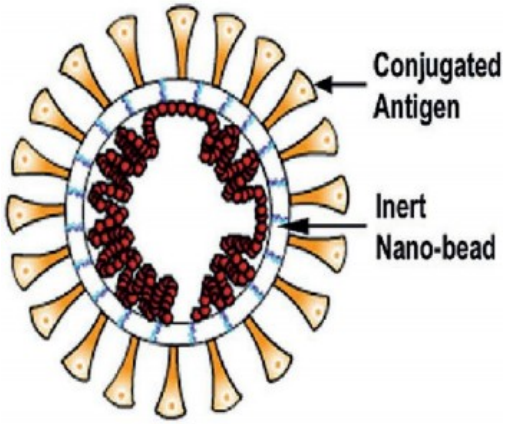
- Nano is very vast field and it can be applied to any area, one of such area is vaccine.
- It provide a different routes of administration of vaccine.
- Nanovaccine can be designed, manufactured and introduced into the human body to improve health, including cellular repairs at the molecular level.
- The nanomaterial is so small that it can easily enter the cell; therefore, nanomaterials can be used in vivo or in vitro for biological applications.

Delivery of Nano-vaccine

Following are the different way to deliver vaccine at nano level

1. **Nanobead:** Inert solid bead Size range 20-200nm.

- When antigens are adsorbed on the surface of bead it has been shown to stimulate CD8 -T cell response.
- The size of the bead play a major role in eliciting a combined response of humoral and cell-mediated immunity.
- Antigen covalently linked to inert nano-beads with a size of ~50 nm is preferentially taken up by dendritic cells, thus inducing humoral as well as cell-mediated immune responses.

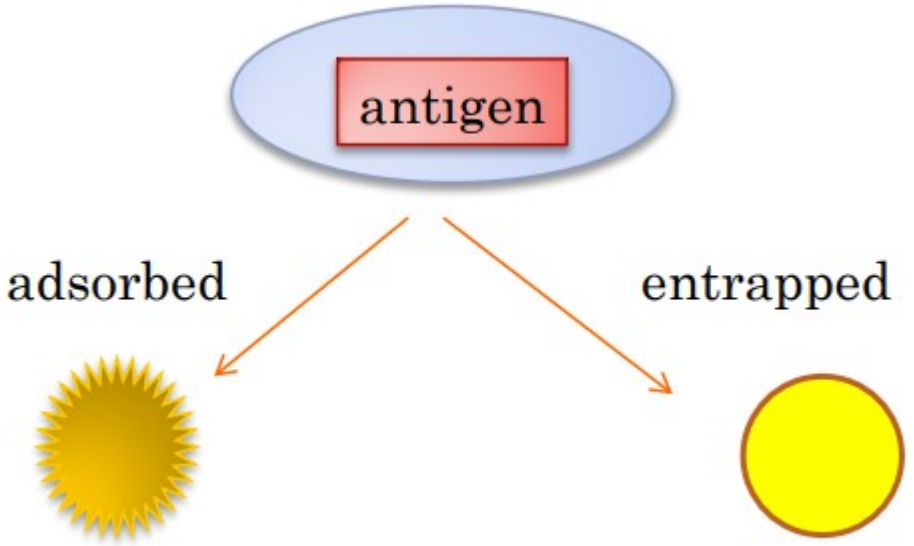


Delivery of Nano-vaccine

2. Polymeric nanoparticles

- Biodegradable, biocompatible polymers have been approved for use in humans.

- Poly(D,L-lactide-co-glycolide) (PLG) and polylactide (PLA).



Delivery of Nano-vaccine

- Polyglycolide (PLG) have been extensively used to encapsulate antigens.

- PLG forms lactic and glycolic acids, After hydrolysis of α-hydroxyl acids, yielding small spherical polymeric particles 1–100 nm in size.

- Adsorbed antigen offer improved stability and activity over encapsulated antigen by avoiding exposure to organic solvents used during formulation and acidic pH conditions caused by degradation of the polymer.

Delivery of Nano-vaccine

3. Nanoemulsion

- Size of globule (100-400nm)

- Nanoemulsion vaccine does not require refrigeration and is stable for 6 months.

- Nanoemulsion is non-toxic, pain free and avoids the risk of spreading needle-borne infections.

- Nanoemulsion of hepatitis B antigen, has been reported to be a safe and effective hepatitis B vaccine.

References

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- Peek Laura et.al, Nanotechnology in vaccine delivery, Advanced Drug Delivery Reviews 60 (2008) 915–928,

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Nano-Vaccines

Learning Objectives

Adjuvants

Advantages of Nano-vaccines

Limitations

Future prospects

Adjuvants

- These substance used in combination with a specific antigen that produced a more robust immunity than the antigen alone.

- It improve the immune response to vaccine antigen by different way:

  1. Increasing immunogenicity of weak antigen.

  2. Enhancing the speed and duration of the immune response.

  3. Modulating antibody avidity, specificity, isotope or subclass distribution.

Adjuvants

Examples

- MF59
- Monophosphoryl lipid A
- Montanide™
- Calcium phosphate nanoparticles.
- Immunostimulating complexes
- Cholesterol-bearing hydrophobized pullulan
- nanoparticles (CHP)

MF59

- MF59 is the only nano-sized vaccine adjuvant approved for human use.
- MF59 is an oil-in-water emulsion (≤ 250 nm droplets)
- Its showed a 34-fold increase in antibody titers when immunized with glycoprotein D of herpes simplex virus (HSV) in guinea pigs.
- The mechanism of adjuvanticity of MF59 is believed to be through direct stimulation of cytokine production.

MONOPHOSPHORYL LIPID A

- Monophosphoryl lipid A (MPL®) is an immunostimulating TLR-4 receptor agonist composed of detoxified lipopolysaccharide (LPS) from Salmonella minnesota R595.

- It is a versatile vaccine adjuvant that may be included in aqueous formulations or in an oil-inwater emulsion for a more dynamic response.

MONTANIDE™

- There are several different types of Montanide™, including ISA 50V, 51, 206 and 720.
- ISA 50V, 51 and 720 are water-in-oil emulsions while ISA 206 is a water-in-oil-in-water emulsion.
- ISA 206 and 50V only used in veterinary vaccine formulations.
- SA 51 & ISA 720 are under investigation for use in humans.
- Emulsions of Montanide™ ISA 51 and 720 are composed of a metabolizable squalene-based oil with a mannide monooleate emulsifier.

MONTANIDE™

- The immune enhancement produced by the Montanide™ emulsions is believed to be due to the formation of a depot at the site of injection.

- The emulsion vaccines against malaria, HIV and various cancer have been in phase I and/or II clinical trials.

- A phase I trial of a trivalent malaria vaccine containing ISA 720 induced both humoral and cellular immune responses

Calcium phosphate nanoparticles

- Calcium phosphate nanoparticles are less than 1000nm in diameter.
- Nanoparticles can be generated by combining (while stirring) calcium chloride, sodium phosphate and sodium citrate.
- It produces longer duration of response as compare to Al salt.
- Phase I study showed that CaP is safe and non-toxic when administered subcutaneously.
- Vaccines utilizing CaP in preclinical studies include anthrax, HBV, flu (H5N1 avian and seasonal) and HSV-2.

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Cholesterol-bearing hydrophobized pullulan nanoparticles (chp)

- Pullulan is the most popular polysaccharide to which cholesterol can be Conjugated. It render the molecules amphiphilic.
- Such molecules have been shown to self assemble with and without proteins into 30–40 nm colloidally stable nanoparticles.
- Its size and density can be modified by altering the degree of substitution of cholesterol groups on the polysaccharide.
- A preclinical study in mice showed that immunization with a complex of the HER2 oncoprotein and CHP induced both humoral and CD8 responses.

Advantages of nano-vaccine

- Nano-vaccine have potential to deliver safe and more effective vaccine.
- Nanobead covalently coupled with antigen offer distinct advantages – a low dose of antigen is required, efficient processing by antigen-presenting cells and stability during storage.
- Encapsulated nanoparticles easily deliver antigen, protects the antigen from degradation and is found to be effective with a single dose due to slow release of the antigen.
- Many of the nano-vaccines are non-invasive, delivered by the oral or nasal route, diffusion patches or microneedle arrays, thus allowing pain-free delivery with minimal damage. This is an advantage over conventional vaccines, which are usually multi injection, multi-dose delivery systems.

Disadvantages of nano-vaccine

- Cost of production.
- Nanomaterials can change size, shape but not composition, which may change their toxicity.
- Small nanoparticles are cleared quickly from the body, large counterparts may accumulate in vital organs causing toxic problems.

Future prospects

- Carbon nanotubes may be used to deliver vaccine.
- Peptide–nano-bead based vaccine approach may be beneficial, especially for highly variable pathogens such as FMDV(foot and mouth disease virus).
- Nanoemulsion may deliver smallpox, influenza, anthrax and HIV vaccines.
- Nanoemulsion against GP 120, one of the major binding proteins, may induce mucosal and cellular immunity, and neutralize antibody to various isolates of HIV.

References

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Quantum dots

Learning Objectives

Introduction

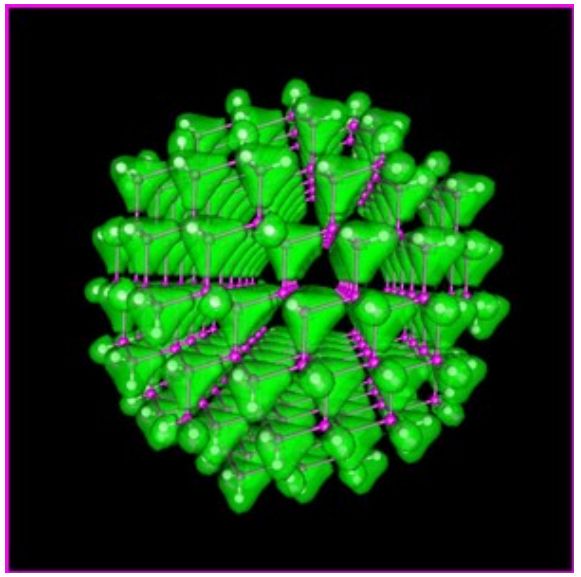
Properties

Methods of synthesis

Introduction

Definition:

- Quantum dots (QD) are nanoparticles/structures that exhibit 3 dimensional quantum confinement, which leads to many unique optical and transport properties.



Lin-Wang Wang, National Energy Research Scientific Computing Center at Lawrence Berkeley National Laboratory. <http://www.nersc.gov>

GaAs Quantum dot containing just 465 atoms.

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Introduction

- Quantum dots are usually regarded as **semiconductors** by definition.
- Similar behavior is observed in some metals. Therefore, in some cases it may be acceptable to speak about metal quantum dots.
- Typically, quantum dots are composed of groups II-VI, III-V, and IV-VI materials**.**
- QDs are **bandgap tunable** by size which means their optical and electrical properties can be engineered to meet specific applications.

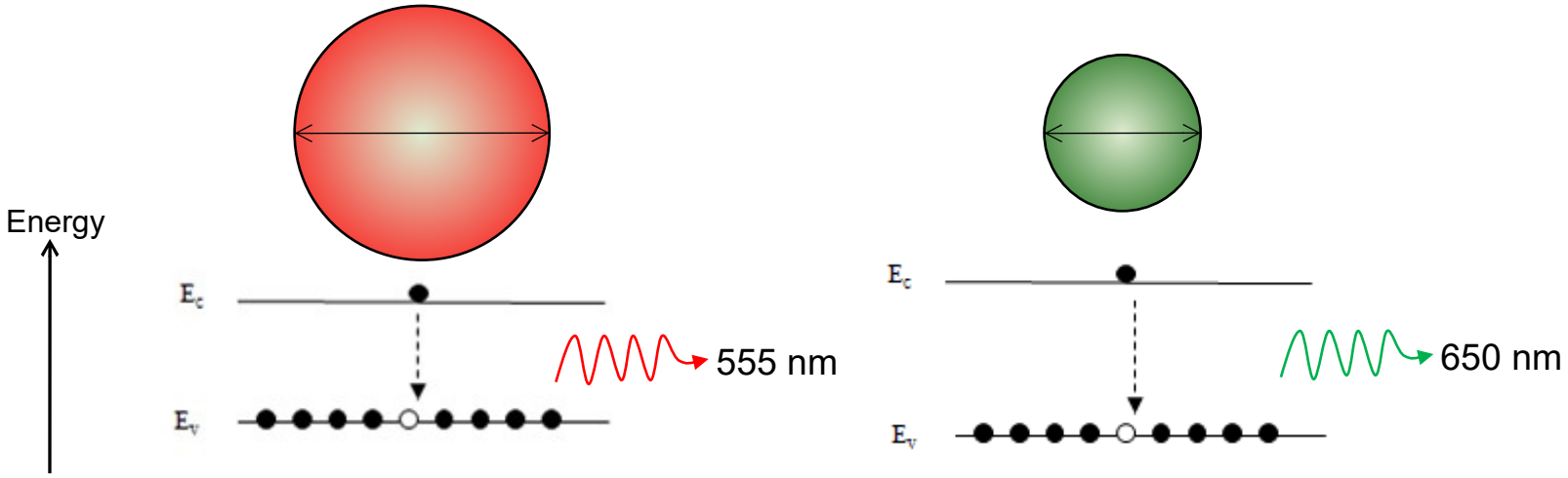
Quantum Confinement

Definition:

- Quantum Confinement is the spatial confinement of electron-hole pairs (excitons) in one or more dimensions within a material.
  - 1D confinement: Quantum Wells
  - 2D confinement: Quantum Wire
  - 3D confinement: Quantum Dot
- Quantum confinement is more prominent in semiconductors because they have an energy gap in their electronic band structure.
- Metals do not have a bandgap, so quantum size effects are less prevalent. Quantum confinement is only observed at dimensions below 2 nm.

Quantum Confinement

- What does this mean?
  - Quantum dots are **bandgap tunable** by size. We can engineer their optical and electrical properties.
  - Smaller QDs have a large bandgap.
  - Absorbance and luminescence spectrums are blue shifted with decreasing particle size.



QD Synthesis: Colloidal Methods

- Example: CdSe quantum dots
- 30mg of Elemental Se and 5mL of octadecene are used to create a stock precursor Se solution.
- 0.4mL of Trioctylphosphine oxide (TOPO) is added to the Se precursor solution to disassociate and cap the Se.
- Separately, 13mg of CdO, 0.6mL of oleic acid and 10mL of octadecene were combined and heated to 225°C
- Once the CdO solution reaches 225°C, room-temperature Se precursor solution was added. Varying the amount of Se solution added to the CdO solution will result in different sized QDs.

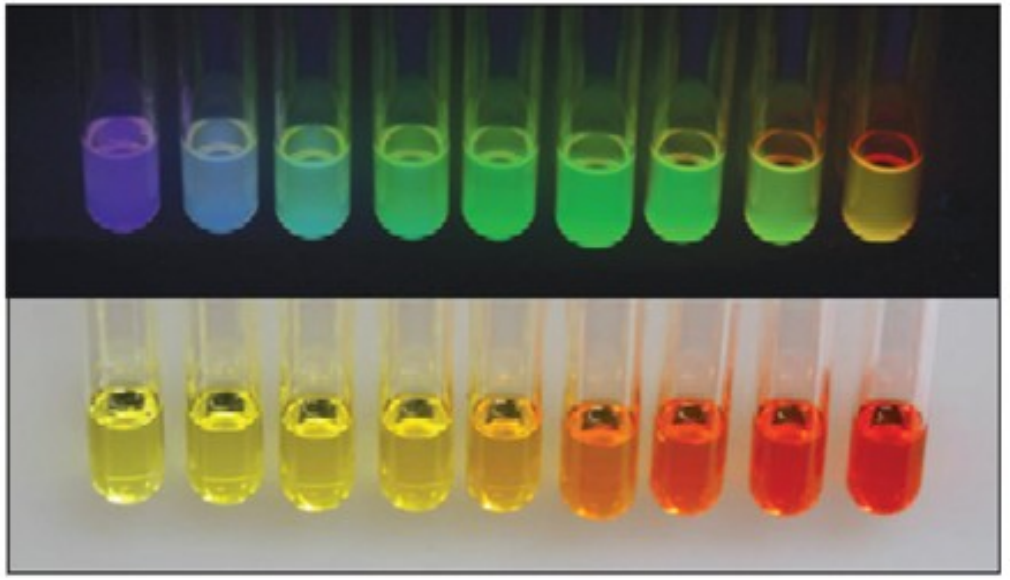


Figure: Colloidal suspensions of CdSe quantum dots of increasing size from left (approximately 1.8-nm diameter) to right (approximately 4.0-nm diameter). Bottom: Samples viewed in ambient light vary in color from green–yellow to orange–red. Top: The same samples viewed under long-wave ultraviolet illumination vary in color from blue to yellow.

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Synthesis of CdSe/ZnS (Core/Shell) QDs

Step 1

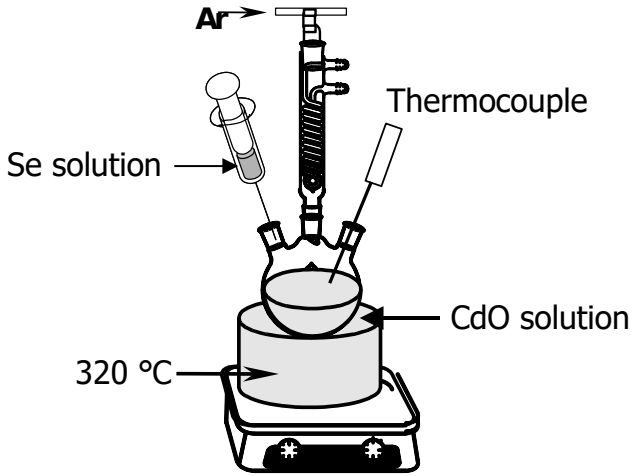
$CdO + Se \longrightarrow CdSe$

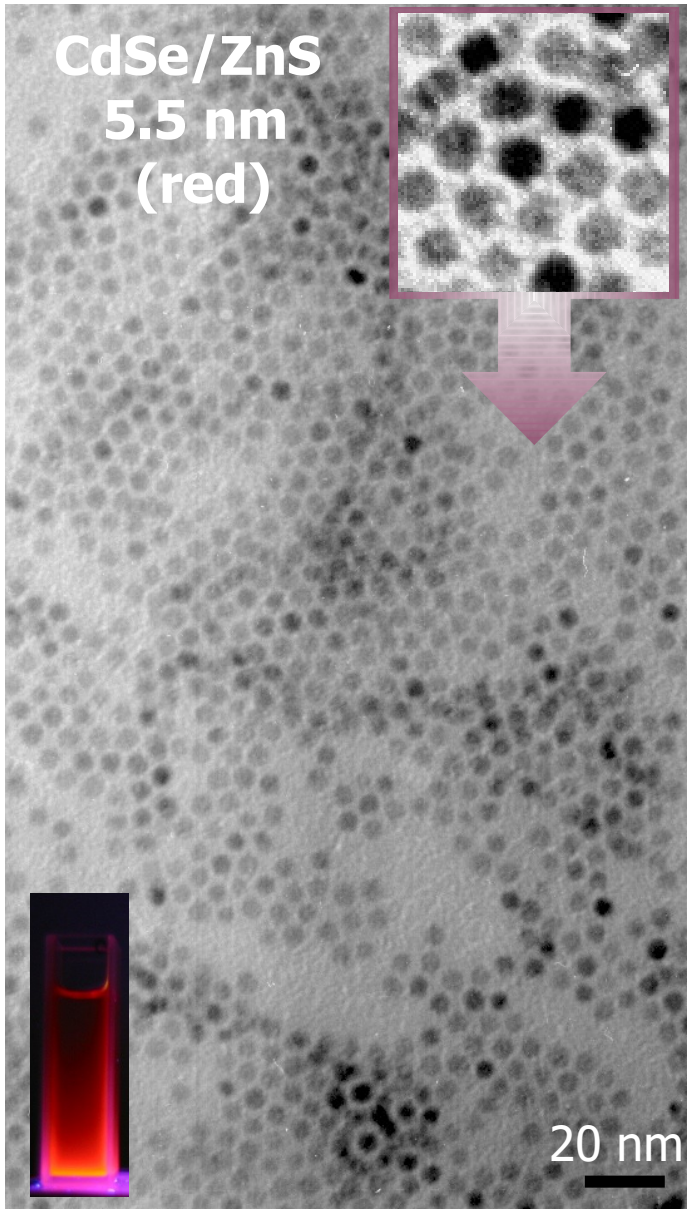
Solvent : TOPO, HAD, TOP
Surfactant : TDPA, dioctylamine

Step 2

$ZnEt_2 + S(TMS)_2 \longrightarrow$ ZnS/CdSe

Growth temperature
→ **140 °C (green)**
200 °C (red)





Bawendi *et al. J. Am. Chem. Soc.* (1994)

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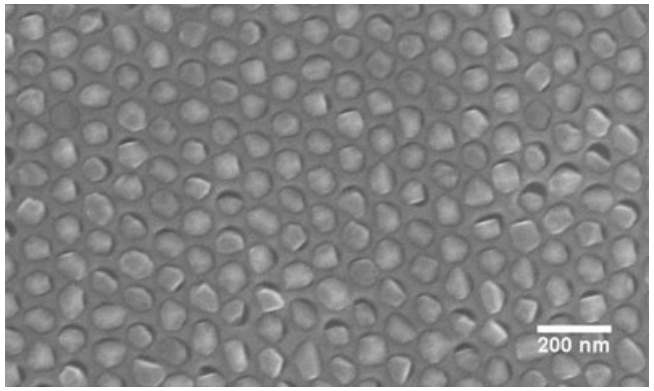
QD Synthesis: Epitaxial Growth

- Epitaxial growth refers to the layer by layer deposition/growth of **monocrystalline** films.
- A liquid or gaseous precursor condenses to form crystallites on the surface of a substrate.
- The substrate acts as a seed crystal. Its lattice structure and crystallographic orientation dictate the morphology of epitaxial film.
- Epitaxial growth techniques can be used to fabricate QD **core/shell structures** and **QD films**.

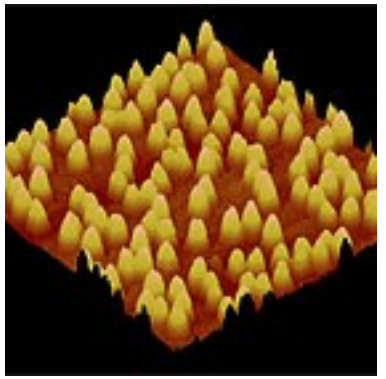
QD Synthesis: Epitaxial Growth

Quantum Dot Films

- QD Film – thin film containing small localized clusters of atoms that behave like quantum dots.
- QD films can be highly ordered quantum dot arrays or randomly agglomerated clusters with a broad size distribution.
- The structure of choice (arrayed or disordered) depends on the particular application.



SEM image of highly order InAs QD array



AFM image of QD film containing random agglomerated clusters of InAs QDs.

QD Synthesis: Epitaxial Growth

Core/Shell Structures:

- Core/shell quantum dots are comprised of a luminescent semiconductor core capped by a thin shell of higher bandgap material.
- The shell quenches non-radiative recombination processes at the surface of the luminescent core, which increases quantum yield (brightness) and photostabilty.
- Core/shell quantum dots have better optical properties than organically passivated quantum dots and are widely used in biological imaging.



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QD Synthesis: Epitaxial Growth

- There are a variety of epitaxial methods, which each have their own sub-techniques:
  - Laser Ablation
  - Vapor Phase Epitaxy (VPE)
  - Liquid Phase Epitaxy (LPE)
  - Molecular Beam Epitaxy (MBE)

Biological applications of Quantum dots

Learning Objectives

Imaging

Biological Tagging

Labelling

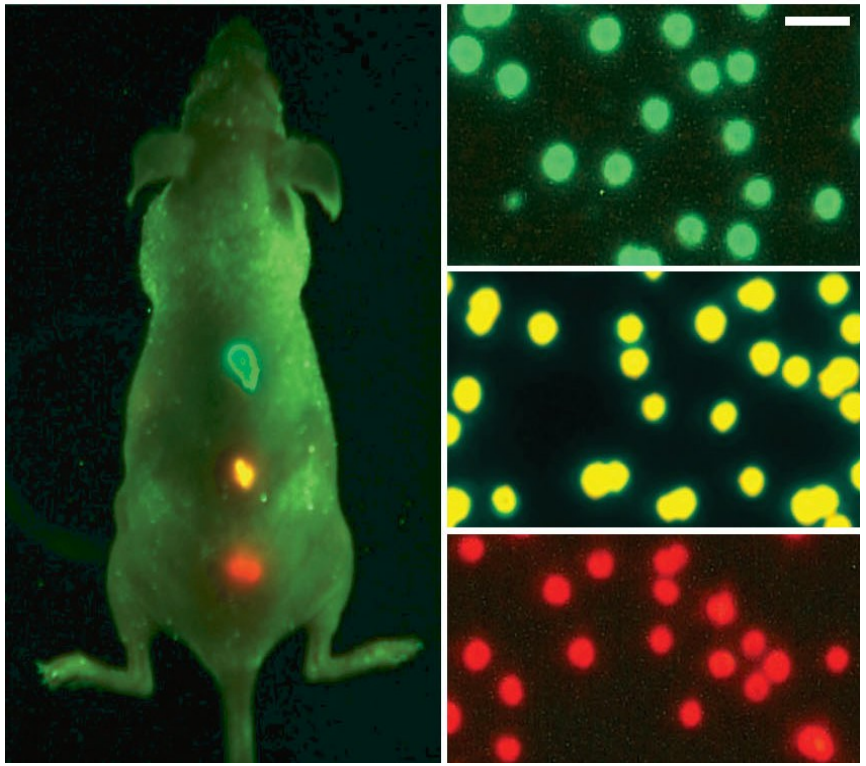
Bioconjugated QDs

Applications of QDs: Biological

- The unique optical properties of quantum dots make them suitable for biological tagging and labeling applications.
- QDs are excellent fluorophores.
  - **Fluorescence** is a type of luminescence in which the absorption of an incident photon triggers the emission of a lower energy or longer wavelength photon.
  - Quantum dots absorb over a broad spectrum and fluoresce over a narrow range of wavelengths. This is tunable by particle size.
  - So, a single excitation source can be used to excite QDs of different colors making them ideal for imaging multiple targets simultaneously.

Applications of QDs: Biological

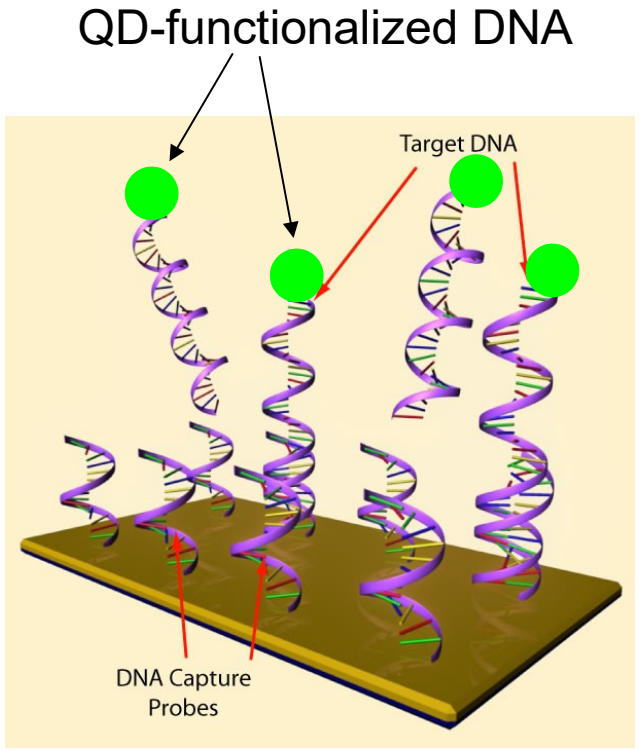
- A broad absorption and narrow emission spectrum means a single excitation source can be used to excite QDs of different colors making them ideal for imaging multiple targets simultaneously.



CdSe/ZnS QDs used to image cancer cells in a live mouse.

Applications of QDs: Biological

DNA assays and microarrays



BioMems Applications Overview
SCME: www.scme-nm.org

Each pixel contains a different DNA sequence
Fluorescence observed if sample binds

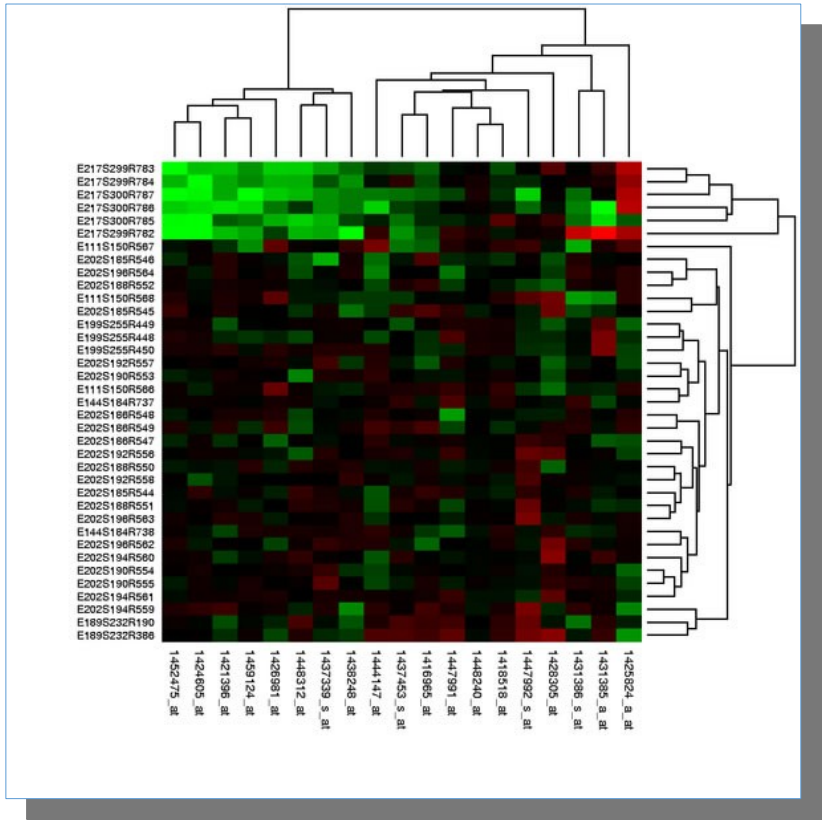


Image source: Wikipedia: Gene Expression Profiling

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Applications of QDs: Biological

- Biological Tagging and Labeling
  - Biological assays and microarrays
  - Labeling of cells and intracellular structures
  - *in vivo* and *in vitro* imaging
  - Pathogen and Toxin detection

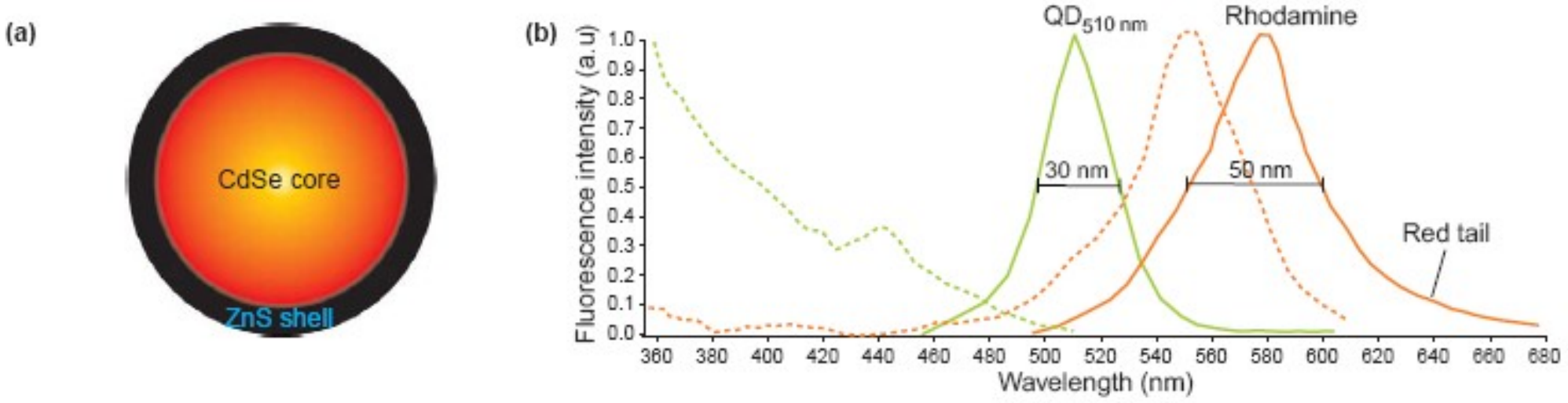
Applications of QDs: Biological

- Biological Tagging
  - Organic fluorophores such as genetically encoded fluorescent protein, like GFP, or chemically synthesized fluorescent dyes have been the most common way of tagging biological entities.
  - Some limitations of organic fluorophores:
    - do not continuously fluoresce for extended periods of time
    - Degrade or photo-bleach
    - are not optimized for multicolor applications

Applications of QDs: Biological

Absorption and emission Spectra of CdSe/ZnS QDs compared to Rhodamine, a common organic die

- The absorption spectrum (dashed lines) of the QD (green) is very broad, whereas that of the organic die (orange) is narrow.
- Conversely, the emission spectrum (solid lines) of the QD is more narrow than that of the organic die



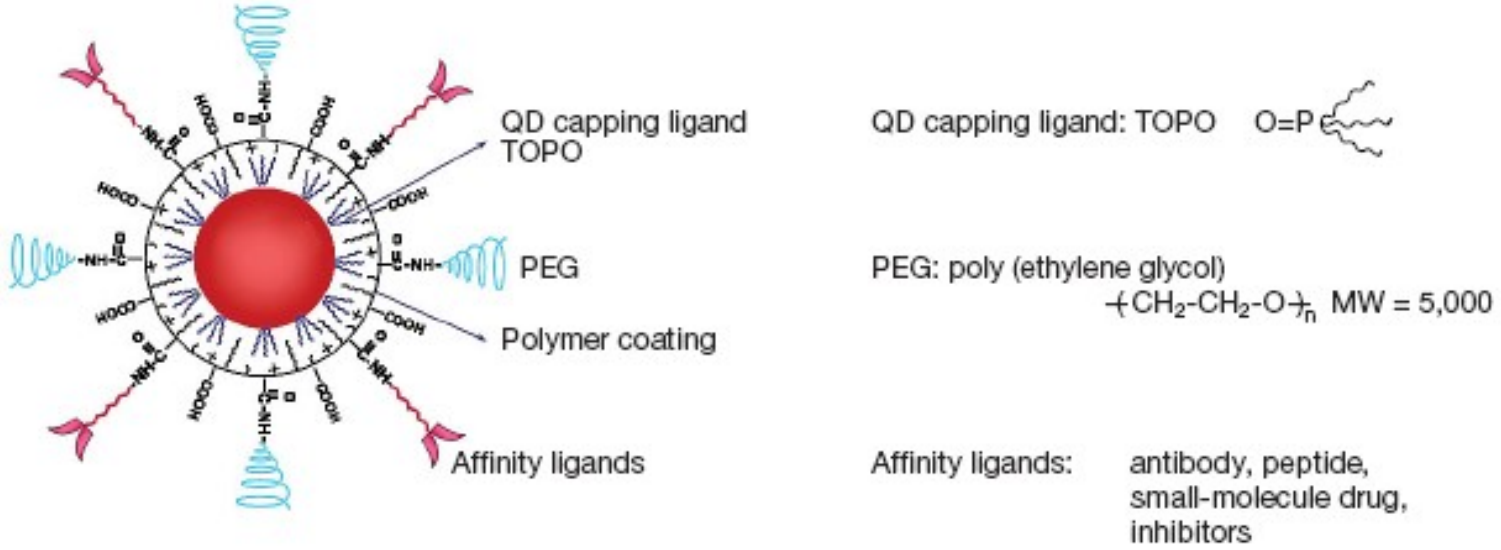
Applications of QDs: Biological

- Quantum dots are an attractive alternative to traditional organic dies because of their high quantum yield and photostability.
- Quantum Yield = # emitted photons / # absorbed photons.
  - Quantum dots have a high quantum yield because they have a high density of energy states near the bandgap.
  - A higher quantum yield means a brighter emission. The quantum yield of some QDs is 20 times greater than traditional organic fluorophores.
- Photostability is a fluorophore's resistance to photobleaching or photochemical degradation due to prolonged exposure to the excitation source.

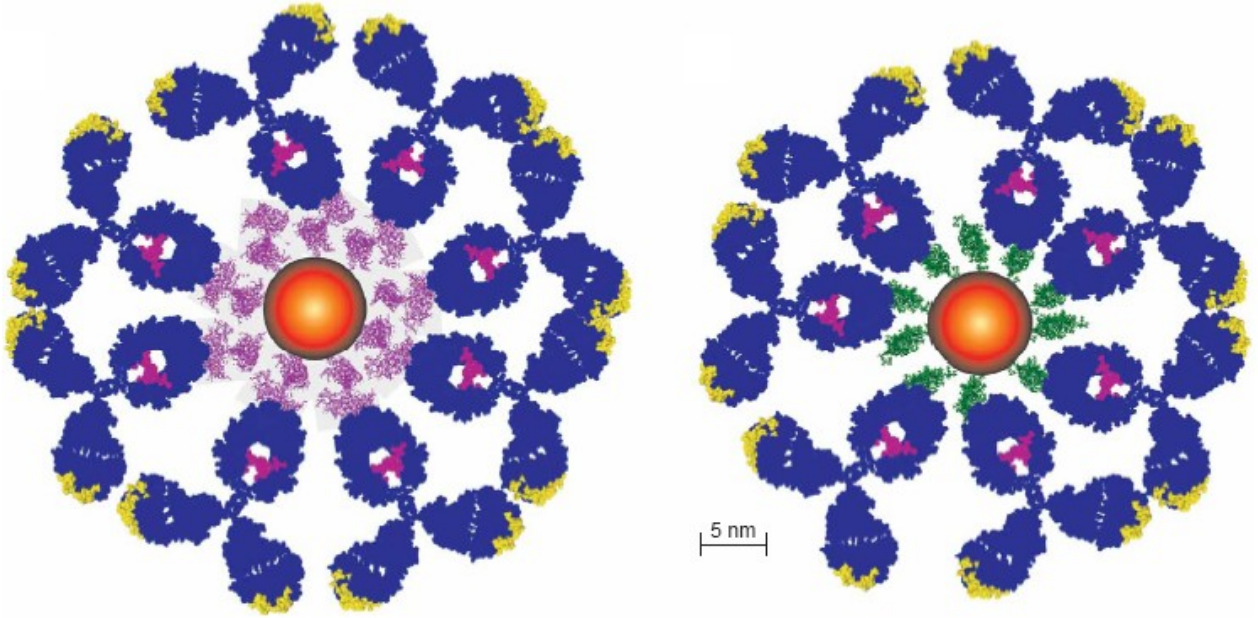
Applications of QDs: Biological

Bioconjugated QDs:

- The surface of QDs can be functionalized with **affinity ligands:** antibodies, peptides, or small-molecule drug inhibitors, to target specific types of cells for *in vivo* or *in vitro* imaging.
- The affinity ligands are not bound directly to the surface of the quantum dots. They are usually connected to a linker molecule referred to as a **capping ligand** or **coordinating ligand**.
- Polymers such as poly ethylene glycol (PEG) may be introduced to reduce nonspecific binding of the affinity ligands.



Applications of QDs



QDs conjugated with antibody molecules (blue) by using avidin (purple) or protein A (green) as linkers. Between 10 and 15 linker molecules can be attached covalently or electrostatically to a single QD, which facilitates the binding of many or a few antibody molecules on each QD.

Jyoti K. Jaiswal and Sanford M. Simon. **Potentials and pitfalls of fluorescent quantum dots for biological imaging.** TRENDS in Cell Biology Vol.14 No.9 September 2004

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DNA-AU Assembly

Learning Objectives

Introduction

Properties

Characterization

DNA-Au Assemblies

- Conjugates of AuNPs-oligonucleotides are of great current interest because of the potential use of the programmability of DNA base-pairing to organize nanocrystals in space.
- Provide multiple ways of providing a signature for the detection of precise DNA sequences.
- Applications in the fields of biosensors, disease diagnosis, and gene expression.
- DNA as a linker to form macroscopic assemblies consisting in discrete 13-nm-diameter particles. The DNA attached to the nanoparticles retained its ability to hybridize with complementary DNA, and the annealing process was thermally reversible and non destructive. The reaction was sequence-specific

Chemical Reviews, 2004, Vol. 104, No. 1

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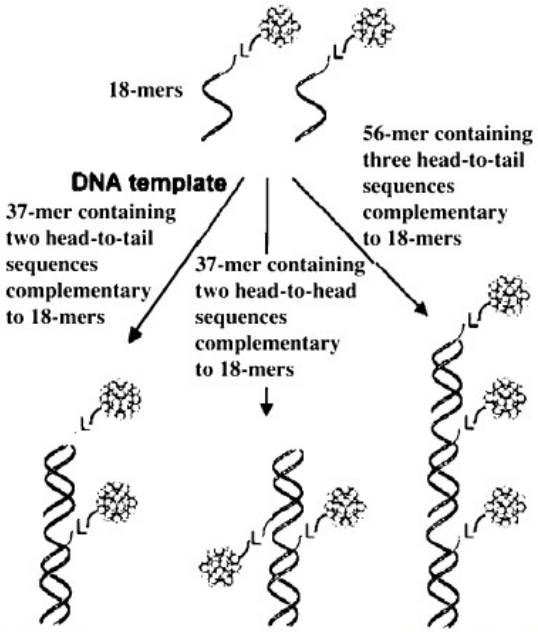
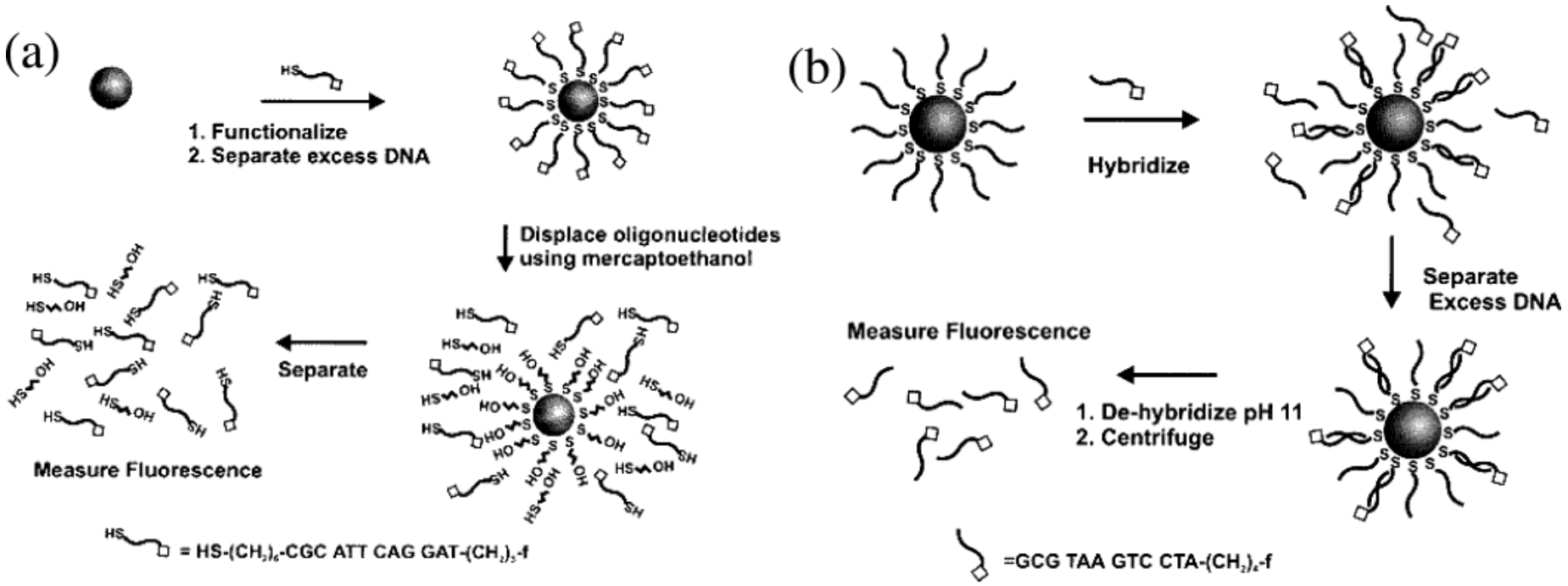


Figure 56. Preparation of "nanocrystal molecules" consisting of two or three DNA modified Au particles attached to a complementary DNA template, using phosphine-stabilized 1.40nm AuNPs modified with a single thiol-capped oligonucleotide and two different DNA template lengths and sequences. Reprinted with permission from ref 509 (Mirkin's group). Copyright 1997 Kluwer.

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Fluorescence-based method for determining (a) the surface coverage and (b) the hybridization efficiency of thiol-capped oligonucleotides bound to gold thin films and AuNPs.

Chemical Reviews, 2004, Vol. 104, No. 1

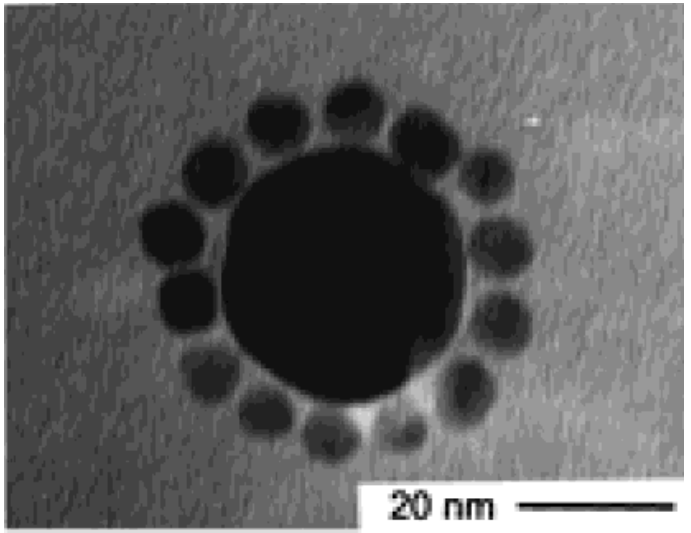
DNA-Au Assemblies

- DNA as a template to prepare nanocrystal molecules consisting of two or three 1.4 nm-diameter particles on a single oligonucleotide strand.
- DNA-driven assemblies of AuNPs have indeed attracted considerable interest, and a new colorimetric technique based on the sensitivity of the SPB to monitor DNA modification.
- In this strategy, AuNPs are used as building blocks, allowing the assembly of alkanethiol-capped oligonucleotides such as single-stranded DNA and complementary linker oligonucleotide (DNA) strands.

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DNA-Au Assemblies

- Aggregation of AuNPs linked by the oligonucleotides provokes a red-to-blue color change (red shift from 520 to 600 nm of the SPB) that is most useful for this DNA-sensing method.
  - Factors involved are: AuNP composition, periodicity, aggregate thermal stability, ; these all can be controlled to influence optical mechanical and electrical properties of AuNPs. (SPB of AuNPs from 13-17 nm is very sensitive and its optical properties are being used for many diagnostic applications). Also the length of interparticle distance affect the SPB frequency as well



TEM images of the binary AuNPs network materials supported on holey carbon grids: a AuNP satellite structure obtained from the reaction involving 120/11-modified 8-nm particles/2-modified 31-nm AuNPs and linking oligonucleotide.

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Characterization of DNA-Au assemblies

- Fluorescence- Fluorospectrophotometery
- TEM, AFM, SEM
- FRET
- SPB
- Quartz Crystal Microbalance-Dissipation
- Raman spectroscopy

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Liposomes

Learning Objectives

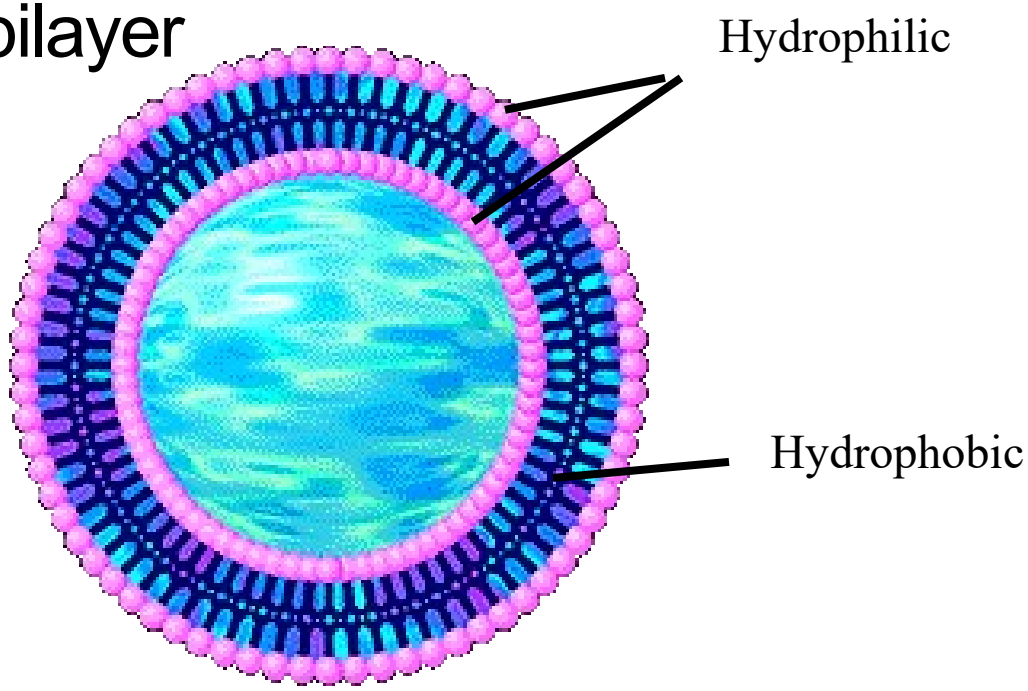
Introduction

Types

Methods of synthesis

What is Liposome

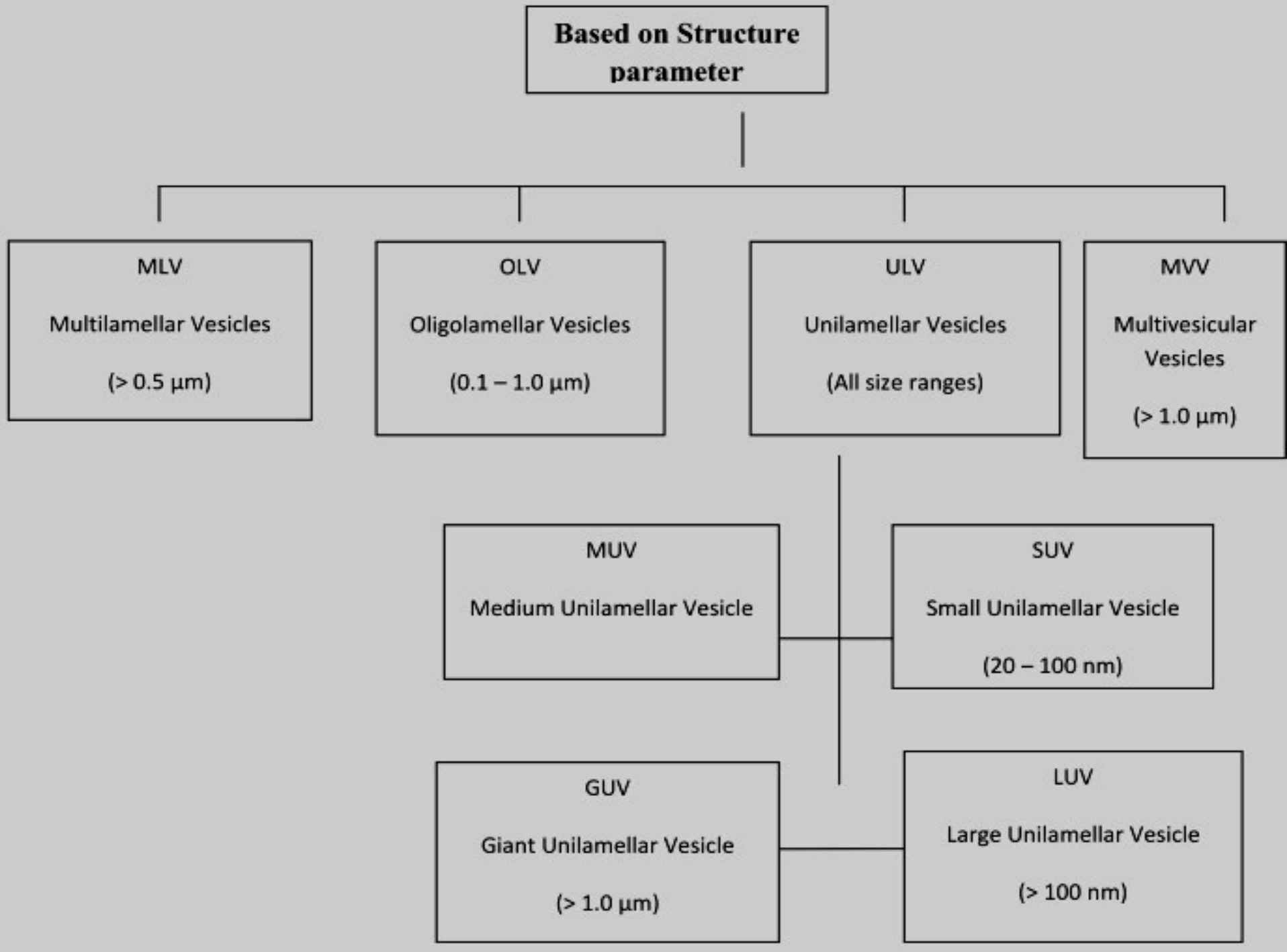
– Spherical (concentric) vesicles with a phospholipid bilayer

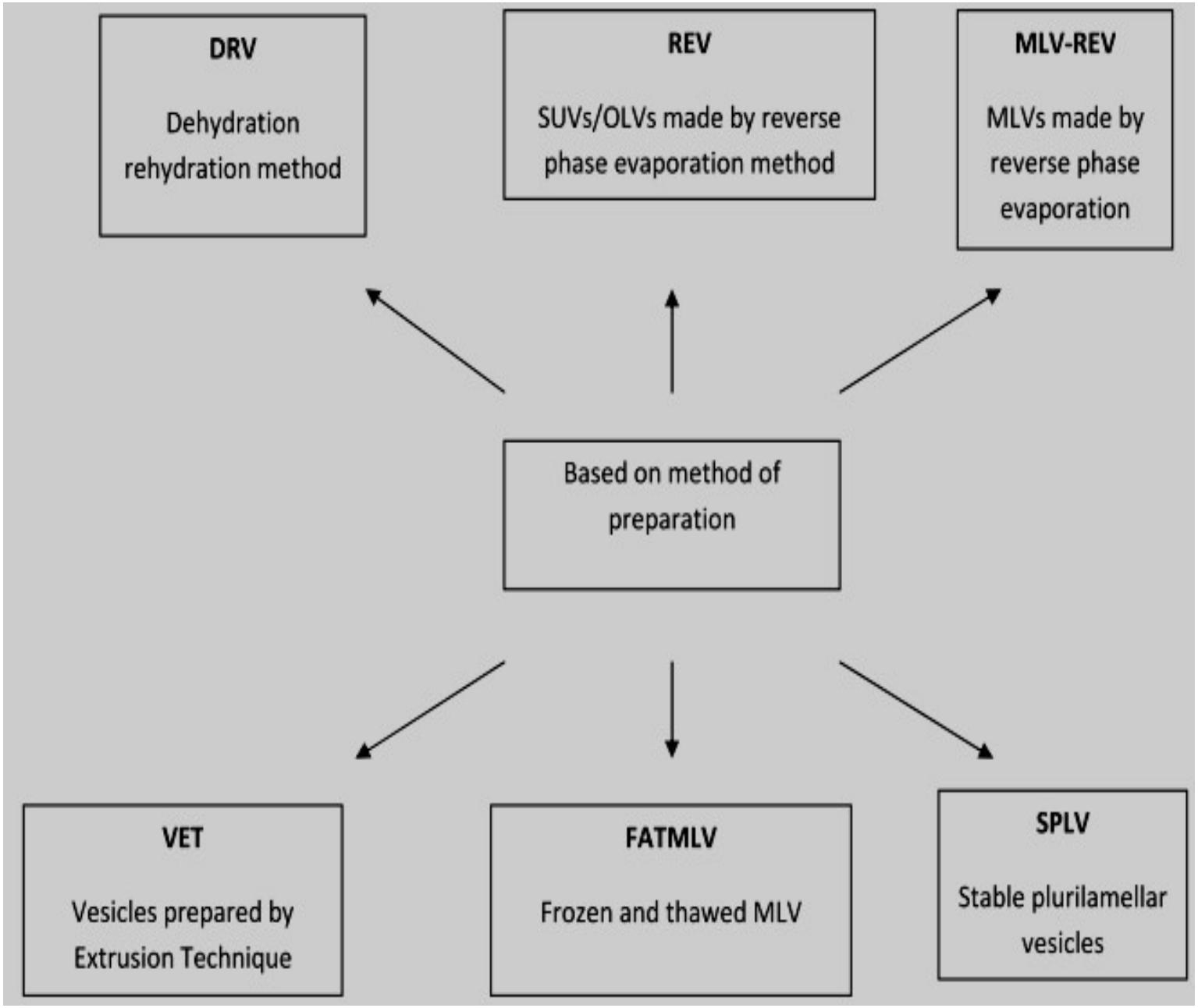


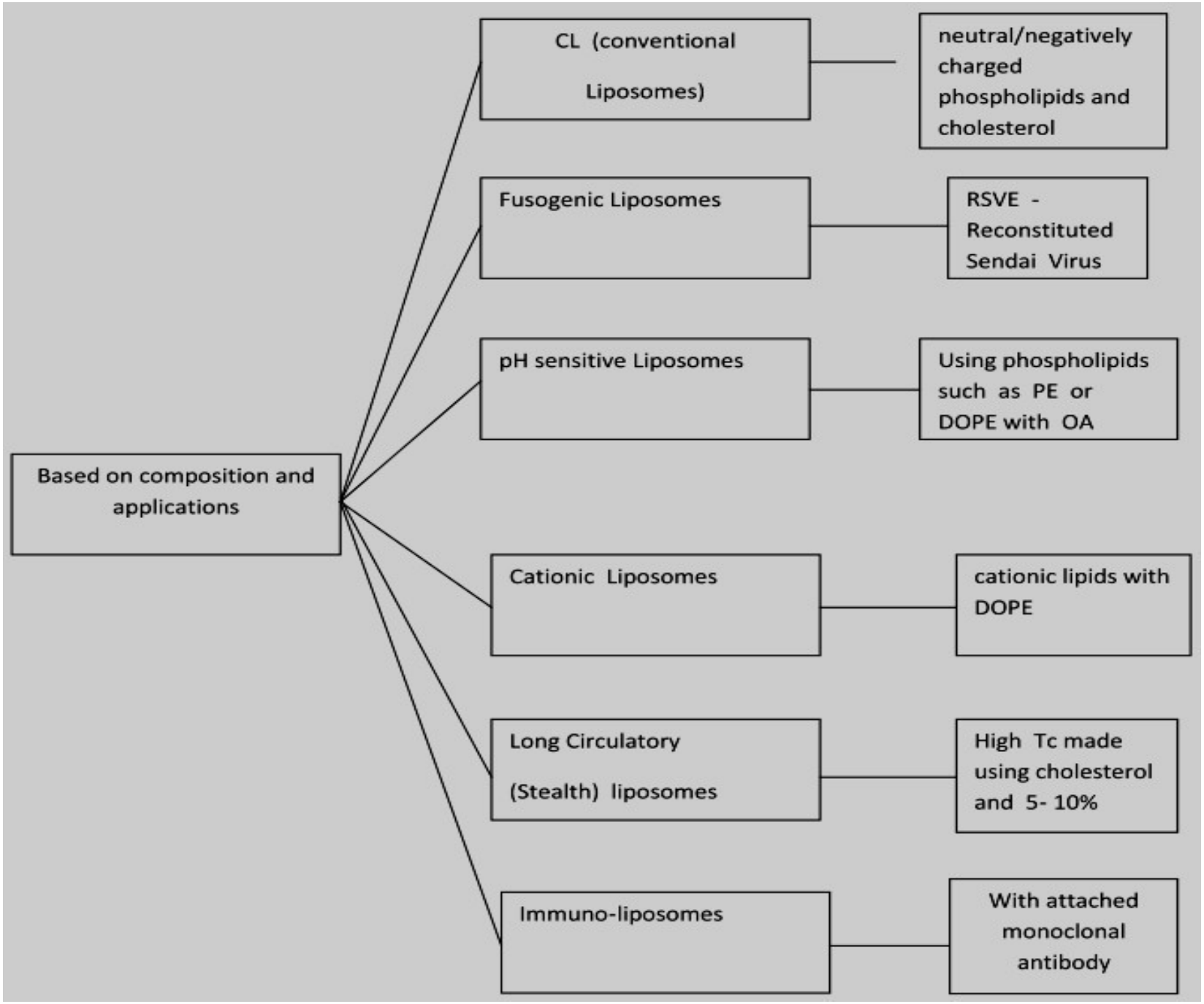
Types of Liposomes

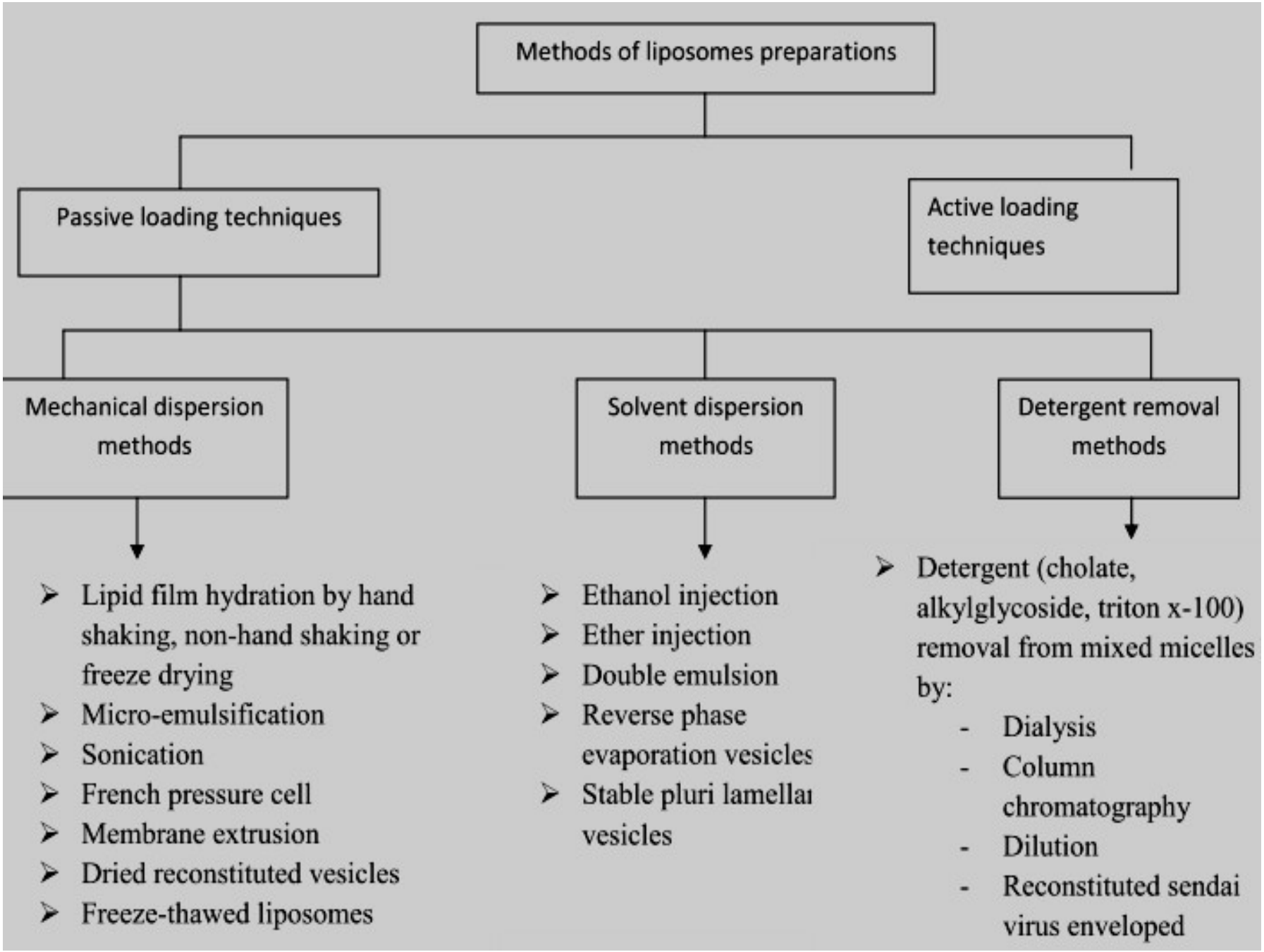
Liposomes are classified on the basis of

- Structural parameters
- Method of preparation
- Composition and applications


Based on Structure parameter
MLV
Multilamellar Vesicles
(> 0.5 µm)
OLV
Oligolamellar Vesicles
(0.1 – 1.0 µm)
ULV
Unilamellar Vesicles
(All size ranges)
MVV
Multivesicular Vesicles
(> 1.0 µm)
MUV
Medium Unilamellar Vesicle
SUV
Small Unilamellar Vesicle
(20 – 100 nm)
GUV
Giant Unilamellar Vesicle
(> 1.0 µm)
LUV
Large Unilamellar Vesicle
(> 100 nm)


DRV
Dehydration rehydration method
REV
SUVs/OLVs made by reverse phase evaporation method
MLV-REV
MLVs made by reverse phase evaporation
Based on method of preparation
VET
Vesicles prepared by Extrusion Technique
FATMLV
Frozen and thawed MLV
SPLV
Stable plurilamellar vesicles


Based on composition and applications
CL (conventional Liposomes)
neutral/negatively charged phospholipids and cholesterol
Fusogenic Liposomes
RSVE - Reconstituted Sendai Virus
pH sensitive Liposomes
Using phospholipids such as PE or DOPE with OA
Cationic Liposomes
cationic lipids with DOPE
Long Circulatory (Stealth) liposomes
High Tc made using cholesterol and 5- 10%
Immuno-liposomes
With attached monoclonal antibody


Methods of liposomes preparations
Passive loading techniques
Active loading techniques
Mechanical dispersion methods
Solvent dispersion methods
Detergent removal methods
Lipid film hydration by hand shaking, non-hand shaking or freeze drying
Micro-emulsification
Sonication
French pressure cell
Membrane extrusion
Dried reconstituted vesicles
Freeze-thawed liposomes
Ethanol injection
Ether injection
Double emulsion
Reverse phase evaporation vesicles
Stable pluri lamellar vesicles
Detergent (cholate, alkylglycoside, triton x-100) removal from mixed micelles by:
- Dialysis
- Column chromatography
- Dilution
- Reconstituted sendai virus enveloped

Parameters for Liposomes preparation

The correct choice of liposome preparation method depends on the following parameters:

1) physicochemical characteristics of the material to be entrapped and those of the liposomal ingredients;

2) the nature of the medium in which the lipid vesicles are dispersed;

3) the effective concentration of the entrapped substance and its potential toxicity;

4) additional processes involved during application/ delivery of the vesicles;

5) optimum size, polydispersity and shelf-life of the vesicles for the intended application.

6) batch-to-batch reproducibility and possibility of large-scale production of safe and efficient liposomal products.

Handling of Liposomes

- The lipids used in the preparation of liposomes are unsaturated and hence susceptible to oxidation.
- Also volatile solvents such as chloroform which are used will tend to evaporate from the container.
- Thus liposomes must be stored in an inert atmosphere of nitrogen, and in the dark, in glass vessels with a securely fastened cap.

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Applications of Liposomes

Applications of Liposomes

- Liposomes are used for drug delivery due to their unique properties.
- Liposomes that contain low (or high) pH can be constructed such that dissolved aqueous drugs will be charged in solution.
- As the pH naturally neutralizes within the liposome (protons can pass through some membranes), the drug will also be neutralized, allowing it to freely pass through a membrane.
- Liposomes are also used as models for artificial cells.

Applications of Liposomes

- Another strategy for liposome drug delivery is to target endocytosis events.
- Liposomes can be made in a particular size range that makes them viable targets for natural macrophage phagocytosis.
- These liposomes may be digested while in the macrophage's phagosome, thus releasing its drug.
- Liposomes can also be decorated with opsonins and ligands to activate endocytosis in other cell types.
- The use of liposomes for transformation or transfection of DNA into a host cell is known as lipofection

Applications of Liposomes

Protection against enzyme degradation of drugs

Liposomes are used to protect the entrapped drug against enzymatic degradation whilst in circulation.

The basis is that the lipids used in their formulation are not susceptible to enzymatic degradation; the entrapped drug is thus protected while the lipid vesicles are in circulation in the extracellular fluid.

Drug targeting

The approach for drug targeting via liposomes involves the use of ligands (e.g., antibodies, sugar residues, apoproteins or hormones), which are tagged on the lipid vesicles.

The ligand recognises specific receptor sites and, thus, causes the lipid vesicles to concentrate at such target sites.

Applications of Liposomes

Enhanced antimicrobial efficacy

Antimicrobial agents have been encapsulated in liposomes for two reasons.

First, they protect the entrapped drug against enzymatic degradation. For instance, the penicillins and cephalosporin are sensitive to the degradative action of β-lactamase, which is produced by certain microorganisms.

Secondly, the lipid nature of the vesicles promotes enhanced cellular uptake of the antibiotics into the microorganisms, thus reducing the effective dose and the incidence of toxicity as exemplified by the liposomal formulation of amphotericin B.

Applications of Liposomes

Topical drug delivery

The application of liposomes on the skin surface has been proven to be effective in drug delivery into the skin.

Liposomes increase the permeability of skin for various entrapped drugs and at the same time diminish the side effect of these drugs.

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Dendrimers

Learning Objectives

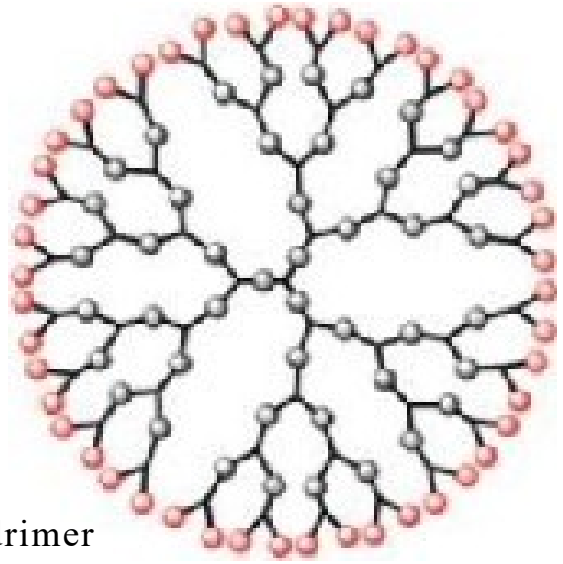
Introduction

Properties

Characterization

Introduction

- Dendrimers are repetitively branched molecules Which are also called as "arborols/ cascade molecules".
- They are family of nanosized, highly branched three dimensional molecules.
- Dendrimers are highly defined nanoparticles



Basic unit of dendrimer

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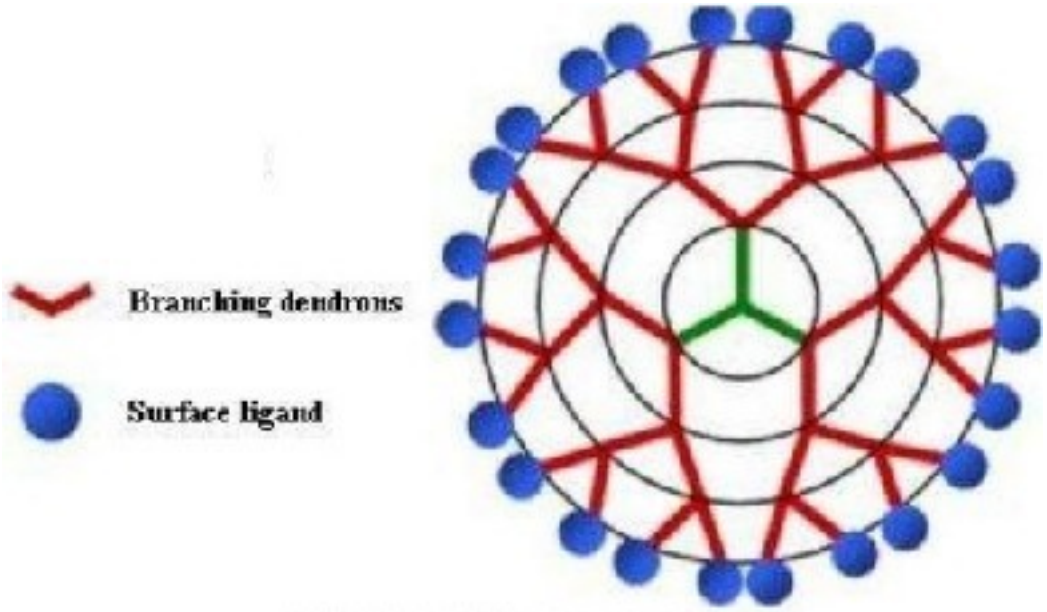
Structure of Dendrimers

- Dendrimers are built from a starting atom, such as nitrogen, to which carbon and other elements are added by a repeating series of chemical reactions that produce a spherical branching structure.
- As the process repeats, successive layers are added, and the sphere can be expanded to the size required by the investigator.
- The result is a spherical macromolecular structure whose size is similar to albumin and hemoglobin

Structure of Dendrimers

Basic dendrimer components are

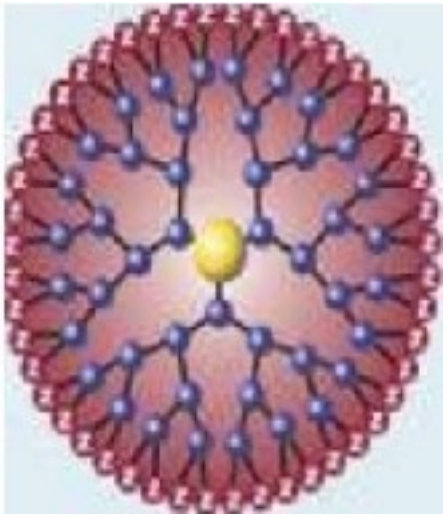
- An interior core
- Interior layers composed of repeating units radically attached to cores.
- Exterior layer (terminal functionality) attached to interior generations

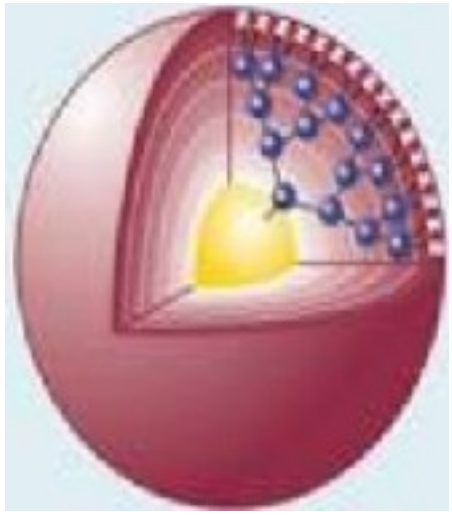


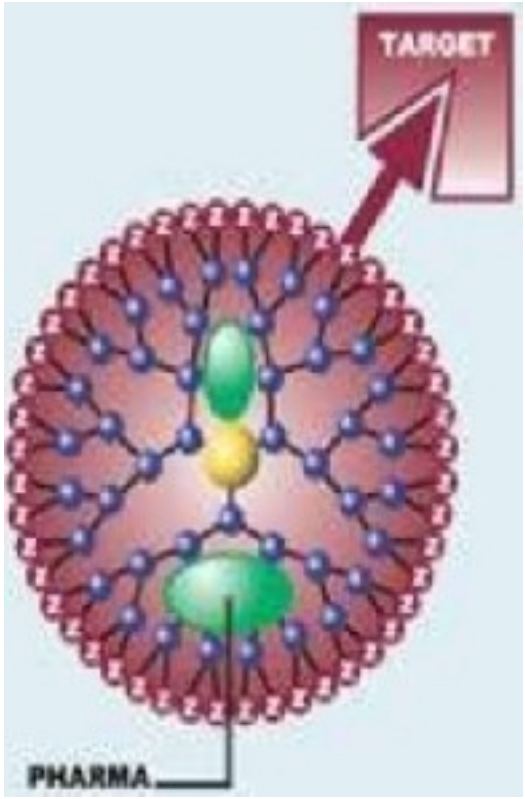
Basic dendrimer components

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3D Structure of Dendrimer







Ideal Properties of Dendrimer

- Inert and non-toxic.
- Biodegradable.
- Non-immunogenic.
- Able to cross barriers such as intestine, blood-tissue barriers, cell membranes.
- Able to stay in circulation for the time needed to have a clinical effect.
- Able to target to specific structures
- Compatible with guest molecules.
- Must protect the drug until it reaches to the desired site of action and release the drug.

Physicochemical Properties of Dendrimer

- Nanoscale size
- High solubility, miscibility and reactivity due to the presence of many chain ends.
- These carriers have significantly lower viscosity than linear polymers.
- Size and molecular weight can be controlled during synthesis.

Characterization of Dendritic Polymers

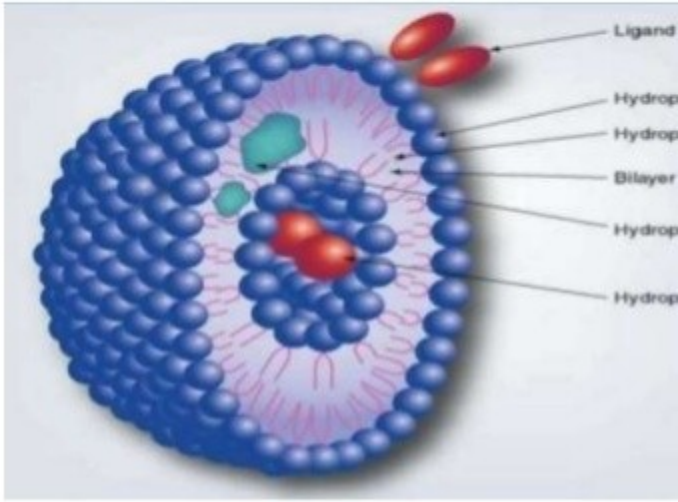
- Spectroscopy (UV-Visible, IR,NMR)
- Microscopy (TEM , SEM)
- Physical properties (Scattering techniques and Electrical techniques)
- Solubility
- Determination of Melting Point
- Elemental Analysis (Nitrogen)

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NIOSOMES

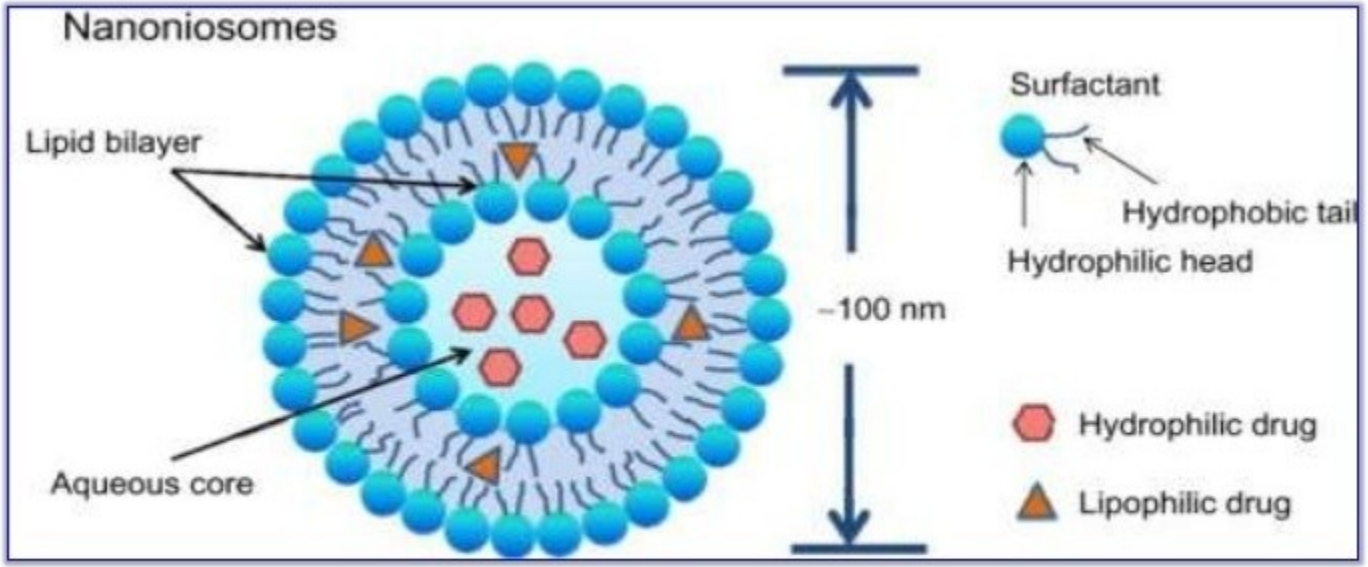
Introduction

- Niosomes are non-ionic surfactant based unilamellar or multilamellar bilayer vesicles upon hydration of non ionic surfactants with or without incorporation of cholesterol .
- The niosomes are very small, and microscopic in size. Their size lies in the nanometric scale.
- Niosomes are a novel drug delivery system, in which the medication is encapsulated in a vesicle.
- Both hydrophilic & lipophilic drugs ,entrap either in the aqueous layer or in lipid layer.

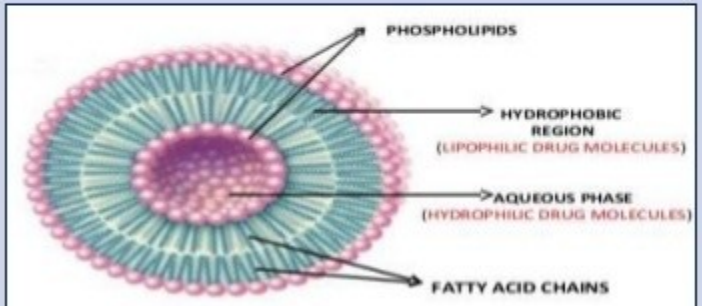
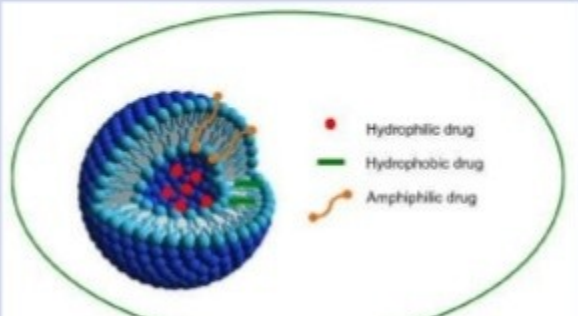


Structure of Niosomes

- These vesicular systems are similar to liposomes that can be used as carriers of amphiphilic and lipophilic drugs.
- It is less toxic and improves the therapeutic index of drug by restricting its action to target cells.



Niosomes Vs. Liposome

Liposomes	Niosomes
Vesicles made up of concentric bilayer of phospholipids	Vesicles made up of surfactants with or withou incorporation of cholesterol.
Size ranges from 10-3000nm	Size ranges from 10-100nm
Comparatively expensive	Inexpensive
Special storage condition are required	No such special requirement
Phospholipids used are unstable	Non-ionic surfactants are stable
Comparatively more toxic	Less toxic
	

Advantages

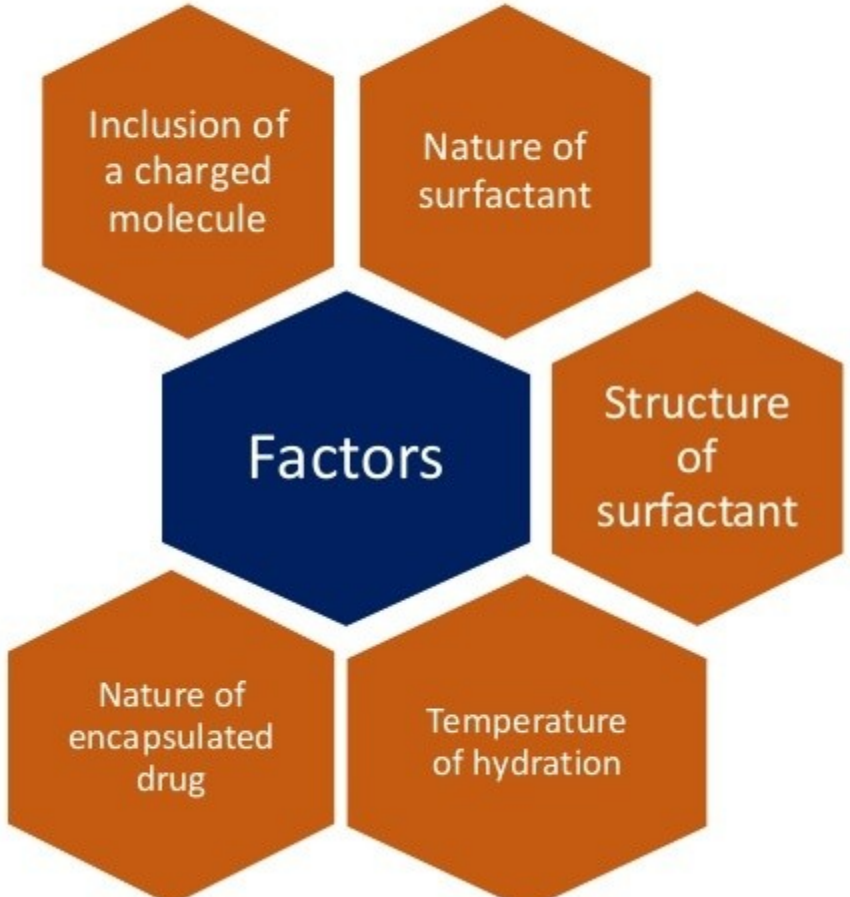
- They are osmotically active and stable.
- They increase the stability of the entrapped drug.
- The vesicle suspension being water based offers greater patient compliance over oil ba systems
- Since the structure of the niosome offers place to accommodate hydrophilic, lipophilic well as ampiphilic drug moieties, they can be used for a variety of drugs.
- The vesicles can act as a depot to release the drug slowly and of controlled release.
- Biodegradable, non-immunogenic and biocompatible.

Disadvantages

- Aggregation
- Fusion
- Leaking of entrapped drug
- Hydrolysis of encapsulated drugs which limiting the shelf life of the dispersion.

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Factors affecting Niosomes formation



Application of Niosome

Localized Drug Action

- Drug delivery through Niosomes is one of the approaches to achieve localized drug action, since their size and low penetrability through epithelium and connective tissue keeps the drug localized at the site of administration.
- Localized drug action results in enhancement of efficacy of potency of the drug and at the same time reduces its systemic toxic effects e.g. Antimonia encapsulated within niosomes are taken up by mononuclear cells resulting in localization of drug, increase in potency and hence decrease both in dos and toxicity.
- The evolution of niosomal drug delivery technology is still at an infancy stage, but this type of drug delivery system has shown promise in cancer chemotherapy and anti-leishmanial therapy.

Application of Niosome

Transdermal delivery of drugs by niosomes

- An increase in the penetration rate has been achieved by transdermal delivery of drug incorporated in niosomes as slow penetration of drug through skin is the major drawback of transdermal route of delivery for other dosage forms.
- The topical delivery of erythromycin from various formulations including niosomes has studied on hair less mouse and from the studies, and confocal microscopy, it was found that nonionic vesicles could be formulated to target pilosebaceous glands

Application of Niosome

Niosome as a carrier for Hemoglobin

- Niosomal suspension shows a visible spectrum superimposable onto that of free hemoglobin so can be used as a carrier for hemoglobin.
- Vesicles are also permeable to oxygen and hemoglobin dissociation curve can be modified similarly to non-encapsulated hemoglobin

Application of Niosome

Delivery of peptide drugs

- Niosomal entrapped oral delivery of 9-desglycinamide, 8- arginine vasopressin was examined in an in-vitro intestinal loop model and reported that stability of peptide increased significantly.
- Immunological applications of niosomes For studying the nature of the immune response provoked by antigens niosomes have been used.
- Niosomes have been reported as potent adjuvant in terms of immunological selectivity, low toxicity and stability

Summary of Niosomes

- Niosomes provide incorporating the drug into for a better targeting of the drug at appropriate tissue destination .
- They presents a structure similar to liposome and hence they can represent alternative vesicular systems with respect to liposomes
- Niosomes are thoughts to be better candidates drug delivery as compared to liposomes due to various factors like cost, stability etc.
- Various type of drug deliveries can be possible using niosomes like targeting, ophthalmic, topical, parenteral etc.

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Aptamers

Learning Objectives

Introduction

Properties

Applications

Introduction

- Aptamer **(aptus: "to fit"& mer: "smallest unit of repeating structure").**
- Single stranded folded oligonucleotides and peptide that bind to molecular (protein) targets with high affinity and specificity.
- They are similar to monoclonal antibodies.
- Provide opportunities for structure-based drug design strategies relevant to therapeutic intervention.
- Can be prepared using SELEX(Systematic Evolution Ligands by Exponential Enrichment).

Introduction

- Based on their three-dimensional structures, Aptamers can well-fittingly bind to a wide variety of targets from single molecules to complex target mixtures or whole organisms.

- They range in size from 20 to 80 bases(6-26 kDa).

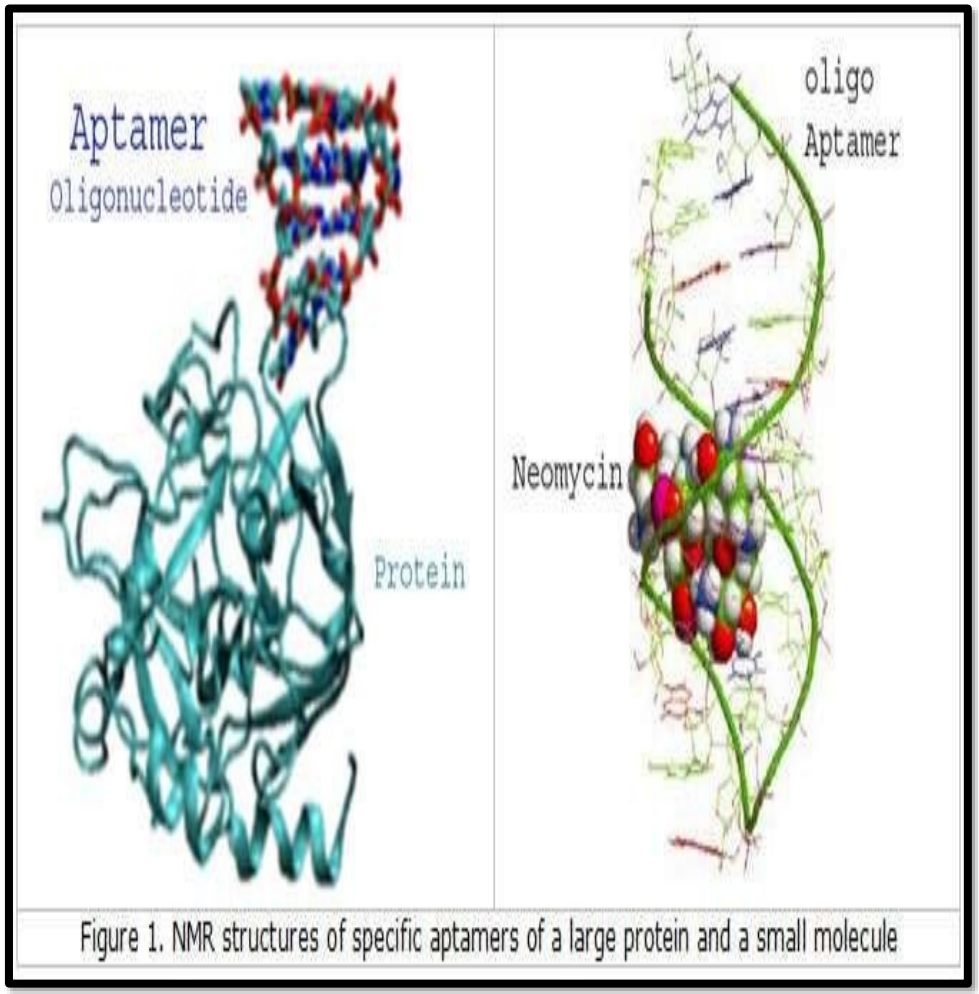


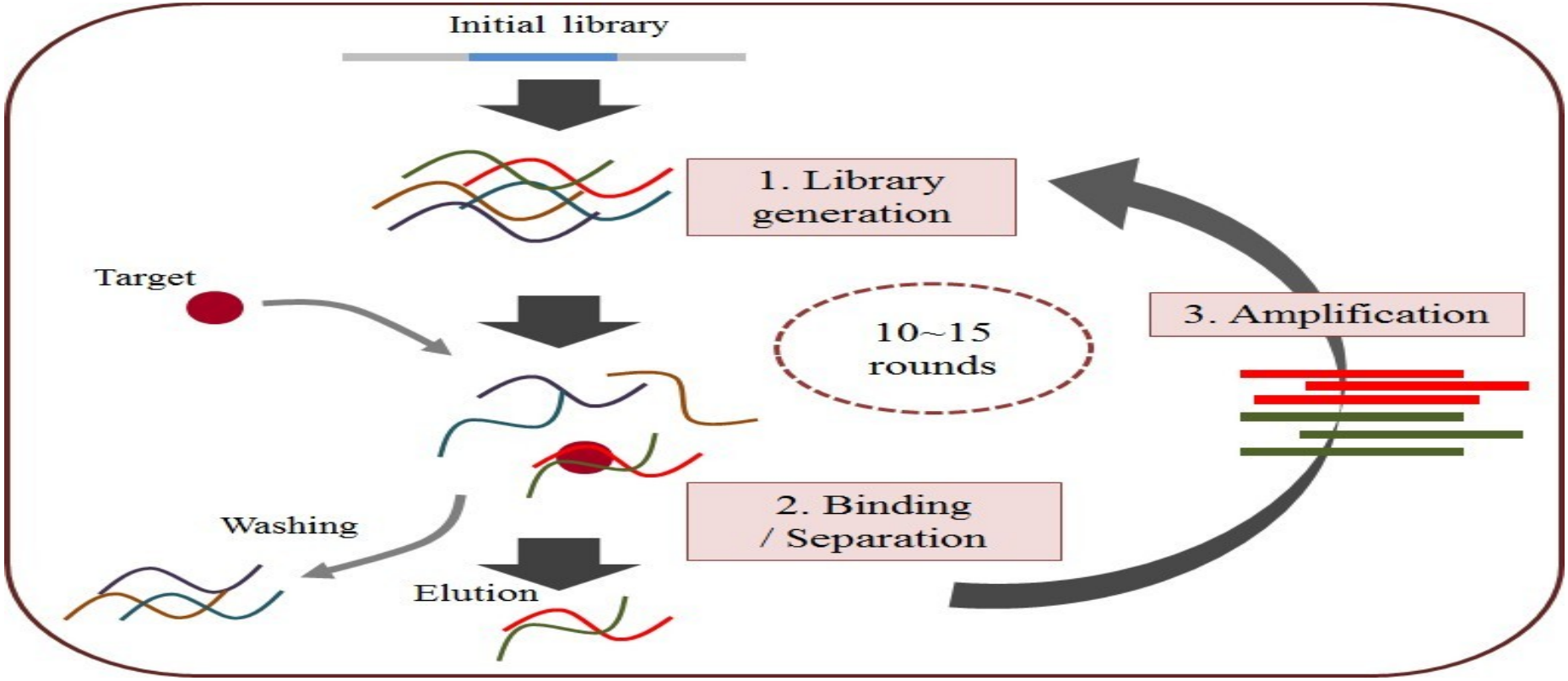
Figure 1. NMR structures of specific aptamers of a large protein and a small molecule

Aptamers production method

- Aptamer are produced by an in vitro selection or systematic evolution of ligand by exponential enrichment(SELEX).
- It will produce ssRNA with specific binding target.
- The first step is to synthesize a large pool of nucleic acid molecules made from average of 15-40 bases of random sequences that are flanked by primers 5' and 3' end.
- The pool of DNA is transcribed into an RNA pool and it subsequently exposed to the target ligand of interest.
- Affinity column chromatography removes unbound sequences identify the strongest binding sequences and bound sequences are eluted and amplified by RT-PCR .

Aptamers production method

- Affinity column chromatography removes unbound sequences identify the strongest binding sequences and bound sequences are eluted and amplified by RT-PCR .



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Aptamers

Learning Objectives

Properties

Applications

Properties of Aptamers

- RNA/DNA or Protein oligonucleotide molecules.
- Ability to bind to proteins with high affinity and specificity.
- Adopt wash number of 3-D shapes- which enables them to bind tightly to specific molecular target.
- Usually very small in size but can easily distinguish between very closely related protein.
- Highest specificity and affinity to the target.

Properties of Aptamers

- Ability to disrupt interaction between proteins.
- High specificity and high affinity to the target and are capable of penetrating tissues then antibodies because of their smaller size.
- Low or non immunogenic and non toxic molecules.
- Production of aptamers is done in-vitro using automated synthesis.
- Less cost of oligonucleotide synthesis.
- Unlimited shelf life.

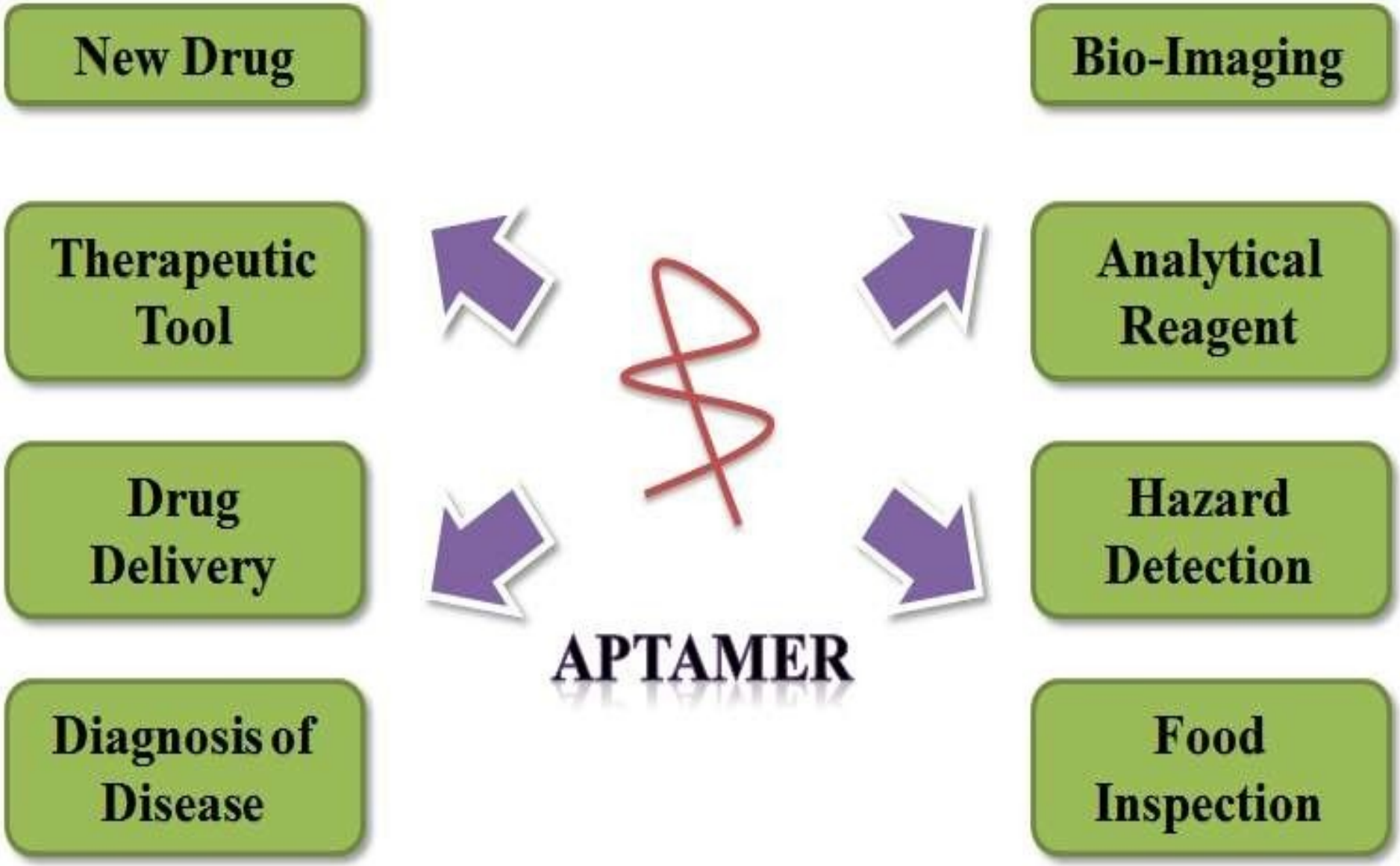
Different types of Aptamers

RNA APTAMER	DNA APTAMER	PEPTIDE APTAMER
From complex secondary and tertiary structure	From complex secondary and tertiary structure	Structure constrains by scaffoled
From diverse 3d structure	Less diverse 3d structure than rna aptamer	3d structure constraints by scaffold
Bind target with the entire sequence	Bind target with the entire sequence	Bind target variable region only
Biosenseor, diagonostic, therapeutics applications	Biosenseor, diagonostic, therapeutics applications	Biosenseor, diagonostic, therapeutics applications

Aptamers vs Antibodies

Aptamer	Antibodies
Aptamer are oligonucleotide and protein	Antibodies are protein in nature.
Uniform activity regardless of batch	varies from batch to batch.
Investigator determines target site of protein	Immune system determines target site of protein.
Wide variety of chemical modifications to molecule for diverse functions	Limited modifications of molecule
No evidence of immunogenicity.	Significant immunogenicity
They are more stable at high temperature and they can be regenerated easily after denaturation.	Temperature sensitive

Applications



Fluorescence(Forster) Resonance Energy Transfer (FRET)

Learning Objectives

Introduction

Properties

Application Examples

FRET background

- FRET, or Fluorescence Resonance Energy Transfer, was first described more than 50 years ago. The availability of new dyes and detection technology has resulted in a much wider use of the application in recent years, especially in biomedical R&D.
- In simple terms FRET is dependent on the distance dependent transfer of energy from a donor molecule to an acceptor molecule, resulting in the loss of fluorescence of the donor and gain of fluorescence of the acceptor – either of which may be measured.
- FRET is therefore an extremely powerful tool for identifying molecular interactions. It can be used in a variety of base techniques, including microscopy, flow cytometry and ELISA.

Conditions influencing FRET

- DISTANCE – donor and acceptor molecules must be in close proximity to each other, generally between 10-100 Å
- SPECTRA – the absorption spectrum of the acceptor must overlap the emission spectrum of the donor
- DIPOLE ORIENTATION – donor and acceptor molecule dipole orientations should be approximately parallel

Measurement of FRET

FRET is effectively measured in one of two ways:-

1. The increase in fluorescence of the acceptor molecule
2. The decrease in fluorescence of the donor molecule

"Acceptor photobleaching" is a widely used variant, whereby bleaching of the acceptor molecule leads to an increase in donor emission, showing that FRET was occurring in the system.

FRET experimental design (1)

- Selection of appropriate donor / acceptor pairs based both on their inherent suitability as a pair, and your equipment.

- Choose suitable filters to maximize excitation of your donor, whilst minimizing any direct excitation of the acceptor molecule.

- Consider concentration – are there likely to be enough detectable events?

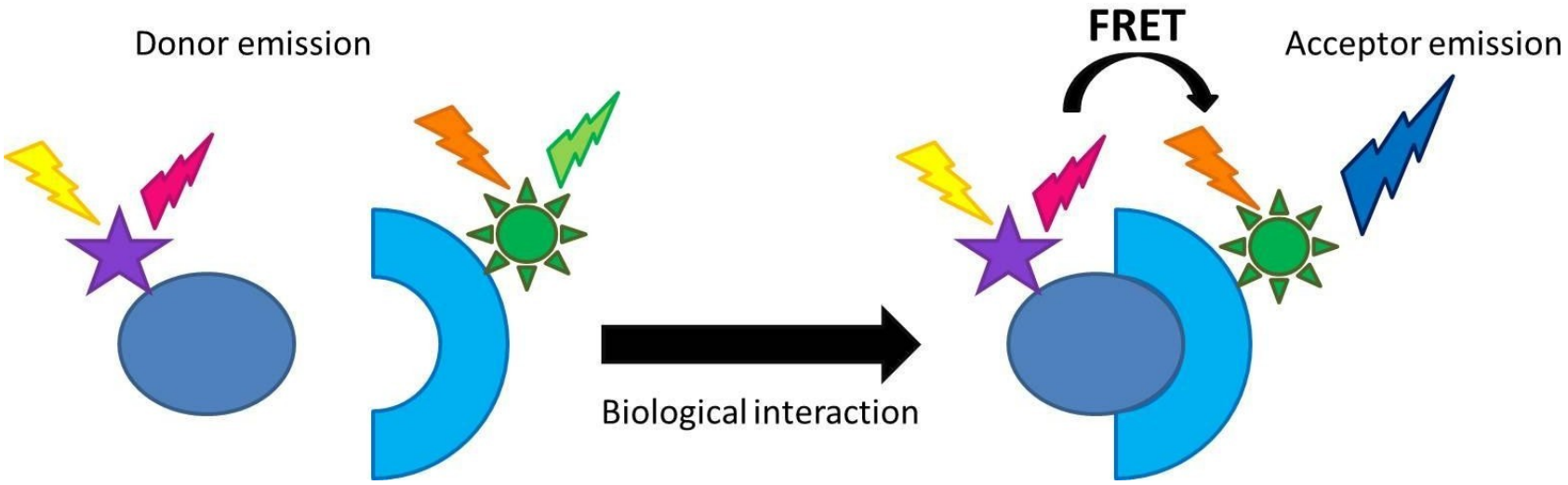
- High concentrations of both donor and acceptor might not be enough to generate a signal if the interaction is very low – remember that it is the interaction that is being measured.

FRET Applications

- To analyze biological interactions
- To predict structural changes within the biomolecules.
- To analyze drug binding with molecular targets
- To analyze protein modifications
- Biosensor development
- Cell cycle analysis

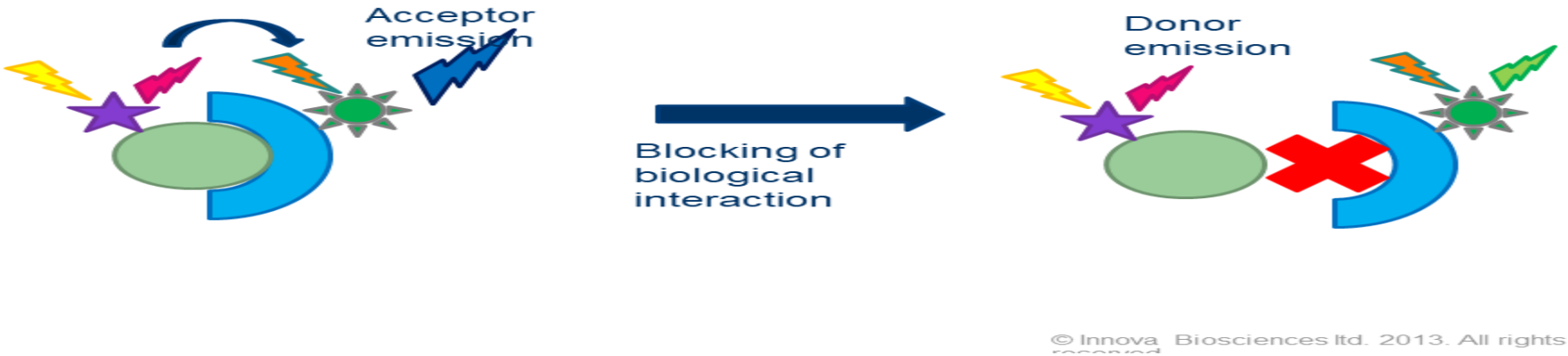
FRET examples

- Typical example of detecting interaction of two molecules – e.g. drug binding to active site or antibody binding to target.



FRET examples

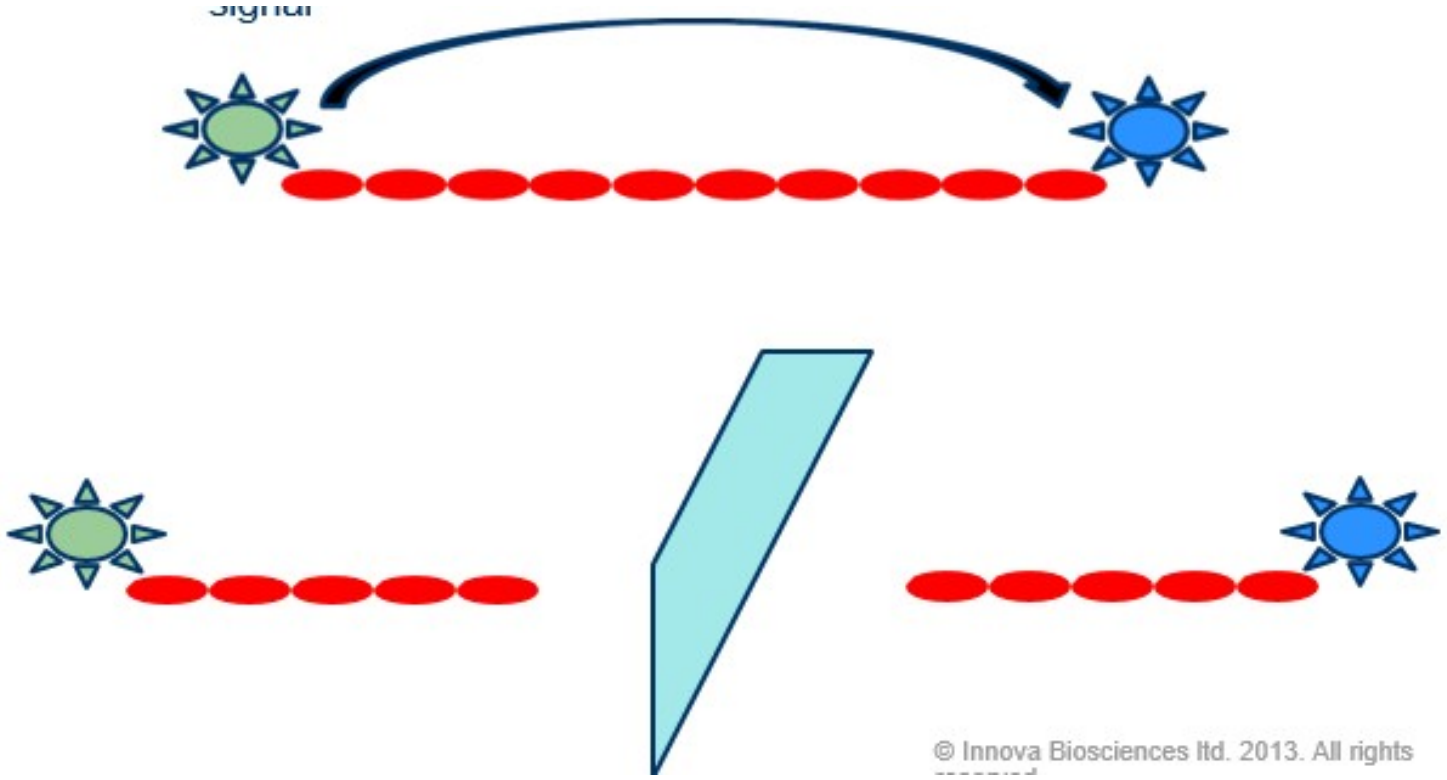
- Example of detecting a compound that blocks interaction of two molecules



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FRET examples

- Example of protein cleavage detected by loss of FRET signal



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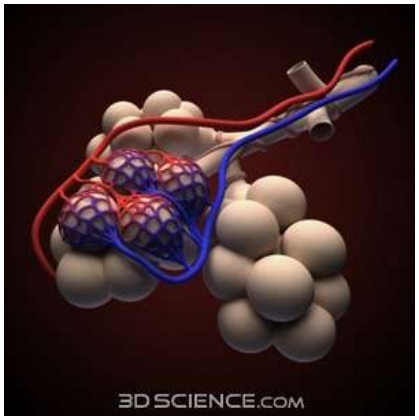
Microfluidics

Learning Objectives

Introduction
Properties
Types

What is Microfluidics?

- It is the science and technology of systems that process or manipulate small (10^{-9} to 10^{-18} litres) amounts of fluids, using channels with dimensions of tens to hundreds of micrometres



← Microfluidics in nature: Lung alveoli

A typical microfluidic system

DNA separation system



- From Agilent-Caliper
- Allow to characterize DNA Fragments with excellent resolution, and in a small time

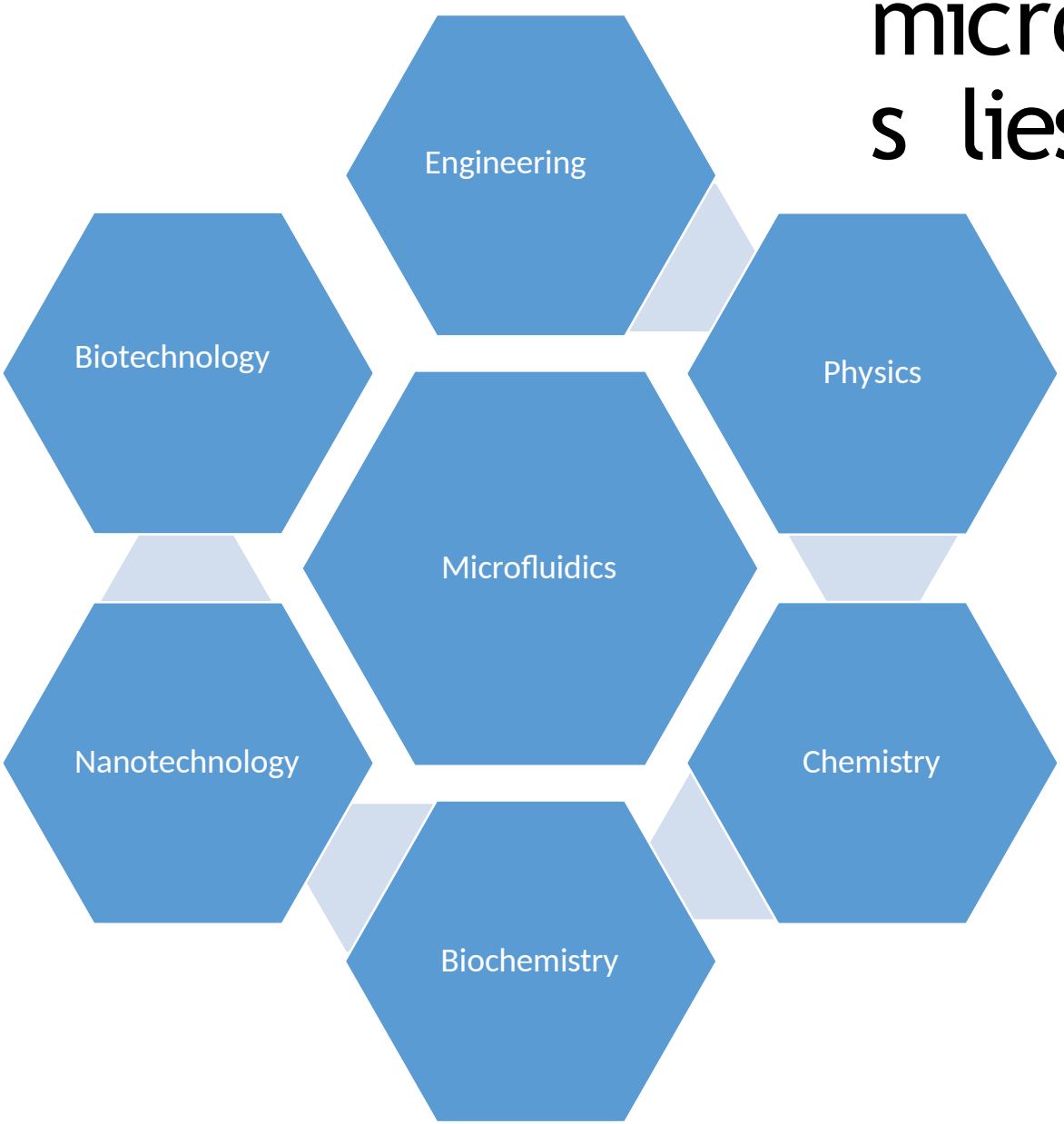
Another example



- An elementary lab-on-a-chip: Diagnoses heart attack within 10 minutes

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Where microfluidics lies



Motivation for miniaturization

- Micro scale = laminar flow
- Laminar flow allows controlled mixing
- Low thermal mass
- Efficient mass transport (speedy diffusion)
- Good (large) ratio of channel surface area: channel volume
- Single cell and molecule manipulations
- Protection against contamination and evaporation
- Kinetics easy to study

Benefits of size reduction

1. Decreased reagent consumption
2. Small economic footprint
3. Rapid heat transfer and catalysis
4. Fast diffusive mixing
5. Automation and integration

Generic components of a microfluidic system

- a method of introducing reagents and samples (as fluids)
- methods for moving these fluids around on the chip, and for combining and mixing them
- methods for moving these fluids around on the chip, and for combining and mixing them

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Typical components

- Microducts
- Micronozzles
- Micropumps
- Microturbines
- Microvalves
- Microsensors
- Microfilters
- Microneedles
- Micromixers
- Microreactors
- Microdispensers
- Microseparators

- **Common fluids used** in microfluidic devices include whole blood samples, bacterial cell suspensions, protein or antibody solutions and various buffers

- Microfluidic devices can be used to obtain a variety of interesting measurements including **molecular diffusion coefficients**, **fluid viscosity**, **pH**, **chemical binding coefficients** and **enzyme reaction kinetics**

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Microfluidics

Learning Objectives

Branches
Applications

Branches of microfluidics

1. CONTINUOUS FLOW MICROFLUIDICS

Continuous flow microfluidics enables to manipulate continuous flow of liquid through micro-channels.

Thanks to devices such as external pressure pumps or integrated mechanical micro-pumps.

Continuous flow processes are used in a wide range of applications like in bioanalytical, chemical, energy and environmental fields.

2. DIGITAL MICROFLUIDICS

Also called droplet microfluidics or emulsion science, digital microfluidics is one of the main application field of microfluidics.

It enables to manipulate autonomous droplets on a substrate using electro-wetting. This allow to generate and control uniform, reproducible droplets over the experiments' parameters.

Droplets generation can be used in a large scale of applications like in synthesis of nanoparticles, single cell analysis, and encapsulation of biological entities. This technology will probably become an important tool for drug delivery and biosensing, by providing new solutions for state-of-the-art diagnostics and therapeutics.

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3. OPTOFLUIDICS AND MICROFLUIDICS

Optfluidics is an emerging fast-growing science resulting from the combination of three fields of science: microphotonics, optics and microfluidics. Optofluidics merges light and liquids into miniaturized optical devices that take advantage of the properties of fluids to generate high precision and flexibility.

Optofluidic applications include lab-on-chip devices, fluid waveguides, deformable lenses, microdroplets lasers, displays, biosensors, optical switches or molecular imaging tools and energy.

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4. ACOUSTO FLUIDICS

Acoustofluidics deals with the use of acoustic fields, mainly ultrasonics onto fluids within microfluidic channels allowing to manipulate cells and particles. It refers to the study and manipulation of acoustic waves on microscale to nanoscale fluidic environments.

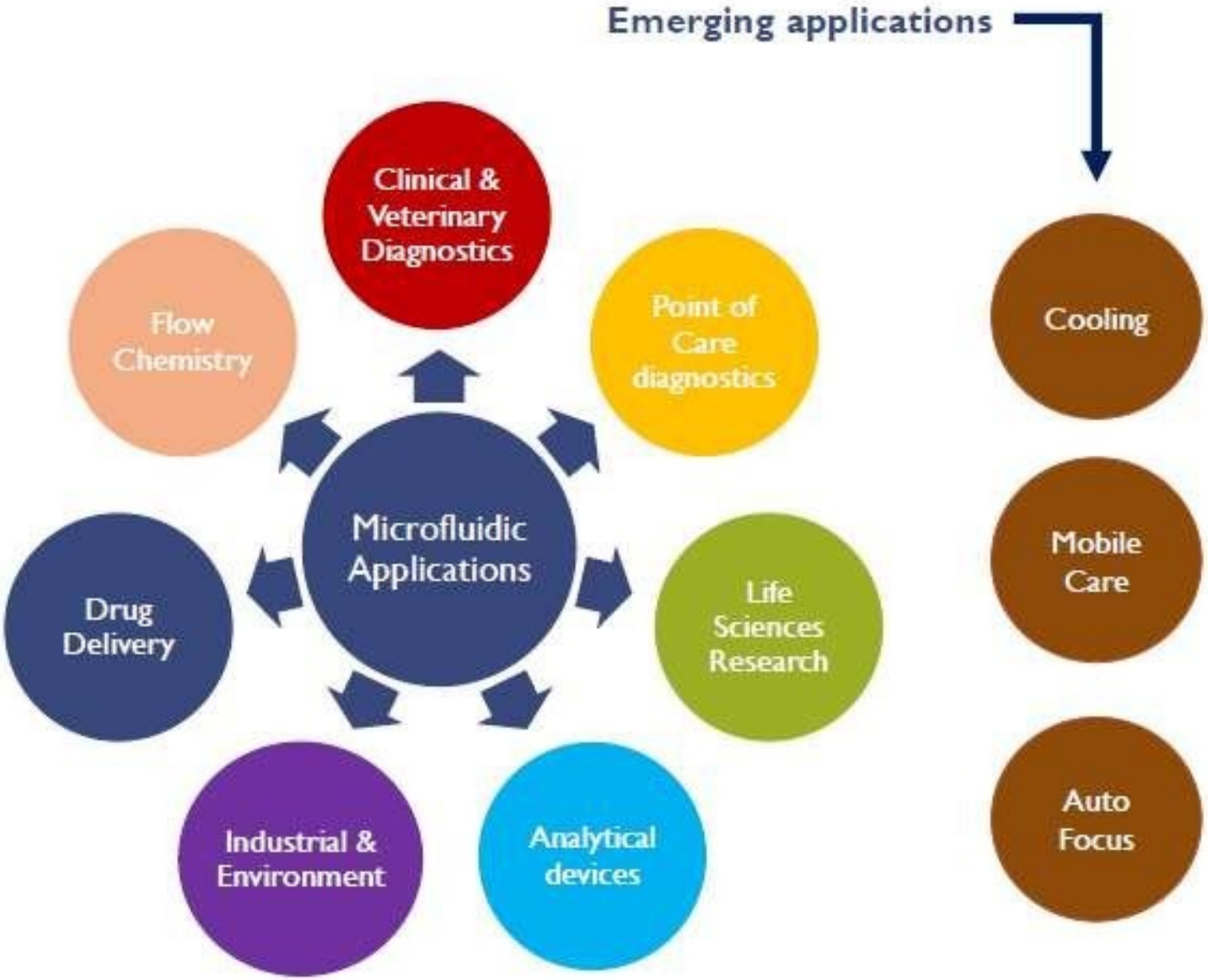
5. ELECTROPHORESIS AND MICROFLUIDICS

Electrophoresis is a technique used in clinical and research laboratories in order to separate molecules based on their size, electrical charge and shape.

An electric current flows through a medium holding the mixture of molecules. Positively-charged ions (cations) proceed towards a negative electrode whereas negatively-charged ions (anions) proceed towards a positive electrode.

This method is used for both DNA and RNA analysis.

Microfluidic Applications



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Key Application Areas

- Polymerase Chain Reactions
- Immunoassays
- Drug Screening
- Electrophoretic separations
- Analysis of unpurified blood samples
- DNA sequencing
- Single Cell manipulation
- Screens for protein crystallization conditions
- Cell culture studies

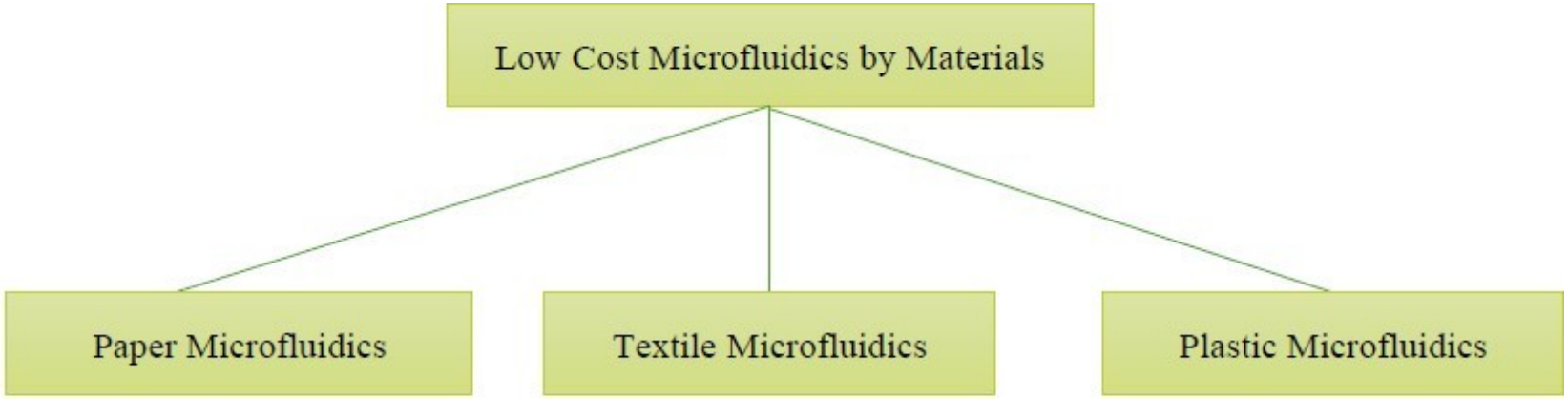
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Microfluidics

Learning Objectives

Low-cost and high-impact Microfluidics

Low-cost and high-impact Microfluidics



	Paper	Textiles	Plastics
Common material	Nitrocellulose, nylon and polyvinylidene fluoride	cotton, polyester, silk	Polyacrylamid, polydymethilsiloxane
Transport	Capillary action	Capillary action	Laminar Bulk Flow
Purpose	Immunoassay, pH detection,	Immunoassay, pH detection	Cell Separation, Cell Culture

Low-cost and high impact microfluidics

1. Paper-based Microfluidics

- Available everywhere and cheap
- Low fabrication cost
- Passive fluid transport through capillary action
- Thin, lightweight
- Easy to stack, store, and transport
- Disposable and Biodegradable

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APPLICATIONS
Paper-based Microfluidics



2. Plastic-based Microfluidics

- Use of materials such as Polydymethilsiloxane (PDMS), acrylic(PMMA), Polystyrene, Cycloolefin
- For variety of applications that cannot be achieved with paper
  -Able to pattern microstructure, microvalves, etc
  -Able to transfer bulk liquid in a micro channel
  -Can be used for cell works (separation, cell culture)
  -Can be used repetitively

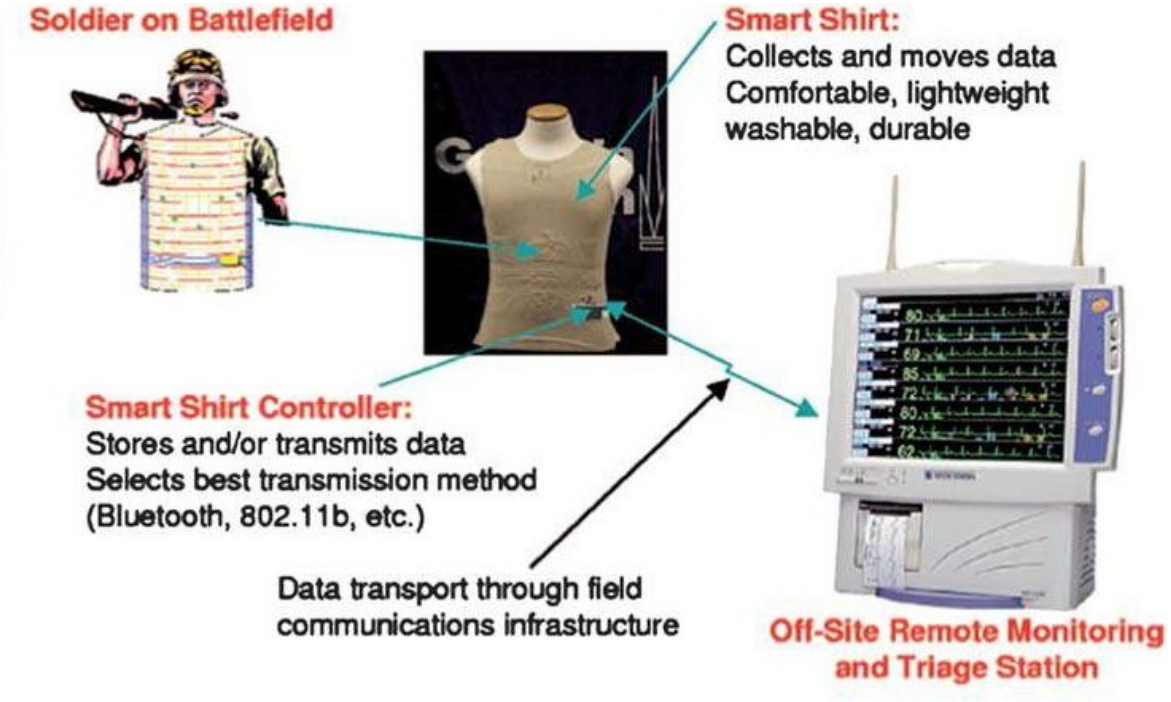
Other applications of plastic-based microfluidics

- Rapid Genotyping of Malaria-Transmitting Mosquitoes
- Circulating Tumor Cells Separation

Low-cost and high impact microfluidics

3. Textile-based Microfluidics

- Sports performance measurement such as real-time sweat pH monitoring
  -Connection via Bluetooth for real-time analysis on smartphones
- Smart shirts, esp for security forces



Lab on a chip

Learning Objectives

Introduction
Properties
Current research
Advantages

Lab-on-a-chip

Start-to-finish systems based on microfluidics

- A lab-on-a-chip is a miniaturized device that integrates onto a single chip one or several analyses which are usually done in a laboratory
- Mainly focuses on human diagnostics and DNA analysis. Less often, on the synthesis of chemicals
- Microfluidic technologies used in lab-on-a-chip devices enable the fabrication of millions of microchannels, each measuring mere micrometers, on a single chip that fits in the palm of your hand.
- Eg. Commerically available chips for glucose monitoring, HIV detection or heart attack diagnostics, A chip which enables security forces to detect as soon as possible biological threats towards troops and civilians.

Lab-on-a-chip

Advantages

- Low cost
- High parallelization
- Ease of use and compactness
- Reduction of human error
- Faster response time and diagnosis
- Low volume samples

Disadvantages

- Not yet ready for industrialization
- Miniaturization increases the signal/noise ratio
- Untrained diagnoses

LAB-ON-A-CHIP: CURRENT RESEARCH FOCUS

- The industrialization of lab-on-a-chip technologies to make them ready for commercialization
- The increase in the maximum number of biological operations on the same chip and the increase in parallelization to achieve the detection of hundreds of pathogens in the same microfluidic cartridge
- Fundamental research on certain technologies with a high potential impact, such as DNA reading through nanopore, which requires more investigation in order to be applicable.

Potential impact on healthcare services

In a near future, lab-on-a-chip devices, with their ability to perform complete diagnosis can change our way of practicing medicine.

- Diagnosis will be done by people with lower qualifications, thus enabling doctors to focus only on treatment.
- Real time diagnosis will increase the chances of survival for patients
- A complete diagnosis will greatly reduce antibio-resistance, which is currently one of the biggest challenges
- In developing countries, lab-on-a-chip will enable healthcare providers to open diagnostics to a wider population

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Photolithography

Learning Objectives

Introduction

Basic steps

Photolithography

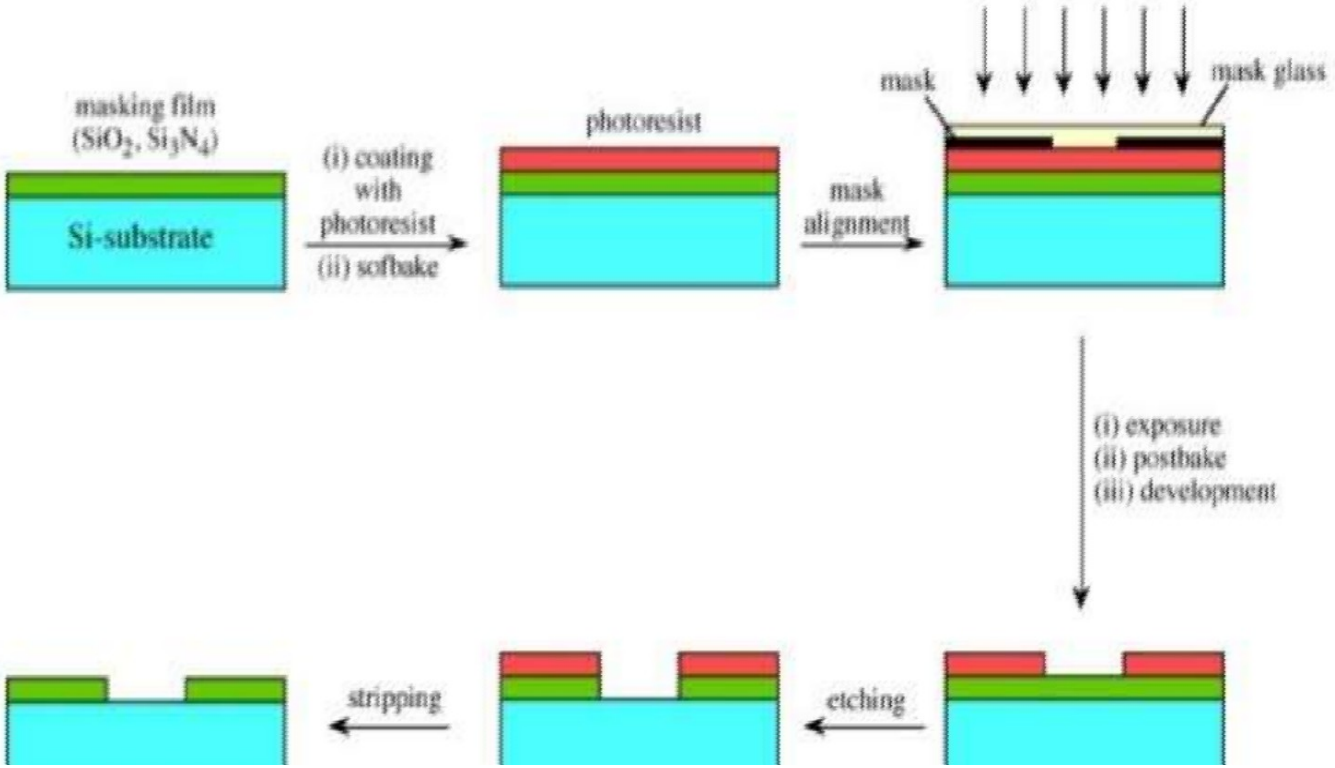
- Photolithography, also termed optical lithography or UV lithography, is a process used in micro fabrication.
- It uses light to transfer a geometric pattern.
- Lithography comes from the Greek word, lithos, means "stone“ and graphein, means "to write”.

Definition

- Photolithography is the process of transferring the geometrical patterns from a mask to the semiconductor material (silicon wafer) using the UV Radiation.

Steps included

- Spin coating with Photoresist.
- Exposure to the Radiation.
- Developing
- Etching
- Stripping



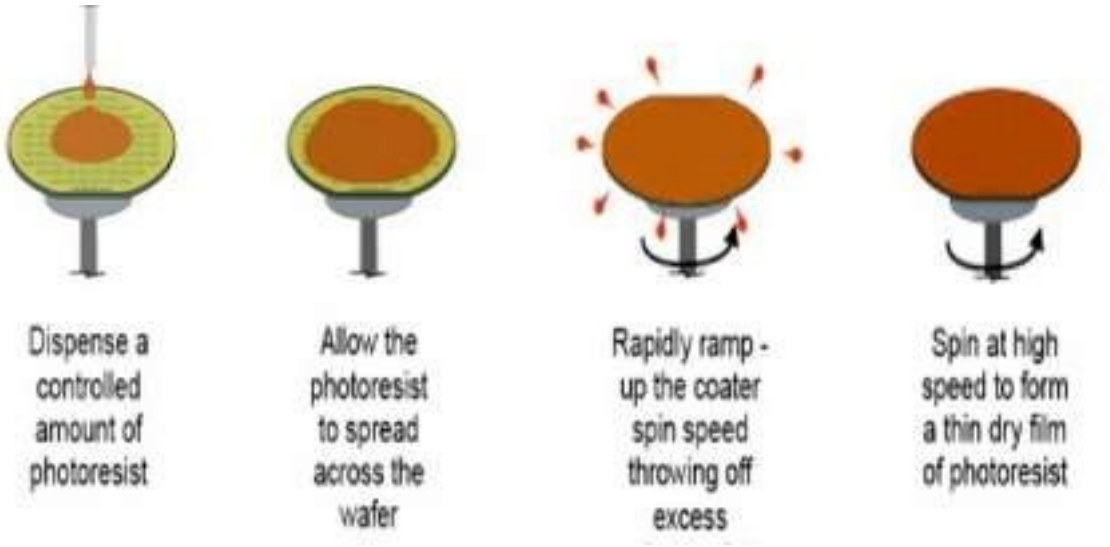
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Photoresist

- Photoresist is an organic polymer which changes its chemical structure when exposed to the sensitive radiation.
- Example: Poly(methyl glutarimide) (PMGI)

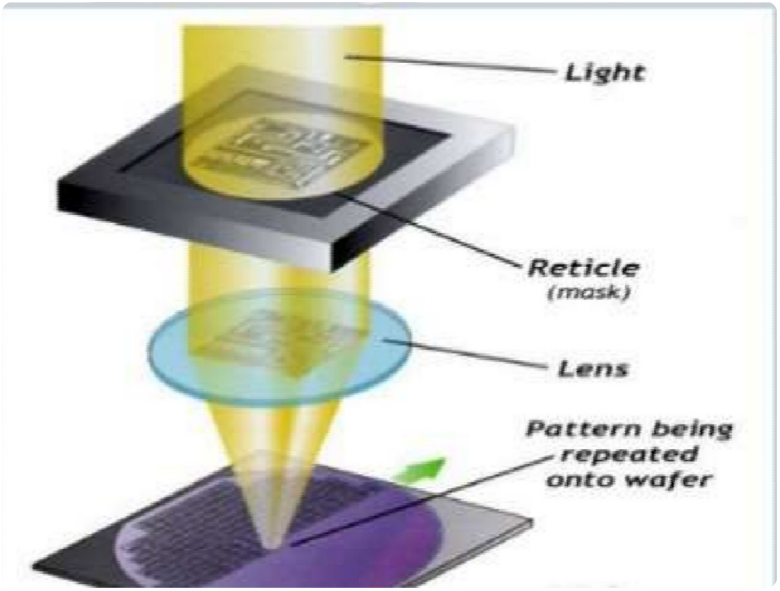
Spin coating

- The Si wafer is placed on the rotating platform or chuck and held via vacuum suction
- Resist is poured onto the wafer, the wafer spins at a high speed (more than thousand rpm's)
- The speed determines the thickness of the coated resist



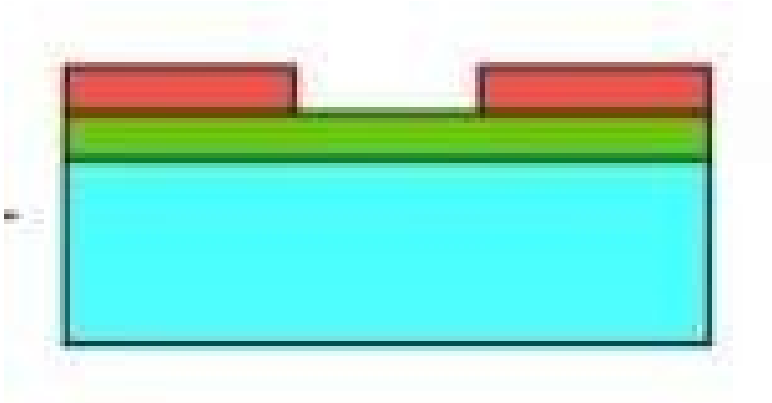
Exposure to the radiation

- The Si wafer is placed on the rotating platform or chuck and held via vacuum suction
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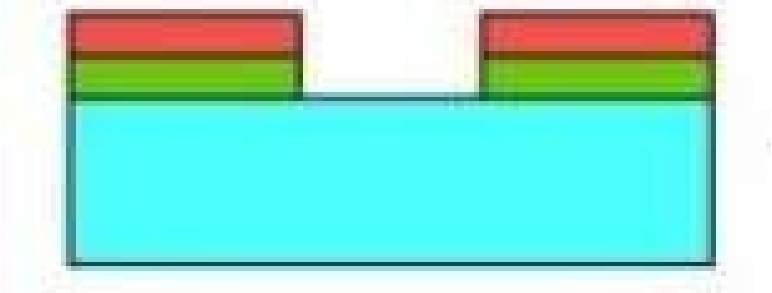
Developing

- After exposure, the wafer or the substrate is soaked in a developer solution
- This helps in removal of the softened regions of the photoresist
- After developing we are left with the photoresist of the same pattern as its on the mask



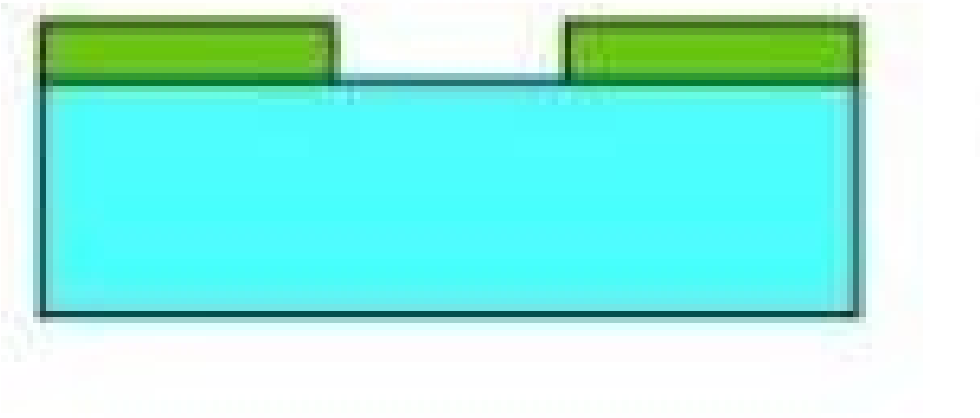
Etching

- After developing, the wafer or the substrate is soaked in a chemical bath or etchant
- This removes the oxide parts of the substrate, from where the Photoresist was removed previously
- Now we are left with the needed pattern, etched on the substrate



Stripping

- Now, remove all the photoresist by soaking the etched Si wafer in a photoresist removal solution.



Photolithography-II

Learning Objectives

Different types

Nanolithography

- Nanolithography concerns with the study and application of nanometre-scale structures.
- Nanolithography is derived from the Greek words “nanos”, meaning (dwarf) “lithos”, meaning (rock or stone) and “graphein” meaning (to write). Therefore the correct translation is "tiny writing on stone.
- Nanolithography is used e.g. during the nanofabrication of semiconductor integrated circuits and nanoelectromechanical systems (NEMS) especially for nanocircuitry.

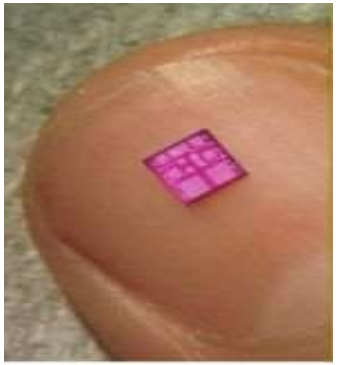


FIG 4 *Nanocircuitry*

Electron Beam Lithography

- Electron Beam Lithography is a specialized technique for creating extremely fine patterns. It is derived from the scanning electron microscope. Electron beams can be focused to a few nanometres in diameter.
- The basic idea behind electron beam lithography is identical to optical lithography. The substrate is coated with a thin layer of resist, which is chemically changed under exposure to the electron beam, so that the exposed/non-exposed areas can be dissolved in a specific solvent.
- Electron beam lithography is the most power full tool for the fabrication of feathers as small as 3nm to 5 nm.

Electron Beam Lithography

Advantages of EBL

- Print complex patterns directly on wafers
- Eliminates the diffraction problem
- High resolution up to 20 nm(photolithography ~50nm)
- Flexible technique

Applications of EBL

- Electron beam Lithography (EBL) is used primarily for two purposes
- Very high resolution lithography.
- Fabrication of masks.

X-Ray Lithography

- X-rays having wavelengths of 0.04nm to 0.5 nm represent another radiation source for high-resolution design reproduction into polymeric resist materials.
- X-ray lithography was first demonstrated to obtain high-resolution designs using X-ray proximity printing by Spears and Smith.
- X-ray lithography can be extended optical resolution of 15 nm.

Essential elements in X-ray lithography

- A mask consisting of a pattern made with an X-ray absorbing material on a thin X-ray transparent membrane.
- An X-ray source of sufficient brightness.
- An X-ray sensitive resist material.

Advantages of X-Ray Lithography

- Short wavelength from X-rays 0.4-4 nm
- No diffraction effect
- Simple to use
- No lens
- Faster than EBL
- Uniform refraction pattern
- High resolution for small feature

Atomic Force Microscope Lithography

- AFM nanolithography is a tool for materials structuring and designing with nanometre precision, AFM was invented in 1986.
- AFM can be used to study both insulating and conducting materials, and can be operated in liquid, air or vacuum
- AFM nanolithography design a wide range of materials including metals, semiconductors, polymers and biological molecules in different media.

Working Principle

The working principle of AFM nanolithography is based on the interaction between the probe and substrate. In AFM a tiny cantilever with a sharp tip is scanned across a surface. The interaction between the surface and the tip cause the cantilever to bend and the bending is monitored using a laser beam. In this way, nanometre changes in height can be measured and used to generate a three dimensional image of the surface.

Applications of AFM Lithography

- AFM is the versatile tool for investigate;
  - ✓ Topography of surface.
  - ✓ Properties of surface.
  - ✓ Properties of single molecules.
  - ✓ Force within molecules.

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DNA Nanotechnology

Learning Objectives

Introduction

Properties

Applications

DNA Nanotechnology

- DNA nanotechnology is the design and manufacture of artificial nucleic acid structures for technological uses.
- In this field, nucleic acids are used as non-biological engineering materials for nanotechnology rather than as the carriers of genetic information in living cells.
- DNA nanotechnology exploits the structural motifs and self-recognition properties of DNA to self assemble pre-designed nanostructures in a bottom-up approach.
- Two and three-dimensional structures have been fabricated using this self-assembly method.

DNA Nanotechnology: Subfields

- DNA nanotechnology is sometimes divided into two overlapping subfields:
- Structural DNA nanotechnology
- Dynamic DNA nanotechnology

DNA Nanotechnology: Subfields

- Structural DNA nanotechnology, sometimes abbreviated as SDN, focuses on synthesizing and characterizing nucleic acid complexes and materials that assemble into a static, equilibrium end state.
- On the other hand, dynamic DNA nanotechnology focuses on complexes with useful non-equilibrium behavior such as the ability to reconfigure based on a chemical or physical stimulus.
- Some complexes, such as nucleic acid nanomechanical devices, combine features of both the structural and dynamic subfields.

DNA Nanostructures Design

- DNA nanostructures must be rationally designed so that individual nucleic acid strands will assemble into the desired structures.
- This process usually begins with specification of a desired target structure or function. Then, the overall secondary structure of the target complex is determined, specifying the arrangement of nucleic acid strands within the structure, and which portions of those strands should be bound to each other.
- The last step is the primary structure design, which is the specification of the actual base sequences of each nucleic acid strand.

DNA Nanotechnology: Applications

- DNA nanotechnology represents one of the latest developments in nanotechnology.
- It has applications for the fabrication of nano-guides (e.g. waveguides), sensors (for diagnostic and imaging), logic gates, drug release, nano-motors and electronics (wires, transistors). It could lead to bottom-up electronics and DNA computing, which could become the computing of the future.

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DNA Origami

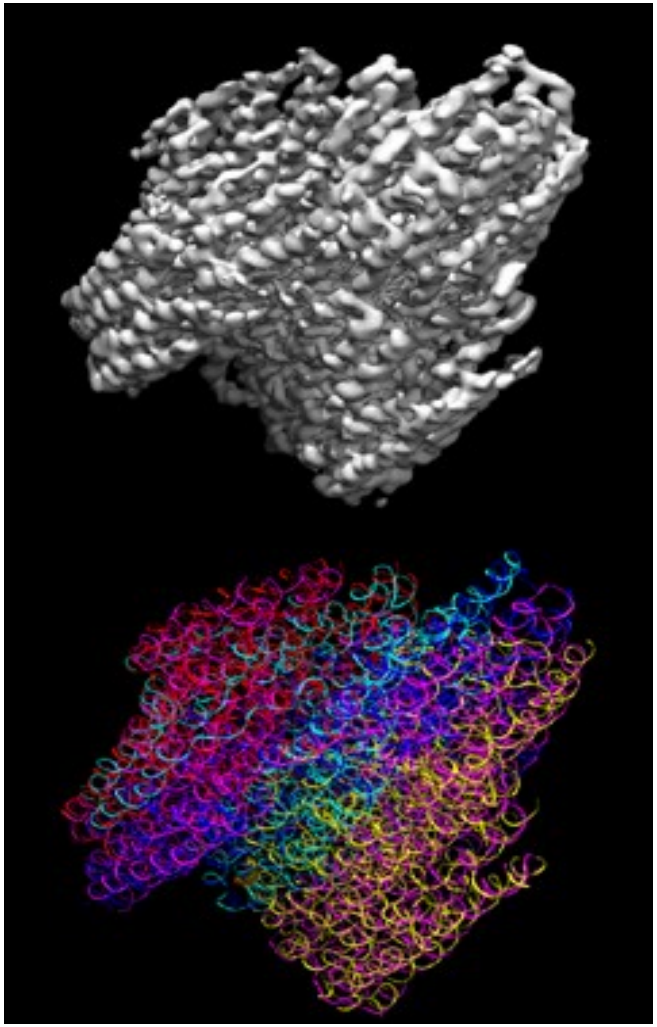
Learning Objectives

Introduction

Applications

DNA origami

- Recently, the revolutionary DNA origami method was developed to build two-dimensional addressable DNA structures of arbitrary shape that can be used as a platform to arrange nanomaterials with high precision and specificity.
- DNA origami is the nanoscale folding of DNA to create non-arbitrary two- and three-dimensional shapes at the nanoscale.
- The specificity of the interactions between complementary base pairs make DNA a useful construction material, through design of its base sequences.



DNA Origami

DNA origami: Applications

- Many potential applications have been suggested in literature, including
- Enzyme immobilization
- Drug delivery systems
- Nanotechnological self-assembly of materials.

DNA origami: Applications

- Researchers in the National Center for Nanoscience and Technology in Beijing and Arizona State University reported a DNA origami delivery vehicle for Doxorubicin, a well-known anti-cancer drug.
- The drug was non-covalently attached to DNA origami nanostructures through intercalation and a high drug load was achieved.
- The DNA-Doxorubicin complex was taken up by human breast adenocarcinoma cancer cells (MCF-7) via cellular internalization with much higher efficiency than doxorubicin in free form.
- The enhancement of cell killing activity was observed not only in regular MCF-7, more importantly, also in doxorubicin-resistant cells.

DNA origami: Applications

- In a study conducted by a group of scientists from iNANO center, researchers were able to construct a small multi-switchable 3D DNA Box Origami.
- The proposed nanoparticle was characterized by AFM, TEM and FRET.
- The constructed box was shown to have a unique reclosing mechanism, which enabled it to repeatedly open and close in response to a unique set of DNA or RNA keys.
- The authors proposed that this "DNA device can potentially be used for a broad range of applications such as controlling the function of single molecules, controlled drug delivery, and molecular computing

DNA origami: Applications

- Nanorobots made of DNA origami demonstrated computing capacities and completed pre-programmed task inside the living organism.
- As a proof of concept, the research team at Harvard University injected various kinds of nanobots (the curled DNA encasing molecules with fluorescent markers) into live cockroaches.
- By tracking the markers inside the cockroaches, the team found the accuracy of delivery of the molecules (released by the uncurled DNA) in target cells.

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NanoPCR

Learning Objectives

Introduction

Properties

History

NanoPCR

- Polymerase Chain Reaction (PCR) is one of the most common technologies used in many laboratories to produce millions of copies of targeted nucleic acid under in vitro conditions.
- However, PCR faces multiple challenges including limited availability of DNA in the sample, high GC contents of the template, low efficiency, and specificity in amplification.
- Moreover, some DNA fragments are very difficult to amplify due to their secondary structure and high melting temperature requirement.

NanoPCR

- To overcome these challenges, many approaches including the application of PCR additives in PCR mixture; change in instrument design; optimization of PCR system by using the accurate concentration of magnesium ions, primers, and cycle number; enzyme modification; and setting up the new touchdown and nested PCR strategies have been adopted.
- Although these approaches have enriched the output of PCR, they are not all-purpose and optimization can be case dependent.

NanoPCR

- Nanometer-sized materials (nanomaterials) have offered a possible solution to these problems as these materials have exceptional physio-chemical properties as compared to macroscopic materials.
- Among these nanomaterials, silicon-based materials, carbon-based materials, semiconductor quantum dots (QDs), and some metals are well-known PCR enhancer.
- Hence, new PCR has been designed to utilize the unique properties of nanomaterial and is known as nanomaterial-assisted PCR or simply nanoPCR.

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History of Nanomaterials in PCR

- In 2003, Cui et al. used scanning electron microscopy (SEM), high-resolution transmission electron microscopy (HRTEM) and X-ray photoelectron spectroscopy (XPS) to examine the effect of single-walled carbon nanotubes (SWCNTs) on the PCR by measuring quantitative PCR product.
- The images obtained from SEM and HRTEM revealed that the DNA templates and Taq enzymes are attached to bundles of SWCNTs in PCR products.
- Based on these results it was concluded that SWCNTs lower than the concentration of 3 g L−1 might increase the efficiency of PCR and could also act as a catalyst in other biochemical reactions.

History of Nanomaterials in PCR

- In 2005, another group of researchers proposed the use of citrate-stabilized gold nanoparticles in PCR which allowed reliable PCR amplification even at lower annealing temperature (25 C), obviating time-consuming “touchdown” PCR.
- They proposed that citrate stabilized gold nanoparticles act like single-stranded DNA binding protein (SSB) and enhanced the specificity of two-round PCR by binding with ssDNA at the error-prone stage.

NanoPCR-II

Learning Objectives

Role of CNTs

Gold nanoparticles

Quantum dots

Mechanism of action

Carbon Nanotubes and PCR

- Both single walled and double walled carbon nanotubes are evaluated for their efficiency in PCR.
- Dispersion of CNTs in different phases enhances the thermal conductivity of suspensions as compared to the dispersed phase of the working fluid. Therefore, CNTs have the potential role as a PCR catalyst.
- It was found that SWCNTs could be used as an alternative to Mg2+ because, in the absence of Mg2+, SWCNTs can produce similar effects as that of Mg2+ on the PCR.
- These CNTs not only interacts with the PCR components but can also achieve a rapid thermal equilibrium through convection heat transfer.

Graphene based nanomaterials and PCR

- Unique physical and thermal properties of graphene substances attract the scientist to use them as PCR enhancers.
- Graphene nanoflakes were evaluated (GNF) for the enhancement of PCR efficiency. The results depicted that the increase in PCR efficiency mainly depends upon the concentration of GNF.
- While at given concentration, the GNF produced an equal amount of PCR product as that of positive control at 65% fewer PCR cycles.
- It was proposed that the GNFs assist the dissipation of heat during the first step of PCR, which results in the easy and early denaturation of DNA strands. At annealing step, these nanoflakes attached to the DNA polymerase due to the action of Van der Waals forces which ultimately facilities the interaction among different PCR's reagents.

Gold nanoparticles and PCR

- Gold nanoparticles provide an excellent working platform for the attachment of biomolecules to exhibit their function both *in vivo* and *in vitro*.
- The use of AuNPs as PCR additive could increase the PCR sensitivity detection by 5- to 10-fold and 104-fold in conventional PCR and high-speed PCR (i.e., real-time PCR), respectively.
- It was proposed that AuNPs have excellent heat dissipation property, which enabled these particles to enhance the yield of PCR.

Quantum dots and PCR

- QDs have unique physical and chemical properties due to which it has got much attention in the field of nanobiotechnology research.
- A study reported that CdSe QDs modified with mercaptoacetic acid (MAA) improved the yield and efficiency of PCR system even at a very low annealing temperature that could not be achieved with only MAA.
- The results of this study further elaborated that the increase in the efficiency of PCR was not due to the surface property of QDs but due to QDs itself.
- Moreover, no interaction was detected between the QDs and chromosomes or DNA, which suggested that QDs interact with Taq polymerase to improve the efficiency of PCR.

Summary of Mechanism

- Several mechanisms have been proposed for the improvement of PCR by NPs. Some studies recommend that the heat conductivity of NPs increased the thermal cycling efficiency of PCR.
- Other claimed that NPs interact with DNA in the similar fashion like SSB and improve the efficiency of PCR.
- Convincing evidence supports the view of the interaction of NPs and DNA polymerase in regulating the output of PCR, thus enhance the PCR efficiency.
- Yet some studies assert that PCR components aggregate on the surface of NPs. This aggregation brings the components together and increases their interaction, which enhances the PCR efficiency.

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Theranostics and Nanotechnology

Learning Objectives

Introduction

Properties

Examples

Theranostics

- One of the most exciting opportunities that nanotechnologies have brought to the therapeutic field is the possibility of integrating the diagnosis, therapy and follow-up of a disease.

- This is referred to as theranostics, and could be enabled by nanoparticles incorporated inside a drug that can change some property — such as colour — once the drug has reached the target (e.g. quantum dots).

Combination of two words:

- Therapeutic + Diagnostic

Theranostics

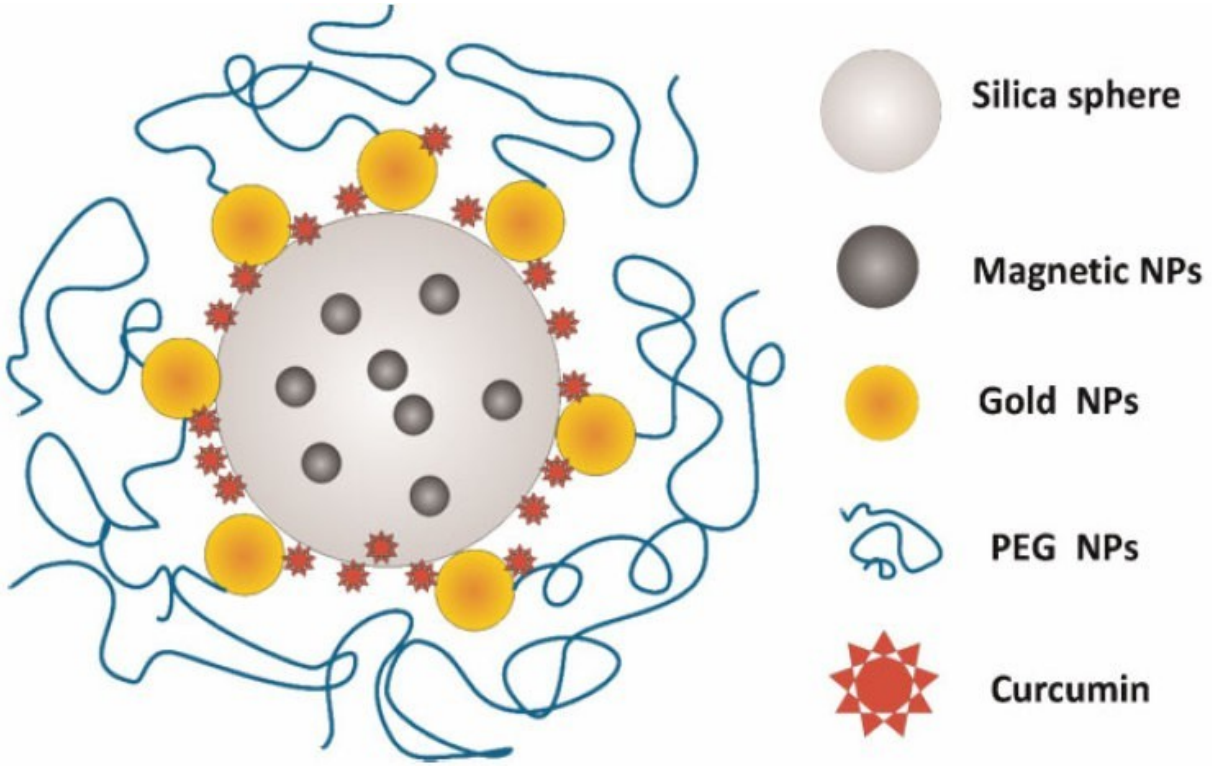
- Theranostics involves the administration of a diagnostic agent:
- To determine localization in the site or disease state under study as a surrogate for a potential therapeutic agent with similar chemical properties;
- To examine its biodistribution as predictive of off-target(adverse) effects of the potential therapeutic agent;
- As an aid in determining the optimal therapeutic dosage or activity to be administered, based on the anticipated tumoricidal doses measured in the tumor site;
- To monitor the response to this treatment

Theranostics

- Drugs could therefore have a feedback action.
- Together with a slow, targeted release system, the nanoparticles could gradually change colour during the drug action, thereby informing doctors of the progress of a therapy.
- This approach is called ‘find, fight and follow’ and could become a reality as a result of the parallel progress in the field of molecular imaging.
- One vision is that, one day, the entire process of diagnosis, pre-imaging, therapy and post-imaging of a specific disease will be integrated.

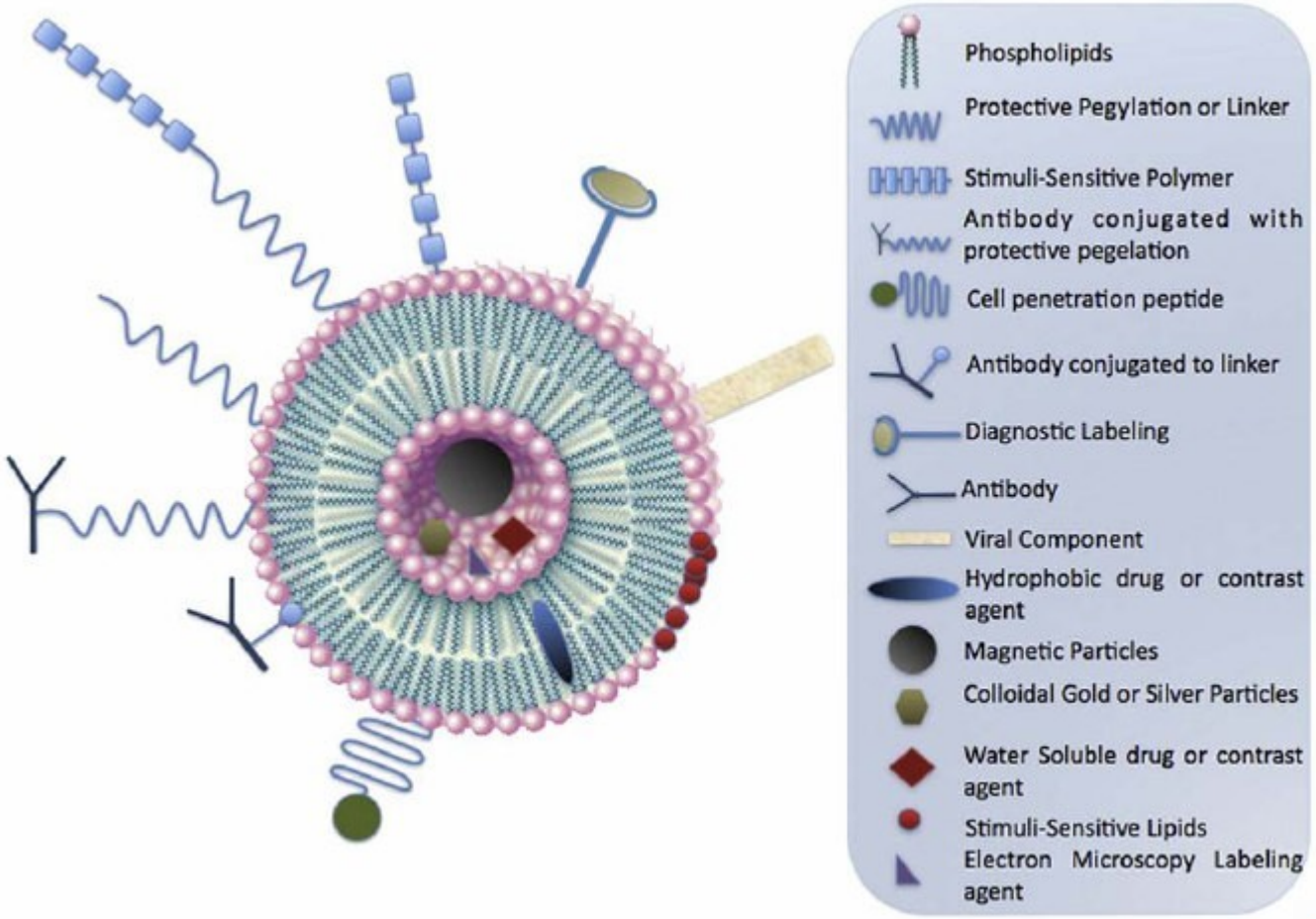
Examples

- An example of theranostics is the use of gold nanoshells for imaging and treating cancer cells at the same time.

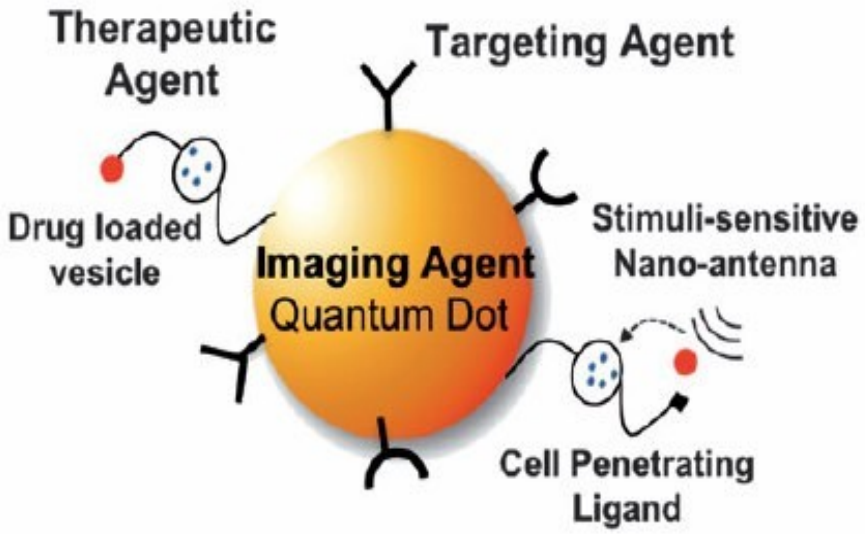


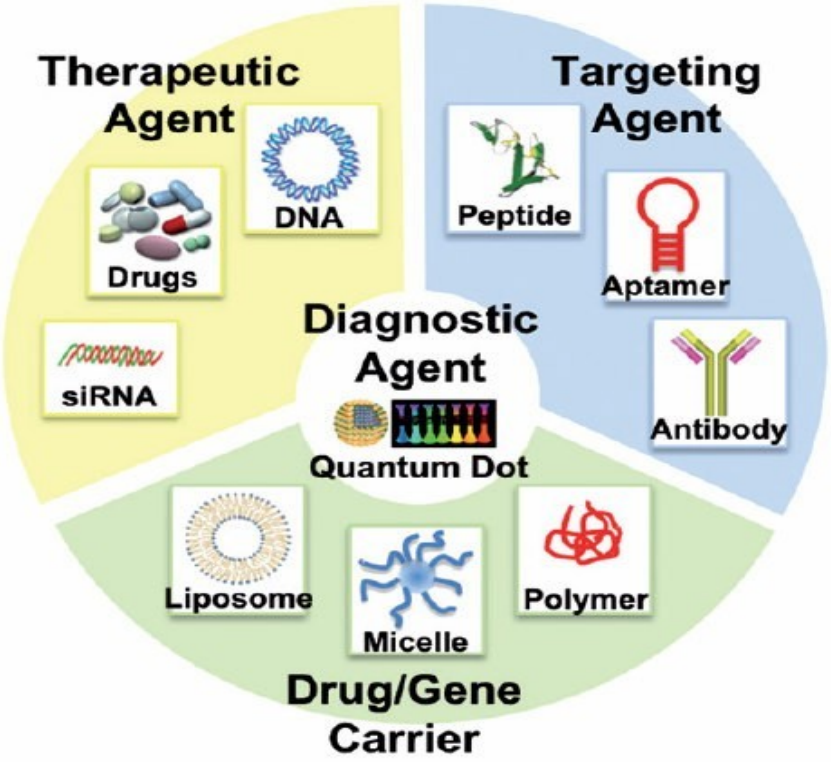
Scheme 1. Schematics of drug-loaded magnetoplasmonic assemblies (MPA) with PEG "camouflage" coating.

Magnetic Theranostic particles



Quantum dots as Theranostics





The possible combinations built around quantum dots

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Nanoengineering

Learning Objectives

Introduction

Properties

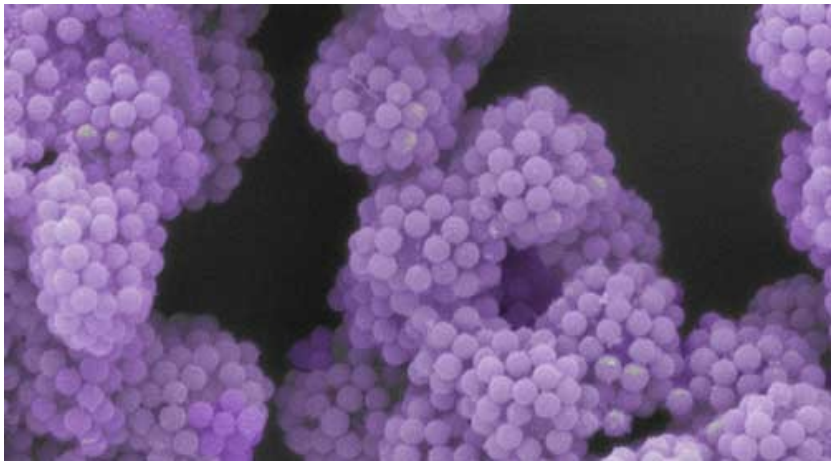
Applications

Nanoengineering

- Nanoengineering is the practice of engineering on the nanoscale

- Nanoengineering is a branch of engineering that deals with all aspects of the design, building, and use of engines, machines, and structures on the nanoscale.

- At its core, nanoengineering deals with nanomaterials and how they interact to make useful materials, structures, devices and systems.



Nanoengineering

- Nanoengineering is not exactly a new science, but, rather, an enabling technology with applications in most industries from electronics, to energy, medicine, and biotechnology.

- Nanoengineeering exploit the unique properties of nanoscale materials (size and quantum effects) in order to design and manufacture devices and systems that possess entirely new functionality and capabilities.

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- Nanoengineeering exploit the unique properties of nanoscale materials (size and quantum effects) in order to design and manufacture devices and systems that possess entirely new functionality and capabilities.

Challenges in Nanoengineering

- Nanoscale objects are difficult to manipulate, as they are too tiny to see directly by eye, far too small to hold, and often have incompatible surfaces for assembling into ordered structures.
- Therefore the fabrication of complex nanoarchitectures requires sophisticated techniques of nanoscale engineering.
- Among the many challenges that researchers have to overcome in developing nanoengineering techniques and processes, the requirement for extremely precise, nanometer-scale control of positioning and shaping of objects is one of the most vexing.

Applications of Nanoengineering

- Tissue Engineering
- Nanomedicines
- Drug delivery
- Regenerative medicine
- Neuron engineering

Nanoengineering: Bone regeneration

Learning Objectives

Introduction

Properties

Artificial Bone concept

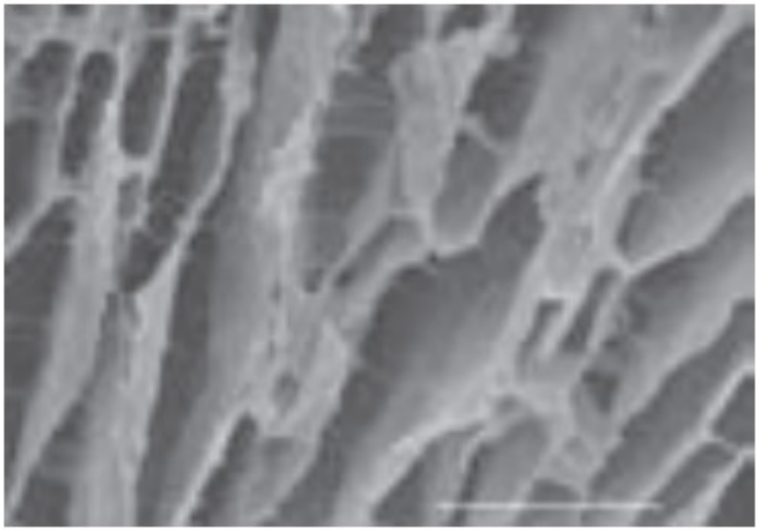
Nanoengineering: Bone regeneration

- Bones and teeth are material that have to bear complex loads of moving bodies, provide a protective cage for vital organs, anchor tendons and muscles, and act as joints and levers.

- Bones have a complex hierarchical structure at the nano, micro and macro-levels, which determines their amazing properties.

Bone regeneration: Classical way

- The ‘classic’ way to promote the regrowth of bone after injury is to provide a scaffold into which bone forming cells can migrate and grow attached (fused) to the scaffold.
- In the past, numerous scaffolds have been used, mainly coated with hydroxyapatite which is a natural bone component.



SEM mage showing a hydrogel scaffold grown to study brain tissue engineering and nerve regeneration.

Nanoengineering: Bone regeneration

- More recently, research has focused on trying to mimic the nanostructure of the scaffold.
- This involves working on the topography of the scaffold (surface nano-roughness) but also at anchoring specific biomolecules to the nano-surface of the scaffold.
- The idea is to mimic the natural fine organisation of natural bone, which is a combination of nano-topography and (bio)chemistry.

Artificial Bone

- A novel approach is the development of ‘artificial bone’, which means using macromolecules that self-assemble into large structures that mimic the natural structure of bone.
- This is a bottom-up approach to bone engineering which leads to materials with nanoscale-level control.
- For example, some researchers have developed a bone scaffold by biomimetic synthesis of nano hydroxyapatite and collagen.

Artificial Bone

- These biomaterials assemble into 3D mineralized fibrils that mimic key features of human bone.
- This material shows some similarities with natural bone in terms of hierarchical micro and nanostructure, and three-dimensional porosity.
- Cells seeded in vitro over this scaffold grew and proliferated well.
- The advantage of this approach is that the ‘building blocks’ are biomimetic macromolecules: once assembled, the final ‘macro’ material can integrate with natural tissues, opening the way to new clinical approaches to bone regeneration.

Artificial Bone

- When these synthetic nanofibres form, they make a gel that could be used as a sort of glue in bone fractures or to create a scaffold for other tissues to regenerate.

- As a result of its chemical structure, the nanofibre gel would encourage attachment of natural bone cells which would help to patch up the fracture. The gel could also be used to improve implants or hip and other joint replacements.

Nanotechnology and Genetic Engineering

Learning Objectives

Introduction

Genetic Nanotechnology

Properties

Nanotechnology and Genetic Engineering

- Genetic engineering represents the direct manipulation of an organism's genome using biotechnology.
- The manipulation usually consists in modifying the genetic background of cells by transferring/inserting or deleting one or more genes, genetic fragments or even
- Altering a single nucleotide, in order to produce improved or novel organisms.
- For instance, it has been recently showed the possibility to engineer stem cells to enhance the anti-tumor immune response and lifespan of animals.

Nanotechnology and Genetic Engineering

- Nanotechnology is the science of engineering on a molecular scale, in effect building matter atom-by-atom from the “bottom-up”.
- Nowadays, genetic nanotechnology (i.e. the combination of genetics with nanotechnology) is employed in agriculture to overcome production-related issues.
- Thereby, a recent method called “nanoinjection”, based on using a MEMS (micro-electromechanical system) chip with a moveable, nanometer-sized lance that is capable of holding DNA based on electrical charge, was reported to be more effective (i.e. higher survival rates through similar DNA integration rates) than microinjection to deliver DNA and fertilize mouse eggs

Nanotechnology and Genetic Engineering

- Nanotechnology also offers disturbing parallels to GMOs (i.e. Genetically Modified Organisms).
- Indeed, nanotechnology would contribute on novel environment friendly crop improvement strategies to overcome food insecurity and malnutrition worldwide.
- After all, DNA is a complex biochemical polymer with properties that make it an ideal platform for engineering new nanotechnologies (e.g. advanced synthesis, sequencing, communication and digital data storage tools).

Nanotechnology and Genetic Engineering

- Furthermore, likewise for proteomics, advances in nanotechnology have begun to allow researchers to identify low-abundance nucleic acids in samples through techniques that rely upon both nanoparticles and nanoscale devices.
- One day, nanotechnology combined with genetic engineering and computerized artificial intelligence may allow people to choose their feelings, increase their intelligence, or lengthen their life spans.
- Eventually, genetic engineering and nanotechnology fit like hand to glove, and should significantly contribute to faster translational medicine and personalized medicine.

Nanotechnology and Genetic Engineering II

Learning Objectives

Biotransformation

Nanocarriers for Biotransformation

Applications

Biotransformation

- Genetic alterations are intended for improving the overall quality of crops.
- A general routine of two distinct steps are followed during genetic engineering of plants:

(1) Transient transformation—the introduction of desired DNA into plant cells, and

(2) Stable transformation—subsequent integration of foreign DNA with plant genome.

- Though each step poses its own challenges, the proper integration of the transgene into the host genome is the most crucial part to produce transgenic plants with desired traits

Biotransformation

- Introduction or transfer of genes has been conventionally achieved by

(1) Natural method—using natural agents as Agrobacterium species, viruses, etc.,

(2) Direct method—protoplast fusion, transfection, etc.,

(3) Physical method—electroporation, particle bombardement, microinjection, etc., and

(4) Chemical method—gene transfer using plasma membrane destabilizing/precipitating agents as PEG, PVA, $NaNO_3$ and DMSO.

- These conventional methods were plagued with limited transformation rates due to various reasons.

Nanotechnology in Biotransformation

- Nanotechnology promises augmentation of agricultural productivity by highly specific and reliable genetic transformation methods at the cellular level.
- Transformation with nanomaterials offer many advantages like

(1) A thousand times less amount of DNA is required for the detection of expression than the conventional methods.

(2) Transient DNA-free genome editing (including gene silencing) is enabled in a controlled fashion.

(3) Co-delivery of multiple biomolecules (DNA activator, DNA proteins, different genes), simultaneously to the target, is facilitated.

(4) The surface of the gene nanocarriers can be easily functionalized using certain bio-recognition entities for specific, targeted delivery.

Nanotechnology in Biotransformation

- Some other benefits include
- Faster gene transfer than Agrobacterium-based process
- DNA encapsulated in protective layers therefore resistant to damage/inactivation.
- Gene mutation due to energy-induced DNA damage (typically in physical methods) is greatly reduced.
- Greater control over transformation process.
- Amalgamation of nano- and physical methods is possible leading to effective and improved gene transfer.

Nanocarriers for Plant Biotransformation

- NP-mediated genome editing to deliver green fluorescent protein (GFP) gene-containing plasmid and functional recombinase into plant tissues has been a great success.
- Different NPs for the delivery of biomolecules (nucleotides, activators, proteins) to plants has been explored and they include
- Silica Nanoparticles
- Calcium-Based Nanoparticles
- Magnetic Nanoparticles
- Dendrimers
- Liposomes
- Gold Nanoparticles
- Carbon Nanotubes

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Nano Agriculture

Learning Objectives

Introduction

Importance

Applications

Nano Agriculture

- Increasing global food security triggered by man-made and natural causes has been exerting tremendous pressure on the agricultural sector.
- There is a need of new system that can manage the burden on agriculture sector and bring revolution.
- Nanomaterials prepared by eco-friendly and green methods could increase agriculture potential for improving the fertilization process, plant growth regulators and pesticides.
- Nanotechnology applications in agriculture sector have the potential to overcome issues in agriculture sector.

Nano Agriculture Applications

- Nanotechnology provides new agrochemical agents and new delivery mechanisms to improve crop productivity, and it promises to reduce pesticide applications.
- Nanotechnology can increase agricultural production, and its applications include:

(1) Nanoformulations of agrochemicals for applying pesticides and fertilizers for crop improvement

(2) The application of nanosensors in crop protection for the identification of diseases and residues of agrochemicals.

Nano Agriculture Applications

(3) Nanodevices for the genetic engineering of plants

(4) Plant disease diagnostics

(5) Animal health, animal breeding, poultry production

(6) Postharvest management

(7) Nanoparticle-mediated gene or DNA transfer in plants for the development of insect-resistant varieties, food processing and storage and increased product shelf life.

Nanotechnology in pesticides and fertilizers

- The recent developments in agriculture cover the applications of NPs for more effective and safe use of chemicals for plants.
- The effects of different NPs on plant growth and phytotoxicity were reported by several workers including
- Magnetite (Fe_3O_4) nanoparticles on plant growth
- Alumina, zinc, and zinc oxide on seed germination
- Silver nanoparticles can stimulate wheat growth and yield.

Nanotechnology in pesticides and fertilizers

- In recent years, the use of nanomaterials has been considered as an alternative solution to control plant pathogens.
- Scientists have synthesized nanoparticles of magnesium oxide (MgO) and tested the effect of different concentrations on the green peach aphid (GPA) under the greenhouse conditions.
- The screening of synthesized bio-nanoparticles revealed that these nanoparticles were effective in increasing the mortality percent of green peach aphid.
- There was no any metal accumulation in any of the plant fruits.

Nanotechnology in Plant protection

- Plant protection plays a significant role in achieving the better yield results.
- With the growing demand of pesticide worldwide to control the pathogens and pests, there is an urgent need to tackle the excessive usage of pesticides and fertilizers by finding alternatives.
- Different carbon based nanomaterials including Graphene, Fullerenes and CNTs have been evaluated to overcome the excessive use of Pesticides.
- These nanomaterials are involved in adsorption and controlled release of pesticides to overcome adverse effects leading to sustainable agriculture.

Nano Agriculture-II

Learning Objectives

Nanoparticles as Micronutrients

Importance

Applications

Micronutrients and their importance

- Micronutrients are a group of elements that are required in trace amounts by the plants.
- Micronutrients include iron (Fe), copper (Cu), zinc (Zn), manganese (Mn), molybdenum (Mo), boron (B), chloride (Cl), and nickel (Ni).
- Micronutrient-deficient food causes anemia and reduction in height and decreases cognitive and physical performance in children.
- The current techniques and methods to supply micronutrients in the soil are ineffective and have low crop fertilizer-micronutrient use efficiency.

Nanoparticles as Micronutrients

Iron (Fe) NPs

- Iron is an important micronutrient for crop growth and is involved in chlorophyll and hormone production (such as gibberellic acid, ethylene), photosynthesis, ROS production and scavenging.
- Iron deficiency affects the growth, functioning, and development of the plants and also causes anemia in animals and humans.
- Iron nanoparticles (Fe NPs) have been studied to show dual effects on plants.
- Low concentrations showed positive effects on the growth and development of plants, whereas high concentrations resulted in harmful effects on plants.
- It has been found that Fe NPs promote nutrient absorption and enhance the photosynthetic efficiency in plants.

Nanoparticles as Micronutrients

Manganese (Mn) NPs

- Mn is required in trace amounts by plants, where it is involved in photosynthesis, respiration, nitrogen metabolism, and providing tolerance toward pathogens.
- Mn nanoparticles help to enhance the functioning of the plant in various ways.
- Mn NPs increase the activity of electron transport chain by binding with CP43 protein chain of photosystem II.
- Mn NPs enhance oxygen evolution process and consequently improve the phosphorylation capacity.
- Thus, it improves the photosynthetic ability of the plants, thereby producing more sugar, and can be useful in the agricultural sector for the improvement of plant productivity.

Nanoparticles as Micronutrients

Zinc (Zn) NPs

- Zinc is necessary for chlorophyll production, fertilization, germination, and biomass production.
- The effect of zinc nanoparticles has been studied in many plants with positive effects.
- A study showed that ZnO NPs at 20 ppm produced the best growth results in mung (*Vigna radiata*) seedlings.
- For gram seedlings, ZnO NPs at 1 ppm caused an increase of 53.13% in roots and 6.38% in shoots.
- In rice, nano-form ZnO showed better results by registering an increment of 26 and 25% in fresh weight of shoot biomass and 30 and 23% for root biomass at concentrations of 50 and 100 ppm.
- Higher concentration of ZNO NPs showed some toxic behavior and retardation in growth

Nanoparticles as Micronutrients

Copper(Cu) NPs

- Copper deficiency may lead to necrosis of apical meristem, retarded growth, bleaching, and crop yield loss.
- The use of copper nanoparticles in plants increased shoot: root ratio in lettuce seedlings.
- Copper nanoparticles are found to cause less oxidative stress than free copper ions.
- But, nano-copper had reported to have a negative impact on soil nutrient cycles.
- Therefore, copper nano-fertilizers need to be evaluated further for their toxicity to microbes in the soil.

Conclusion

- Different micronutrients in the form of nanoparticles have been evaluated in plants

- Studies showed that low concentration nanoparticles as nutrients are advantageous to plant growth and survival but higher concentrations are associated with toxicity.

Nano Agriculture-III

Learning Objectives

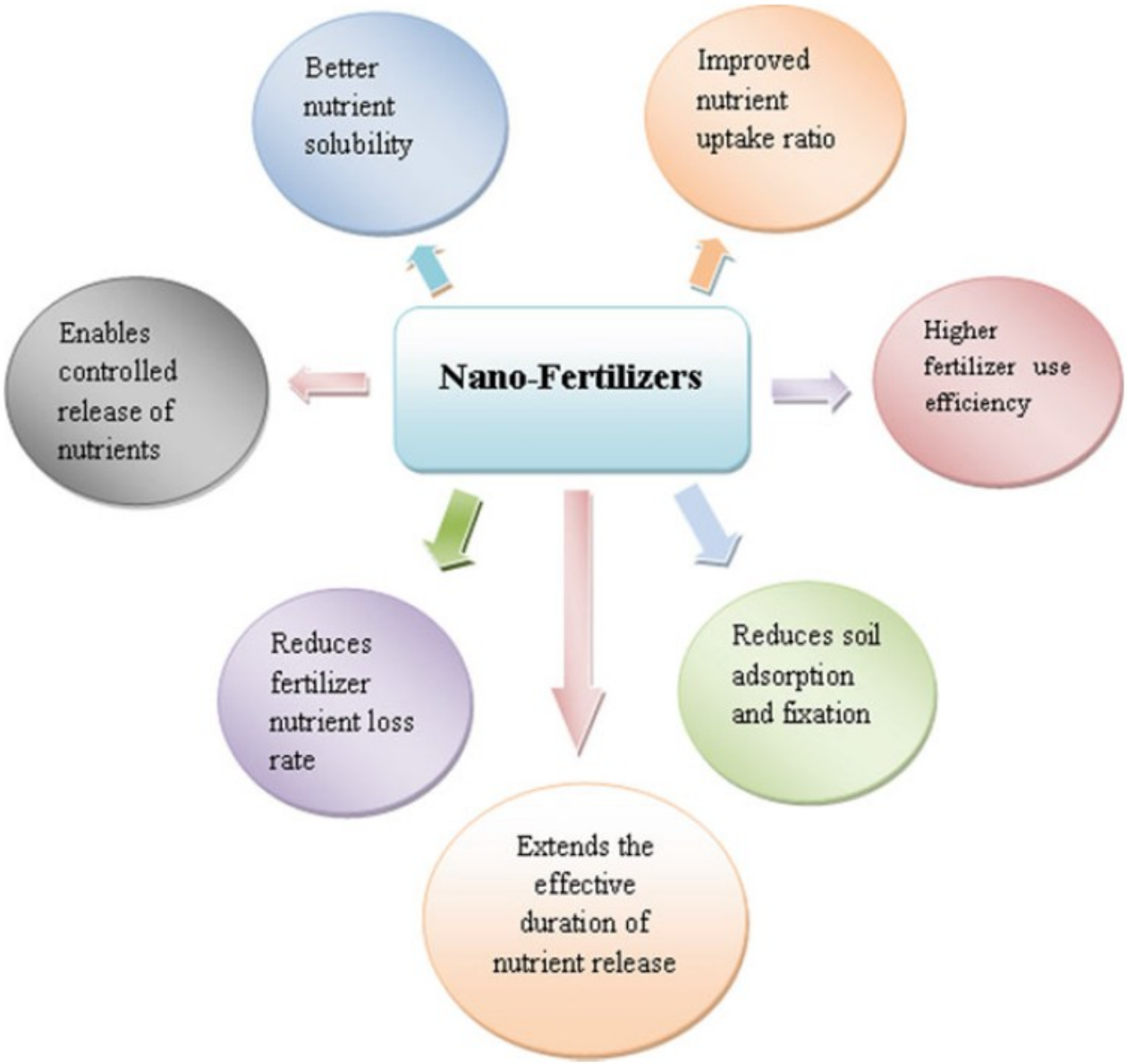
Nanofertilizers

Conventional vs Nano

Applications

Nano-fertilizers

- Nano-fertilizers can be defined as nutrient fertilizers made, in part or whole, of nano-structured formulation(s) that allows targeted delivery to plants, allowing controlled release and efficient uptake of nutrients.
- It is a fertilizer in the nanometer regime.
- By using nanoparticles as filler or reservoir of fertilizer, nanotechnology may help in fine tuning the existing trends of nutrient management and crop productivity.
- Nano-fertilizers have been developed to address the problems of conventional fertilizer applications and offer different advantages.


Better nutrient solubility
Improved nutrient uptake ratio
Enables controlled release of nutrients
Nano-Fertilizers
Higher fertilizer use efficiency
Reduces fertilizer nutrient loss rate
Reduces soil adsorption and fixation
Extends the effective duration of nutrient release

Aim and features of Nanofertilizers

Aim:

- The major aim of Nanofertilizers is to improve nutrient use efficiency (NUE) either by enhanced nutrient bioavailability through reduction of nutrient losses to the environment incurred due to fixation, leaching or by targeted and timely release of nutrients.

Unique Features:

- high surface area-to-volume ratio,
- high mobility and solubility,
- target specificity
- low toxicity

These features of Nanofertilizers are responsible for improved crop performance.

Applications of Nanofertilizers

(1) Nano-fertilizers can initiate significant changes in the site and functioning of photosynthesis via

- expansion of the leaf area.
- increase in the rate of photosynthesis,
- improvement in the light absorption potentials

(2) Productivity can be improved with crop plants exhibiting higher dry matter and improved yield.

Applications of Nanofertilizers

Nano-fertilizers in Precision Agriculture

- **Precision agriculture** involves designing controlled systems that improve the use efficiency of the applied agri-inputs both temporally and spatially.
- For precision agriculture, the nutrient materials/fertilizer may be delivered by various means such as

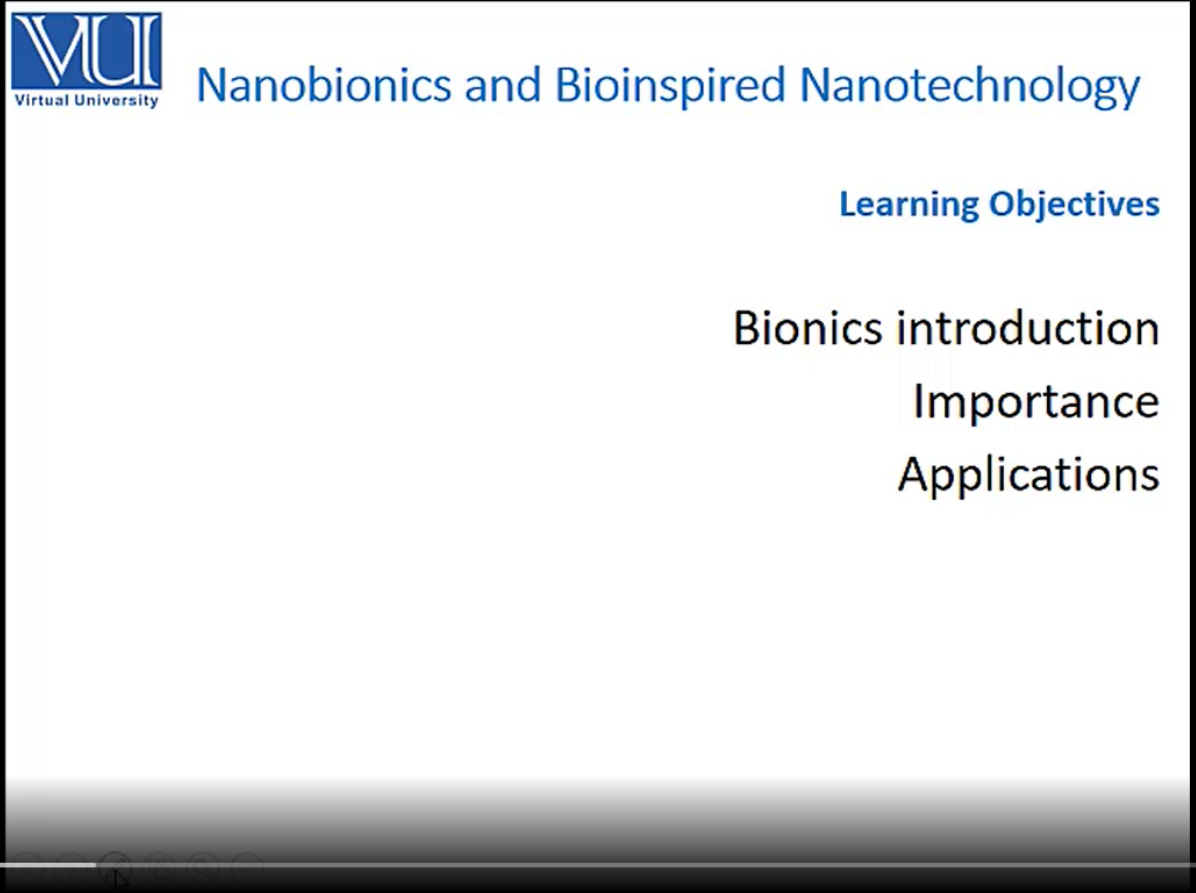
(1) nutrient encapsulation by nanomaterials

(2) encrusted within a thin protective nano-particulate

(3) polymeric film, in the form of nano-emulsions

(4) directly as nutrient nanoparticles

- Gradual release of nutrients can help reduce land and water pollution, prevent eutrophication of water bodies and protect plant from various abiotic and biotic stresses.


Virtual University
Nanobionics and Bioinspired Nanotechnology
Learning Objectives
Bionics introduction
Importance
Applications
0:03 / 12:26

Bionics and Nanobionics

- Bionics is the study and design of engineering system including electronic ones and these systems are based on the biological system.
- Nanobionics is the nanoscale electronic interactions in biological systems.
- Bioinspired nanotechnology works with small-scale biomolecules fabricated with high accuracy.
- Nanobionics finds immense applications in the research field of artificial intelligence, nanobiotechnology, and nanorobotics in addition to medicine and military purposes.

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0:51 / 12:26

Applications of Nanobionics and Bioinspired nanotechnology

Textile industry

- Nano-finished fabrics have some remarkable properties that it possible by the use of nanotechnology and these properties include water repellency, self-cleaning, nano-capsules of moisturizing agents, soil resistance, anti-bacterial, anti-static and UV-protection, and flame retardation.

3:06 / 12:26

Applications of Nanobionics and Bioinspired nanotechnology

Targeting thrombosis

- Thrombosis is a major health threat but biologically inspired nanotechnology have the greater role in diagnosis by molecular imaging and drug delivery to the target site. Thrombus formation can be diagnosed by bioengineered nano-probes formation.

4:24 / 12:26

Applications of Nanobionics and Bioinspired nanotechnology

Environmental Decontamination

- Bioinspired silica nanomaterial's that are porous in nature are used in selective adsorption and pollutants removal from air or water.
- It also has a role in the removal of formaldehyde from contaminated water bodies.
- Bio-inspired silica has the potential for adsorption and environmental remediation of volatile organic compounds.

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5:59 / 12:26

Applications of Nanobionics and Bioinspired nanotechnology

Biocatalysts

- Bioinspired nanotechnology is used for the production of catalysts and has several advantages; mild conditions requirement, one-step procedure, high enzymatic activity, high stability, and fewer waste productions.

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7:30 / 12:26

Plant Nanobionics

- The Plant Nanobionics is an emerging field of bioengineering which alters the functioning of the plant tissue or organelle by introducing nanoparticles into the cells and chloroplasts of living plants.
- Plants are now being exploited for nanobionic purposes due to their exceptional capability to produce energy from sunlight and photosynthesis.
- The trapping of nanomaterials/nanotubes improved chloroplast carbon capture, that is, photosynthesis by improving chloroplast solar energy harnessing and electron transport rate.

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8:59 / 12:26

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Nanopore Sequencing

Learning Objectives

Introduction

Types

Virtual University

DNA Sequencing

- DNA sequencing is the process of determining the precise order of the nucleotides within a DNA molecule
- DNA sequencing may be used to determine the sequence of individual genes, larger genetic regions, full chromosome or entire whole genome.

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What is Nanopore?

- Nanopore = Very small hole
- Electrical current flows through the hole
- Introduction of analyte of interest in the hole Identify analyte by the disruption or block to the electric current.

Introduction

- The fourth-generation DNA sequencing technology
- Studies the interaction between DNA and protein, as well as between protein and protein.
- Have the potential to quickly and reliably sequence the entire human genome for less than $1000, and possibly for even less than $100.
- The detection principle is based on monitoring the ionic current passing through the nanopore as a voltage is applied across the membrane.
- When the nanopore is of molecular dimensions, passage of molecules (e.g., DNA) cause interruptions of the current level, leading to a signal.

Introduction

- Since DNA bases (adenine, thymine, cytosine and guanine) are different from each other in atomic scale, it is essential to collect base-specific information at atomic level to correspond the DNA sequence with the measured signals.
- According to the different types of the signals, the detection methods can be roughly classified into two categories:
  - The electrical detection methods and the optical readout method.

Types of Nanopores Sequencing

- Nanopore technologies can be broadly divided into two categories:
- 1. Biological nanopore sequencing
- 2. Solid-state nanopore sequencing
- More recently, hybrid nanopores have been proposed to take advantage of the features of both biological and solid-state nanopores.

Applications

- Nanopores as single- molecule sensing technologies have great potential applications in many areas, such as
- Analysis of ions, DNA, RNA, Peptides, Proteins, Drugs, Polymers, Macromolecules.

Biological Nanopore Sequencing

Learning Objectives

Biological nanopores

Working Principle

Biological Nanopores

- Also called transmembrane protein channels
- Are usually inserted into a substrate, such as planar lipid bilayers, liposomes, or other polymer films
- Nanopores formed by pore-forming proteins
- Examples:
  - alpha-hemolysin, MspA, phi 29

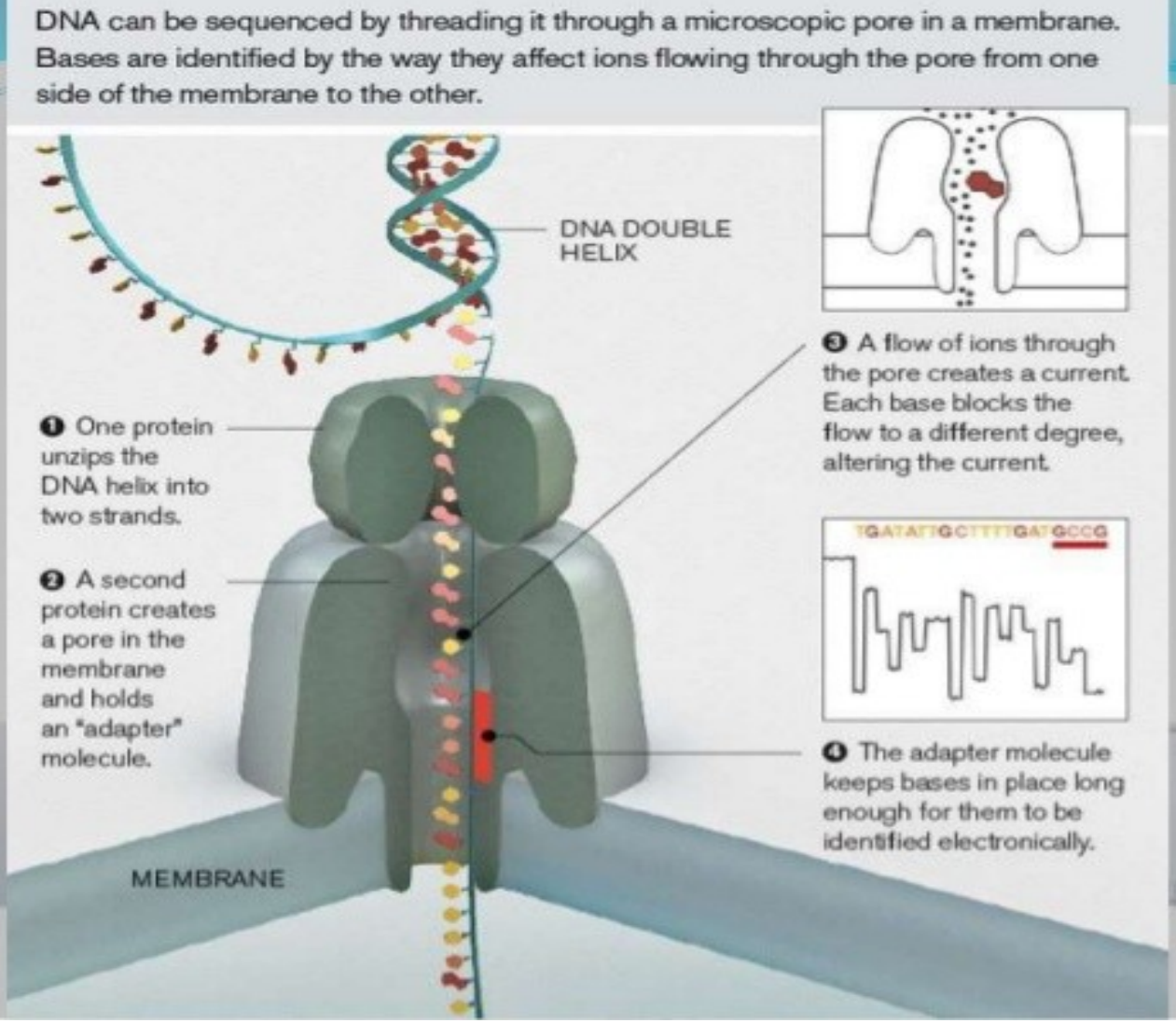
Alpha-hemolysin

- a-Hemolysin (a-HL, also called a- toxin)
- a-HL is an exotoxin secreted by the bacterium Staphylococcus aureus, a human pathogen.
- This mushroom-shaped heptameric protein normally forms pores in the membranes.
- The limited pore size (1.4 nm) has restricted its application in the analysis of ssDNA, RNA, or small molecules.

Mycobacterium smegmatis porin A (MspA)

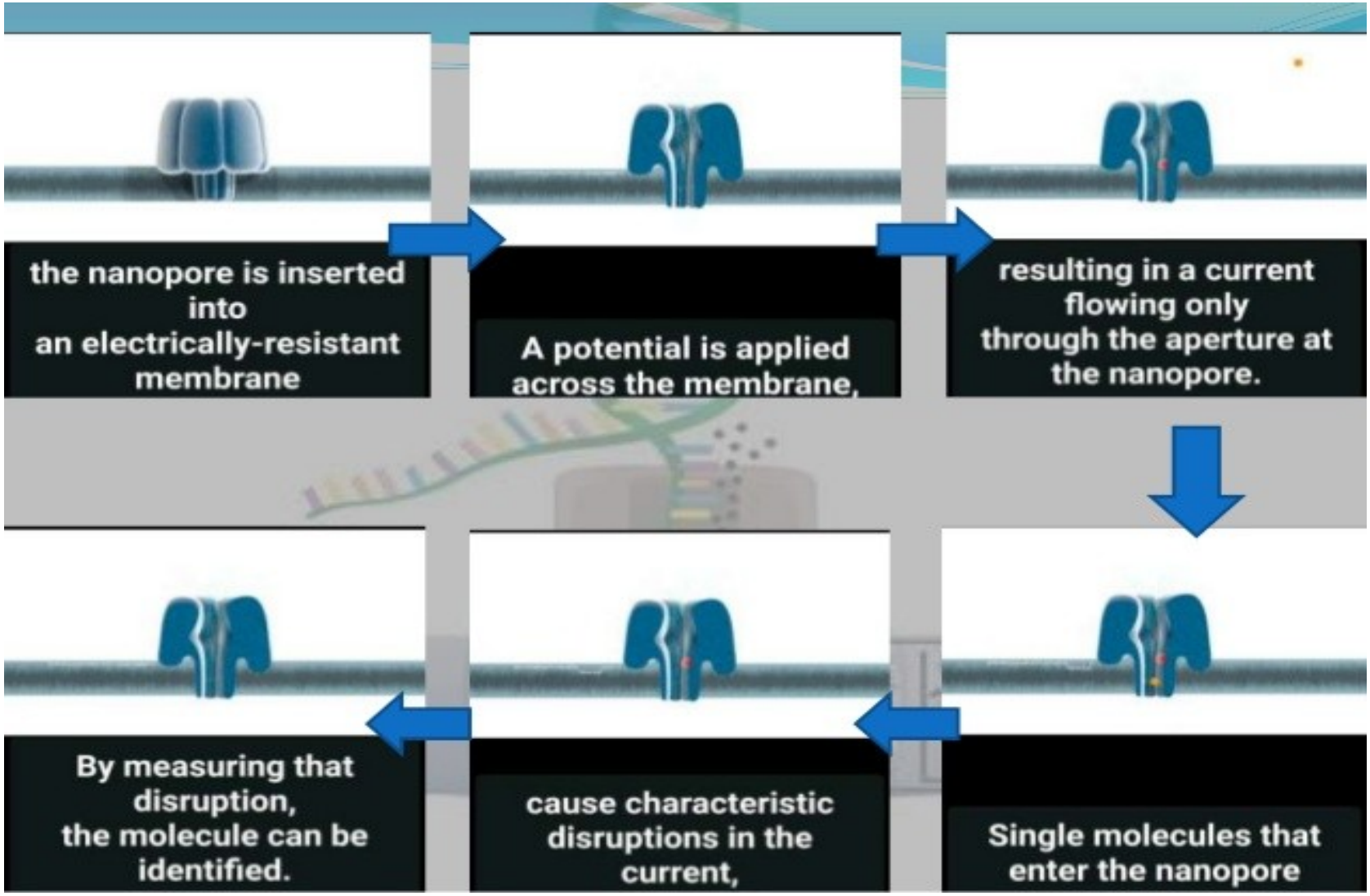
- Powerful nanopore for reading information from four nucleotides simultaneously.
- The channel of the MspA octamer is 1 nm in diameter at the minimal point, which is relatively small and narrow, compared to that of alpha-HL.
- Thus, it can improve the spatial resolution of ssDNA sequencing

Principle


DNA can be sequenced by threading it through a microscopic pore in a membrane. Bases are identified by the way they affect ions flowing through the pore from one side of the membrane to the other.
DNA DOUBLE HELIX
❶ One protein unzips the DNA helix into two strands.
❷ A second protein creates a pore in the membrane and holds an "adapter" molecule.
MEMBRANE
❸ A flow of ions through the pore creates a current. Each base blocks the flow to a different degree, altering the current.
❹ The adapter molecule keeps bases in place long enough for them to be identified electronically.

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Working Process



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Advantages of Biological Nanopores

- Their well-defined and highly-reproducible nanopore size and structure.
- Biological nanopores can be modified easily with modern molecular biology techniques, such as mutating the nucleotide sequence to change the amino acid residue at a specific site

Disadvantages of biological nanopores

- The environmental demands (like temperature, electrolyte concentration and Ph) of nanopores to keep there biological activities.
- The fragility of lipid bilayer makes the biological nanopore breakdown easily

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Nanoinformatics-I

Learning Objectives

Introduction

Importance

Applications

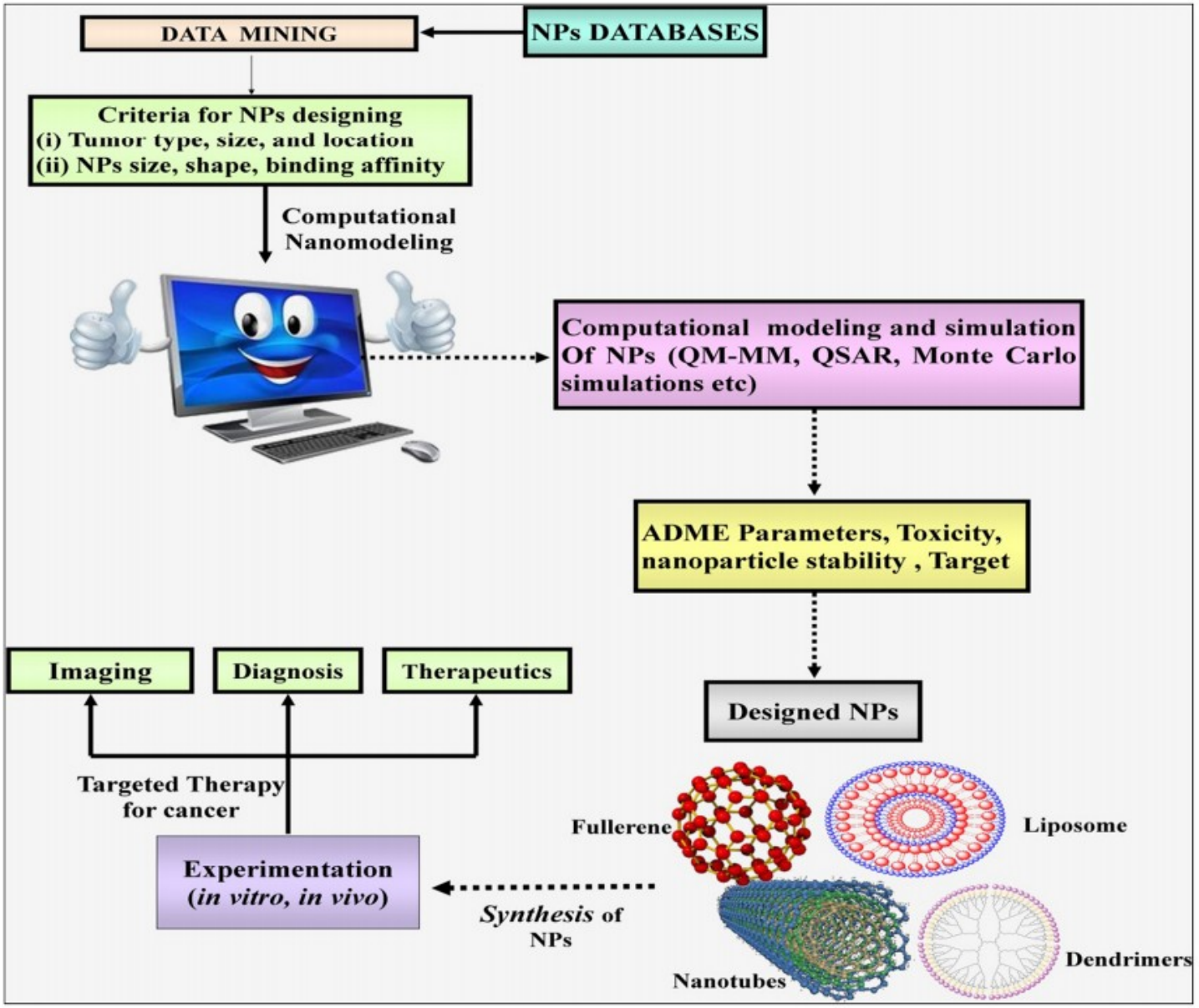
Nanoinformatics

- The need of computational applications at the nano scale and its defined implications has given rise to the field of Nanoinformatics.
- Nanoinformatics provides a useful platform for researchers to in silico design and analyze the effects of nanoparticles (NPs) as medicines, at an aggregated, molecular, cellular, tissue, organ, and organism level orientations.
- It helps in understanding the physiological and biochemical properties of NPs along with its risk assessment and cytotoxic analysis.
- Nanoinformatics studies has significantly helped scientists to overcome different barriers of drug resistance, poor intratumoral retention, circulation time, toxicity to nontumorogenic cells, and poor pharmacokinetics in cancerous study

Nanoinformatics

- Nanoinformatics specifically has helped in storage and computational management of large amounts of structural and property data in the field of nanomedicine, promoting data mining, data modeling, and knowledge discovery studies.
- Nanoinformatics studies have provided an in silico novel archetypes in the design and discovery of nanomaterials, with improved material selection potential.
- This certainly has enhanced NPs product performance at in vitro and in vivo level along with their precise drug delivery mechanism.

Nanoinformatics



Nanoinformatics and biomolecular modeling-based approach

Nanoscale modeling and simulation tools

- Nanoscale modeling and simulation tools are being utilized to develop a refined or high-quality nanobiomodels with limited and specific characteristics.
- Modeling and simulation modules basically involve

(i) computational building of nanomodels: physiochemical delineation, annotating nonmaterial surface, design, and controlled production of process

(ii) pharmacodynamics and pharmacokinetics models

(iii) model of lifecycle of nanoparticles inside the human body: absorption, distribution, metabolism, excretion, phagocytosis, immunogenicity, toxicity analysis.

(iv) model of nano-QSAR

(v) model of nanoparticle and nanotoxicity in interaction with biological environment at the tissue, cellular and atomic level.

Nanoinformatics databases

Learning Objectives

Nanodatabases

Properties

Applications

Nanoinformatics databases

Nanomaterial Registry

- Provides searching, browsing, side-by-side evaluation of nanomaterials, compliance ratings based on the quality and quantity of data, and assists to see the sights for similar nanomaterials; assembling and sharing a broad body of nanomaterial knowledge that contains curated data from additional data sources.

Nanomaterial-Biological interactions (NBI) knowledgebase

- repository for annotated data on nanomaterial characterization[e.g. purity, electronic and photonic properties, size, shape, charge, composition, functionalization, agglomeration state, etc.], synthesis methods, and nanomaterial-biological interactions.

Nanoinformatics databases

Nanoparticle Information Library (NIL]

- a web-based nanoparticle repository, developed by the National Institute for Occupational Safety and Health[NIOSH] and the Oregon Nanoscience and Microtechnologies Institute through the Safer Nanomaterials and Nanomanufacturing Initiative; contains records of nanoparticle's material data such as core element, structure or production method, the nanoparticle's image

Nanotechnology characterization Laboratory (NCL)

- performs and standardizes the pre-clinical characterization of nanomaterials intended for cancer therapeutics and diagnostics developed by researchers from academia, government, and industry/ serves as a national resource and knowledge base for cancer researchers, and aids the development and translation of nanoscale particles and devices for clinical applications.

Nanoinformatics databases

Nanowerk

- Powerful, free tool for the nanotechnology community to research nanomaterials such as carbon nanotubes, nanoparticles, graphene or quantum dots from more than 150 suppliers worldwide. Currently, nanowerk database contains about 4000 nanomaterials.

NanoLINK

- Dedicated to the establishment of high-quality, peer reviewed nanotechnology content to help educators to coach young people about nano-related topics

Nanoinformatics databases

nanoHUB

- It is a virtual toolkit as well as a community where students and faculty contribute to the science of nanotechnology/ uses cyber infrastructure to provide access to scientific tools for research, demonstration, and collaboration, as well as instructional material.

NanoWIKI

- an internal toxicity knowledge base of, metal oxides; carries physicochemical properties and toxicological data for more than three hundred nanomaterials: more than two hundred metal oxides, 80 carbon nanotubes, and a few metal and alloy particles

Nanoinformatics databases

ACTION-Grid

- Creating links between Biomedical Informatics, Grid computing and the innovative field of Nanoinformatics; data and knowledge bases of nanoparticles and biological interactions; formation of nano-ontologies; research on interoperability and integration; data mining from large databases; Development of models for nanoparticles and connection of databases of toxic effects with computerized medical records

Nano Particle Ontology (NPO)

- Characterizes the knowledge fundamental to the description, preparation, and characterization of nanomaterials in cancer nanotechnology research.

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Nanotoxicity-I

Learning Objectives

Introduction

Importance

Potential harms

Nanotoxicology

- **Toxicology =** the science evaluating inherent **dangers** of substances.
- **Risk = exposure x danger**

- *Toxicology is the study of the adverse effects of chemical, physical or biological agents on people, animals, and the environment."*
- *"Nanotoxicology is the study of the adverse effects of engineered nanomaterials (ENMs) on living organisms and the ecosystems, including the prevention and amelioration of such adverse effects."*

Nanotoxicology

- Because of the small size and large surface area of nanomaterials, these materials have unique properties compared with their larger counterparts.

- Most reports find that ultrafine particles are more toxic than equivalent larger-sized particles of a given material at similar doses per gram of body weight.

- Nanotoxicological studies are intended to determine whether and to what extent these may pose a threat to the environment and to human beings.

The Premises, Promises and Problems of Nanotechnology

- Unforseen properties will emerge from nanoscale aggregates of atoms and molecules
- Not predicted by quantum mechanics (individual atoms) or classical physics (bulk materials)
- Novel physicochemical properties will open door to novel commercial applications
- The behavior of nanoscale objects are poorly characterized (esp. in “real-world” applications)

Why nanoparticle might be toxic

- Nanometer range is where life processes happen
- Inhaled ultrafine particles are toxic to lung and cardiovascular system
- Transported easily – go to unexpected places
- Accumulate in cellular organelles
- Some components are toxic as chemicals

Toxicity concerns of nanomaterials

- Very little human data concerning NP toxicity
- In vitro and in vivo animal studies highlight two concerns:

1)Immunotoxicity

2) Oxidative stress

Oxidative stress: surface properties matters

- Production of reactive oxygen species (ROS) a common NP property can reflect NP surface contamination with metals used for *in vitro NP screens*
- In vivo, oxidative damage to lipids, protein, DNA by NP

Challenges in evaluating Toxicology of nanomaterials

- Toxicity of nanomaterials may be different from its elemental forms
- Toxicity may change with exposure to light, pH changes, etc.
- Toxicity is frequently masked by biocoatings which may be stripped at different rates by different cell types
- “Toxicity”needs to assays beyond simple, rapid cell death, including: apoptosis, cell proliferation, cell differentiation, changes in cell function, etc.

Effect on Human health

1. Entry routes
    1. Inhalation : main route
    2. Oral route: via feeding (e.g Food packaging nanoparticles), via ingestion of inhaled particles
    3. Dermal route : via cosmetics...
2. Biological effects
    1. (cellular stress, inflammation, tumours)

Potential harms

- Potential toxicity, mechanism issues
- Harm to the environment and/or ecosystem
- through manufacture, use, and/or disposal
- Unknown transport, transformation and fate
- information of nanomaterials
- Potential bioaccumulation, biotransformation, and
- bioavailibility issues
- Dose/Response

Nanotoxicity-II

Learning Objectives

Human exposure and toxicology studies

ROS and toxicity

Toxicity in medical products

Human exposure and toxicology studies

Study Focus	Examples of specific effects investigated	Nanomaterials Tested
Cytotoxicity	Affinity to cell membranes, oxidative damage, structure-function relationships, mechanisms	aluminum oxide (Al_2O_3), cerium oxide (CeO_2), cupric oxide (CuO) dendrimers, iron oxide (Fe_2O_3), nickel oxide (NiO), silicon dioxide (SiO_2), titanium dioxide (TiO_2), zinc oxide (ZnO)
Dermal toxicity	Dermal absorption, cutaneous toxicity,	cadmium celenide (CdSe), fullerenes, iron (Fe)
General toxicity	Human blood coagulation, induction of inflammatory gene expression, genotoxicity	aluminum oxide (Al_2O_3), cadmium celenide (CdSe), cadmium telluride (CdTe) dendrimers, fullerenes, gallium nitride (GaN)Geranium, lead selenide (PbSe), nanofibers, nanowires, quantum dots, silicon dioxide (SiO_2), quantum dots, titanium dioxide (TiO_2), zinc sulfide (ZnS)
Pulmonary toxicity	Oxidative stress, inflammation, surface coating effects, nano/non-nano effects, new/aged agglomerated effects, clearance mechanisms	aluminum oxide (Al_2O_3), cerium oxide (CeO_2), cupric oxide (CuO) dendrimers, gold (Au), iron oxide (Fe_2O_3), multiwalled nanotubes (MWNT), nickel oxide (NiO), silicon dioxide (SiO_2), single walled nanotubes (SWNT), silver (Ag), titanium dioxide (TiO_2), zinc oxide (ZnO)
Translocation/Disposition	Translocation to sites distant from original exposure, persistence in vivo.	aluminum oxide (Al_2O_3), iron oxide (Fe_2O_3), titanium dioxide (TiO_2), silicon dioxide (SiO_2), zinc oxide (ZnO)

Reactive oxygen species (ROS)

- Reactive oxygen species (ROS) are ions or very small molecules that include oxygen ions, free radicals, and peroxides, both inorganic and organic. They are highly reactive due to the presence of unpaired valence shell electrons.

- ROS form as a **natural byproduct** of the normal metabolism of oxygen and have important roles in cell signaling.

- **Environmental stresses** may induce ROS leading to cell damage. This cumulates into a situation known as oxidative stress. ROS are also generated by exogenous sources such as ionizing radiation.

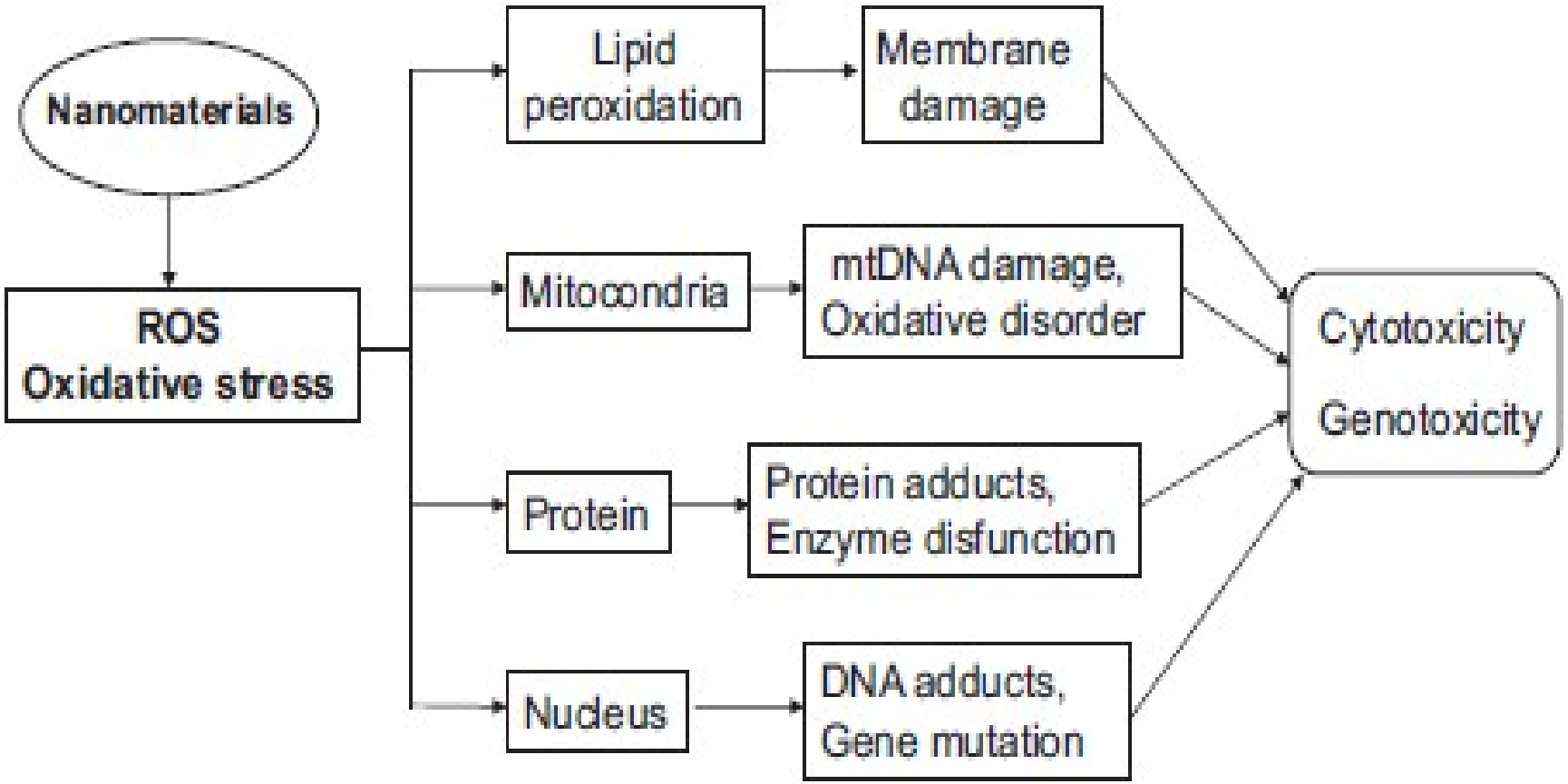
Ag nanoparticles and ROS

- Man-made nanomaterials with their unique physical (size, shape, crystallinity, surface charge) and chemical (surface coating, elemental composition and solubility) attributes, can produce chemical conditions to induce a pro-oxidant environment in the cells, causing an imbalanced cellular energy system dependent on redox potential and leading to adverse biological consequences, ranging from the initiation of inflammatory pathways through to cell death.

- Silver nanoparticles of 15, 30 and 55 nm are known to be biologically active, inducing stress mostly through ROS production initiation

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ROS and cytotoxicity



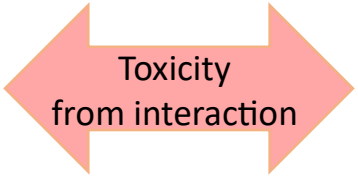
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Fish

Algae

Fungi

Bacteria



Carbon nanotubes

Metal oxides

Gold nanoparticles

Polymeric nanoparticles

Safety concerns of ENMs in medical products

- Routes of exposure –different cellular/tissue targets
- Different kinetics due to size, surface charge
- Higher exposure per unit mass –biological effects correlate with ENM surface area
- Distribution in tissues –surface chemistry/coatings
- Metabolism and clearance

Effects, associated with NPs on human health

- Pulmonary inflammation
- Immune adjuvent effects
- Systemic effects
  - Blood coagulation
  - Cardiovascular effects

Problems/ concerns for particulate drug carriers

- Nanomaterials are entrapped in mononuclear phagocytic systems as in liver and spleen.

Toxicity concern: NPs in drug delivery (Engineered Nanoparticles)

- The primary goals for research of nano-bio-technologies in drug delivery include:
  - More specific drug targeting and delivery
  - Reduction in toxicity while maintaining therapeutic effects
  - Greater safety and biocompatibility, and
  - Faster development of new safe medicines

Pre-requisites for the designing of nanocarrier is to have knowledge on

- Drug incorporation and release
- Formulation stability and shelf life
- Biocompatibility
- Biodistribution and targeting
- Functionality
- Effects of residual materials after drug delivery

Nanotoxicity-III

Learning Objectives

Issues for safety evaluation of ENMs
Nanotoxicology in vivo studies
Biocompatibility

Nanotoxicology: what we don't know outweights what we know

- Characterization/metrology (e.g., size, surface charge and chemistry, agglomeration/aggregation)What happens in biological fluids/tissue

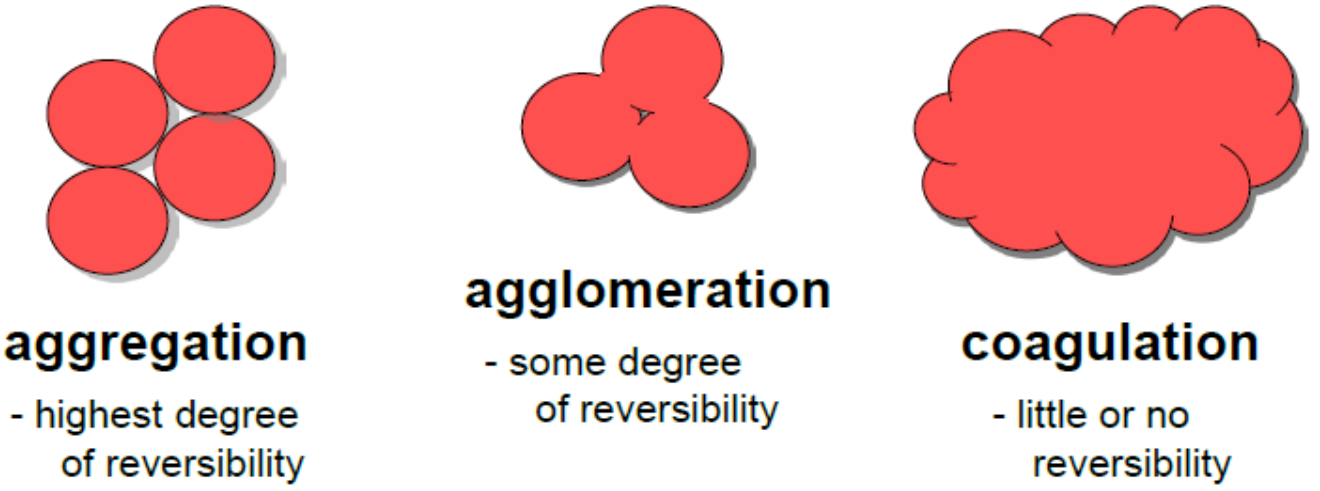
- Toxicokinetics –absorption, tissue distribution, excretion and Detection methods

- Effect of size, surface charge on toxicity
- Dose metrics –concentration (mass), number, surface area?
- Membrane translocation, cellular uptake
- Adequacy of existing toxicity tests (*in vitro*& *in vivo*)
- Interference with laboratory assay kits

Issues for safety evaluation of ENMs

1. Effect of biological media on ENM/ particulate durg nanocarriers properties

- Hazard assessment depends on an understanding of the properties (e.g., size, surface charge) of ENMs that drive biological responses.
- Properties of ENMs can change dramatically with dispersion media, e.g., water vs serum.



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Some factors that influence NP aggregation

- Surface chemistries (surface charge and functionalization, capping agents)

- Particle-particle & particle-matrix (dispersants, vehicles, Biological media) interactions

Nanotoxicology in vivo studies

- There are fewer *in vivo* than *in vitro* studies which address nanomaterial toxicity.
- Of the *in vivo* studies, few address Toxicokinetics - **a**bsorption, tissue **d**isposition**, m**etabolism, and **e**xcretion (ADME).
  - Even fewer ADME and toxicity data are available for ENM exposures (e.g., sc and iv) relevant to medical devices and therapeutics.
  - Most studies have used other routes of exposure: Inhalation Skin penetration

Nanotoxicology in vivo studies

- Very few studies have evaluated nanoparticle exposure in pregnant animals. Public health impact for women receiving Nanomedicine-based therapies during pregnancy

What we want to know

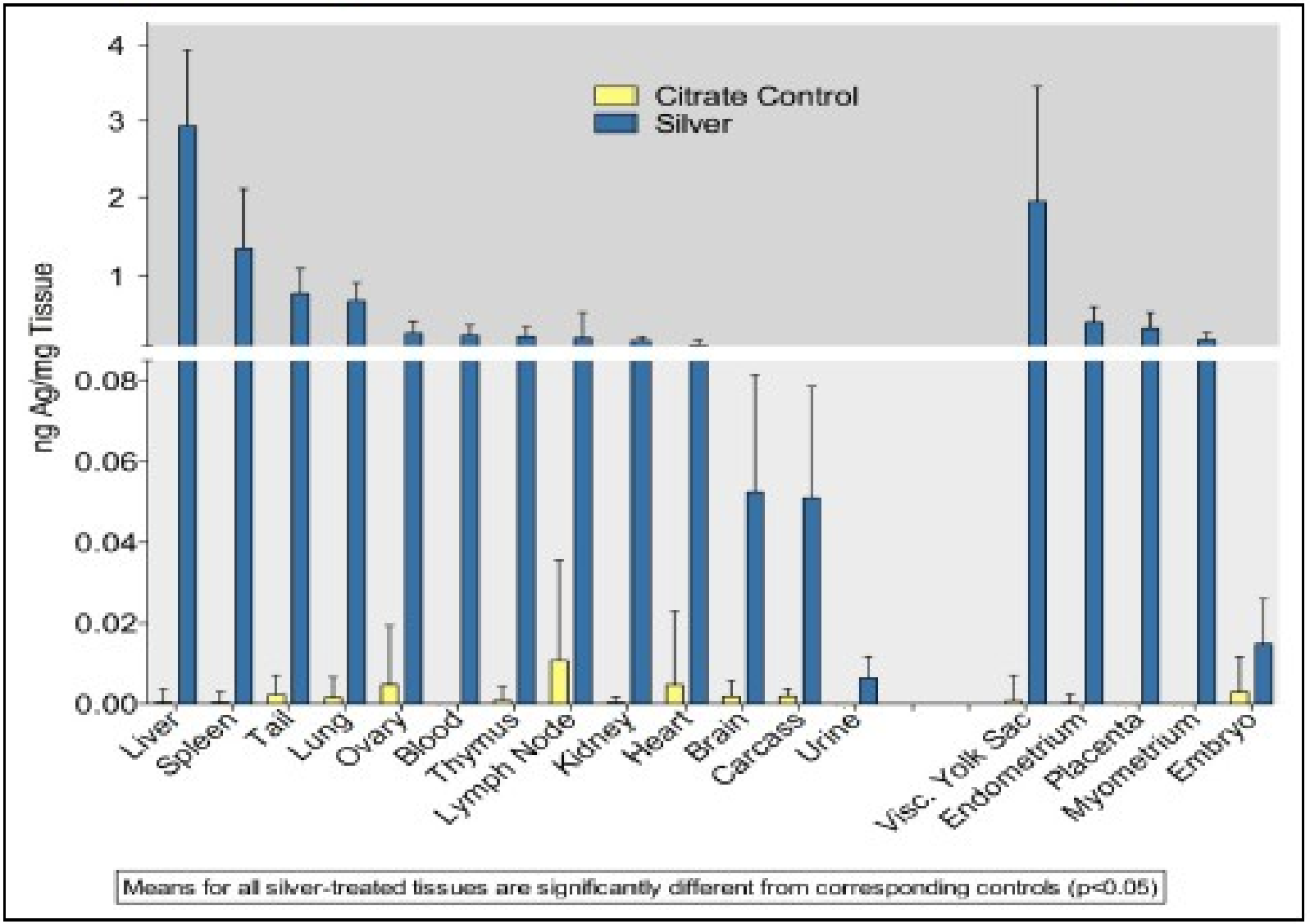
- Where do they go in the body?
- Is there placental transfer to reach embryos?
- Are there adverse effects to the embryos?

Example: Ag NP distribution in pregnant mouse

- Silver (Ag) nanoparticles were injected intravenously into pregnant CD-1 mice on gestation days (GDs) 7, 8, and 9 (critical time of organogenesis). Mice euthanized and tissues were harvested on GD10. Tissue Ag levels were quantitated using Inductively coupled plasma mass spectrometry (ICP-MS).
- Total amount of silver injected: **~9 μg Ag/mouse**
- **Tissues harvested:**

Brain	Lymph nodes	Myometrium of the uterus
Heart	Ovaries	Endometrium of the uterus
Kidneys	Spleen	Chorio-allantoic placentas
Liver	Thymus	Visceral yolk sacs
Lungs	Blood/urine	Embryos

Ag tissue concentrations after 3 daily AgNP injections - ICP-MS analysis



In-vitro evaluation

- ISO 10993 –"Biological evaluation of medical devices"
- Compare the biological responses of nanoparticles and micron-sized particles of the same material in cultured macrophage cells.

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Biocompatibility

- Biocompatibility testing answers two fundamental questions:
  - Is the material safe?
  - Does it have the necessary physical and mechanical properties for its proposed function?
- The extent to which a material needs to be characterized depends on:
  - Type of material,
  - End use of the device (is it a medical device?),
  - Function of the material within the device.
  - Availability of existing data on the material.

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- Biocompatibility needs to be considered at the onset of design:
  - Use known biocompatible materials.
  - Individual components as well as overall packaging are important.
- Material testing is performed to determine toxicity of the material, leachable substances and degradation products.
- The FDA relies on ISO 10993 Standard and has additional recommended steps.

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Characterization methods

- Identification of a materials constituents and:
  - Changes of the material over time,
  - Changes with exposure to different environments,
  - Lot-to-lot consistency for manufacturing purposes.
- Methodologies:
  - Infrared spectral analysis (IR),
  - Thermal analysis,
  - Density analysis,
  - Molecular weight distribution,
  - Mechanical properties,
  - Surface properties,
  - Extract Characterization.

Future Prospects of Nanotechnology

Learning Objectives

Future of biomedical

Future of Nanosensors

Self-healing structures

Tackling Climate change

Future Prospects of Nanotechnology

- The past 70 years have seen the way we live and work transformed by two tiny inventions.
- The electronic transistor and the microchip are what make all modern electronics possible, and since their development in the 1940s they've been getting smaller.
- To keep this progress going we need to be able to create circuits on the extremely small, nanometre scale.

Future Prospects of Nanotechnology

- The real challenge is using such techniques reliably to manufacture working nanoscale devices.
- The physical properties of matter, such as its melting point, electrical conductivity and chemical reactivity, become very different at the nanoscale, so shrinking a device can affect its performance.
- If we can master this technology, however, then we have the opportunity to improve not just electronics but all sorts of areas of modern life including biomedical applications.

Doctor inside your body

- By scaling down this technology, we could go further by implanting or injecting tiny sensors inside our bodies.
- This would capture much more detailed information with less hassle to the patient, enabling doctors to personalize their treatment.
- The possibilities can involved in monitoring inflammation and post-surgery recovery to more exotic applications whereby electronic devices actually interfere with our body's signals for controlling organ function.



Medical nanobots

Future Prospects of Nanosensors

- Sensors rely on newly-invented nanomaterials and manufacturing techniques to make them smaller, more complex and more energy efficient.
- For example, sensors with very fine features can now be printed in large quantities on flexible rolls of plastic at low cost.
- This opens up the possibility of placing sensors at lots of points over critical infrastructure to constantly check that everything is running correctly.
- Bridges, aircraft and even nuclear power plants could benefit.

Future Prospects of Nanosensors

- Cracks do appear, then nanotechnology could play a further role.
- Changing the structure of materials at the nanoscale can give them some amazing properties – by giving them a texture that repels water,
- for example. In the future, nanotechnology coatings or additives will even have the potential to allow materials to "heal" when damaged or worn.



Cracks

Tackling climate change

- The fight against climate change means we need new ways to generate and use electricity, and nanotechnology is already playing a role.
- It has helped create batteries that can store more energy for electric cars and has enabled solar panels to convert more sunlight into electricity.
- In the future, nanotechnology could also enable objects to harvest energy from their environment.
- New nano-materials and concepts are currently being developed that show potential for producing energy from movement, light, variations in temperature, glucose and other sources with high conversion efficiency.

REMEMBER ME IN YOUR PRAYERS
SAY NO TO CORRUPTION
"ADMIN TABIH FATIMA"

STAY SAFE, STAY HAPPY
JOIN OUR FACEBOOK GROUP
"ALL ABOUT BIOTECH"